Problems in Analysis

Real and complex analysis, with solutions

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Introduction

These are problems in real and complex analysis that we worked through together during our graduate studies at Wayne State University, with our solutions. The problems are grouped by topic. The first section collects the standard results that the solutions use; try each problem before you open its solution.

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1 Background

1.1 Inequalities

Proposition (Triangle inequalities). *Let (X, d) be a metric space. For $x, y, z \in X$, we have:*

1. *The triangle inequality: $d(x,z) \leq d(x,y) + d(y,z)$.*
2. *The reverse triangle inequality: $d(x,z) \geq |d(x,y) - d(y,z)|$.*

Theorem (AM-GM inequality). *Let $a, b \geq 0$. Then*

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Proof. Notice that

$$\begin{aligned} 0 &\leq (a-b)^2 \\ &= a^2 - 2ab + b^2 \\ &= a^2 + 2ab + b^2 - 4ab \\ &= (a+b)^2 - 4ab. \end{aligned}$$

Therefore

$$4ab \leq (a+b)^2 \implies \sqrt{ab} \leq \frac{a+b}{2}.$$

□

Theorem (Cauchy-Schwarz inequality). *Let $(V, \cdot)$ be an inner product space. For $u, v \in V$, we have:*

$$|u \cdot v|^2 \leq (u \cdot u)(v \cdot v).$$

Corollary.

1. *Let $V = \mathbb{R}^n$ with the standard dot product. Then for $(u_1, \ldots, u_n)$, $(v_1, \ldots, v_n) \in \mathbb{R}^n$, the Cauchy-Schwarz inequality reads*

$$\left(\sum_{i=1}^{n} u_i v_i\right)^2 \leq \left(\sum_{i=1}^{n} u_i^2\right)\left(\sum_{i=1}^{n} v_i^2\right).$$

2. *Generalizing the previous example, let $V = \ell^2(\mathbb{N})$ be the space of all* square-summable *complex-valued sequences, with the usual dot product*

$$(a_i)_{i\geq 1} \cdot (b_i)_{i \geq 1} = \sum_{i=1}^{\infty} a_i \overline{b_i}.$$

Then the Cauchy-Schwarz inequality reads:

$$\left|\sum_{i=1}^{\infty} a_i \overline{b_i}\right|^2 \leq \left(\sum_{i=1}^{\infty} |a_i|^2\right)\left(\sum_{i=1}^{\infty} |b_i|^2\right).$$

3. *Generalizing further, let $V = L^2(X)$ be the space of all square-integrable complex-valued functions on a space X. Notice this includes all continuous functions with compact support. Then the Cauchy-Schwarz inequality reads:*

$$\left|\int_X f(x)\overline{g(x)}\,dx\right|^2 \leq \left(\int_X |f(x)|^2\,dx\right)\left(\int_X |g(x)|^2\,dx\right).$$

Theorem (Hölder's inequality). *Let (S, μ) be a measure space, let $p, q \in [1, \infty]$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, and let $f \in L^p(S)$ and $g \in L^q(S)$. Then*

$$\int |f(x)g(x)|\,d\mu \leq \left(\int |f(x)|^p\,d\mu\right)^{\frac{1}{p}}\left(\int |g(x)|^q\,d\mu\right)^{\frac{1}{q}}$$

Theorem (Jensen's inequality). *For any $\Omega \subseteq \mathbb{R}^k$ of measure $|\Omega| = 1$, if $f : \Omega \to \mathbb{R}$ is integrable and $\phi : \mathbb{R} \to \mathbb{R}$ is convex, then*

$$\phi\left(\int_\Omega f(x)\,dx\right) \leq \int_\Omega \phi(f(x))\,dx.$$

1.2 Convergence of series

Proposition (D'Alembert criterion). *Given positive series $\sum_{n=1}^{\infty} a_n$. If*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} < 1$$

then $\sum_{n=1}^{\infty} a_n$ converges.

Proposition (Dirichlet's test). *Let $\{a_n\}$ be a sequence of real numbers and $\{b_n\}$ a sequence of complex numbers. If*

- *$\{a_n\}$ is monotonic,*
- *$\lim_{n\to\infty} a_n = 0$, and*
- *there exists an $M > 0$ such that*

$$\left|\sum_{i=0}^{n} b_i\right| < M$$

for all $n > 0$,

then the series

$$\sum_{i=0}^{\infty} a_i b_i$$

converges.

Proposition (Abel's identity). *If we have the following series* $\sum_{k=0}^{\infty} a_k b_k$ *then we have the following identity*

$$\sum_{k=0}^{N} a_k b_k = a_N B_N - \sum_{k=0}^{N-1} (a_{k+1} - a_k) B_k, \ B_n = \sum_{k=0}^{n} b_k$$

Proof. By induction, for $N = 1$ we have

$$R.H.S = a_1 B_1 - (a_1 - a_0) B_0 = a_1(b_0 + b_1) - (a_1 - a_0) b_0 = a_1 b_1 + a_0 b_0 = L.H.S$$

Assume it is true for N, we want to show it for $N + 1$

$$\begin{aligned}
\sum_{k=0}^{N+1} a_k b_k &= a_{N+1} b_{N+1} + \sum_{k=0}^{N} a_k b_k \\
&= a_{N+1} b_{N+1} + a_N B_N - \sum_{k=0}^{N-1} (a_{k+1} - a_k) B_k \\
&= a_{N+1} b_{N+1} + a_{N+1} B_N - a_{N+1} B_N + a_N B_N - \sum_{k=0}^{N-1} (a_{k+1} - a_k) B_k \\
&= a_{N+1}(b_{N+1} + B_N) - B_N(a_{N+1} - a_N) - \sum_{k=0}^{N-1} (a_{k+1} - a_k) B_k \\
&= a_{N+1} B_{N+1} - \sum_{k=0}^{N} (a_{k+1} - a_k) B_k
\end{aligned}$$

□

1.3 Sequences and series of functions

Definition. Let $\{f_n\}$ be a sequence of real-valued functions on a metric space D.

- The sequence is called *uniformly bounded* if there exists an $M > 0$ such that

  $$|f_n(x)| < M$$

  for all $x \in D$ and $n \in \mathbb{N}$.
- The sequence is called *uniformly equicontinuous* if for all $\varepsilon > 0$ there exists a $\delta > 0$ such that, for any $x, y \in D$ such that $d(x, y) < \delta$, we have

  $$|f_n(x) - f_n(y)| < \varepsilon$$

  for all $n \in \mathbb{N}$. (In other words, not only are all the functions uniformly continuous, but there is a single δ that works for all functions at once.)
- The sequence is called *uniformly convergent* if there exists a function f such that for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$ and for all $x \in D$, we have

  $$|f_n(x) - f(x)| < \varepsilon.$$

Theorem (Arzelà-Ascoli). *Let X be a compact metric space. Recall that $C(X)$ is a normed vector space with norm*

$$||f||_\infty = \sup_{x \in X} |f(x)|.$$

Let $F \subseteq C(X)$ be a set of continuous real-valued functions on X. Then F is relatively compact (its closure is compact) if and only if it is bounded with respect to this norm and equicontinuous.

Proposition. *The series of functions $\sum_{n=1}^{\infty} a_n(x)$ converges uniformly on D iff*

$$\forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}, \forall m, n \in \mathbb{N}, m > n \geq n_\varepsilon \Rightarrow \left| \sum_{k=n+1}^{m} a_k(x) \right| < \varepsilon, \forall x \in D.$$

Proposition (Weierstrass M-test). *Suppose that $|f_n(x)| \leq a_n$ for all $n \in \mathbb{N}$ and $x \in D$. Then, if $\sum_{n=1}^{\infty} a_n$ converges, $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly.*

Proposition. *Given $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly to $f(x)$ on $[a, b]$. Then, if for all $n \in \mathbb{N}$, f_n is integrable on $[a, b]$, then f is integrable on $[a, b]$. Furthermore, we have*

$$\int_a^b f(x)\, dx = \int_a^b \left(\sum_{n=1}^{\infty} f_n(x) \right) dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\, dx.$$

1.4 Topology

Definition. Let X be a topological space. A subset $A \subset X$ is called dense in X if $\overline{A} = X$. Equivalently, A is dense iff $A \cap U \neq \emptyset$ for any nonempty open subset $U \subseteq X$.

Lemma (Lebesgue's Covering Lemma / number lemma). *Let (X, d) be a compact metric space, and $\{U_i\}_{i \in I}$ an open cover of X. Then there exists an $\varepsilon > 0$ such that, for any $x \in X$, $B(x, \varepsilon) \subseteq U_i$ for some $i \in I$. (The lemma gets its name because this ε is usually called the* Lebesgue number *for the cover.)*

1.5 Inverse Function Theorem

Theorem. *Let D be an open subset of $\mathbb{R}^n$, $f : D \to \mathbb{R}^n$ is continuously differentiable on D and $a \in D$. If $\det f'(a) \neq 0$, there exists open subset $U \subset D$ containing a, V containing $f(a)$ such*

that $f : U \to V$ is bijective and $f^{-1} : V \to U$ is also continuously differentiable given by

$$(f^{-1})'(y) = \left[f'(f^{-1}(y))\right]^{-1}, \forall y \in V,$$

the inverse of the Jacobian matrix.

1.6 Complex analysis

Theorem (Open Mapping Principle). *Any non-constant analytic function on a domain maps open sets to open sets.*

Theorem (Maximum Modulus Principle). *Let $f : U \to \mathbb{C}$ be a holomorphic function on a domain. If $|f|$ has a local maximum, then f is constant.*

Theorem (Liouville's theorem). *Let $f : \mathbb{C} \to \mathbb{C}$ be entire. If f is bounded, then f is constant.*

Theorem (Residue Theorem). *Let C be a simple closed contour, described in the positive sense. If a function f is analytic inside and on C except for a finite number of singular points $z_k, k = 1, 2, \ldots, n$ inside C, then*

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \mathrm{Res}_{z=z_k} f(z).$$

Corollary. *Suppose that a function f is continuous on a closed bounded region R and that it is analytic and not constant in the interior of R. Then the maximum value of $|f(z)|$ in R, which is always reached, occurs somewhere on the boundary of R and never in the interior.*

Theorem (Jordan's lemma). *Suppose that f is analytic in all points in the upper half plane above circle $|z| = R_0$, and let C_R denote the upper half of the circle $|z| = R$. If, for $R > R_0$ and each $z \in C_R$, we have $|f(z)| \leq M_R$ with $M_R \to 0$ as $R \to \infty$, then*

$$\lim_{R\to\infty} \int_{C_R} e^{iaz} f(z) dz = 0 \quad (a > 0).$$

Theorem (Small half-circles). *Suppose that f has a simple pole at $z = x_0$ on the real line with a Laurent series representation in a punctured disk $0 < |z - x_0| < R_2$ with residue B_0. If C_ρ denote the upper half of a circle $|z - x_0| = \rho$, where $\rho < R_2$, and clockwise oriented, then*

$$\lim_{\rho\to 0} \int_{C_\rho} f(z) dz = -B_0 \pi i.$$

Theorem (Rouché's theorem). *Let C denote a simple closed contour, and suppose that*
(a) Two functions $f(z)$ and $g(z)$ are analytic inside and on C;
(b) $|f(z)| > |g(z)|$ at each point on C.
Then $f(z)$ and $f(z) + g(z)$ have the same number of zeros, counting multiplicities, inside C.

Theorem (Schwarz lemma). *Let $D = \{z \mid |z| < 1\}$ be the unit open disk on the complex plane and $f : D \to \mathbb{C}$ be an analytic function such that $f(0) = 0$ and $|f(z)| \leq 1$ for all $z \in D$. Then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Moreover, if there is some nonzero z such that $|f(z)| = |z|$, or if $|f'(0)| = 1$, then $f(z) = az$ with $|a| = 1$.*

Proof. Since $f(0) = 0$, $f(z)/z$ has a removable singularity at $z = 0$, hence there is some function g analytic on D such that $f(z) = zg(z)$. Then by the Maximum Modulus Principle, for $z \in D$ and $|z| < r < 1$, we have

$$|g(z)| \leq \max_{0<\theta<2\pi} \frac{|f(re^{i\theta})|}{r} \leq \frac{1}{r}.$$

Letting $r \to 1$, we find that $|g(z)| \leq 1$ for all $z \in D$, or equivalently, $|f(z)| \leq |z|$ for all $z \in D$. Moreover,

$$f'(0) = g(0) + 0g'(0) = g(0),$$

so we have $|f'(0)| \leq 1$. If $|g(z)| = 1$ for some $z \neq 0$, or $|g(0)| = 1$, the Maximum Modulus Principle gives that g is constant. Therefore, $f(z) = az$ with $|a| = 1$. □

Theorem (Morera's theorem). *Let $f : D \to \mathbb{C}$ be a continuous function on a domain D. If, for every closed curve γ in D, we have*

$$\oint_\gamma f(z)\, dz = 0,$$

then f is holomorphic on D. (The converse is not true: $f = 1/z$ on the punctured unit disk is a counterexample.)

Theorem (Riemann Mapping Theorem). *If $U \subset \mathbb{C}$ is a simply-connected open subset of $\mathbb{C}$ which is not all of $\mathbb{C}$, then there exists a biholomorphism (a bijective holomorphic map with a holomorphic inverse)*

$$f : U \to \{z \in \mathbb{C} \mid |z| < 1\}.$$

Theorem (Riemann's theorem). *Suppose that a function f is analytic and bounded in some deleted neighborhood $0 < |z - z_0| < \varepsilon$ of a point z_0. If f is not analytic at z_0, then it has a removable singularity there.*

2 Sequences and limits

Problem 2.1. Suppose the sequence is defined recursively by $a_1 = 1$ and

$$a_{n+1} = \sqrt{2a_n}$$

. Show that this sequence converges and determine its limit.

Solution. Note that $a_n < 2$ for any n, we have $a_1 = 1 < 2$, suppose it is true for $a_n < 2$, now for $a_{n+1} = \sqrt{2a_n} < \sqrt{2 \cdot 2} = 2$. In addition

$$a_{n+1} - a_n = \sqrt{2a_n} - a_n = \sqrt{a_n}(\sqrt{2} - \sqrt{a_n}) > 0$$

Thus, a_n is bounded monotone squence so it is convergent assume $\lim a_n = L$. Since $a_n \geq a_1 = 1$, we have $L \geq 1 > 0$, so we may divide by $\sqrt{L}$:

$$a_{n+1} = \sqrt{2a_n} \Rightarrow L = \sqrt{2L} \Rightarrow \sqrt{L} = \sqrt{2} \Rightarrow L = 2$$

□

Problem 2.2. Let $\{x_n\}$ be defined by $x_1 = 1$ and $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$ for $n = 1, 2, \ldots$ Show that $\{x_n\}$ converges and find its limit.

Solution. Firstly, it is obvious that $x_n > 0$ for all n. Then AM-GM inequality tells us $x_{n+1} \geq \sqrt{2}$ for all $n \in \mathbb{N}$. Moreover,

$$x_{n+1} - x_n = \frac{1}{2}(\frac{2}{x_n} - x_n) = \frac{2 - x_n^2}{2x_n} \leq 0, \text{ for all } n \geq 2.$$

Then $\{x_n\}_{n \geq 2}$ is a decreasing sequence, bounded below by $\sqrt{2}$, so the limit exists. Suppose that the limit is $a > 0$, we must have $a = \frac{1}{2}(a + \frac{2}{a})$, or equivalently, $a = \sqrt{2}$. □

Problem 2.3. Let $x_0 = 1$ and $x_{n+1} = \sqrt{x_n + 4}$ for $n \geq 0$. Prove $\{x_n\}$ converges and find the limit.

Solution. By the construction of $\{x_n\}$ and $x_0 = 1$, it is obvious that $x_n \geq 0$ for all n. Let $L = \frac{1 + \sqrt{17}}{2}$, the positive root of $L^2 = L + 4$. We will prove that $x_n < L$ for all n by induction. This holds for $n = 0$ since $x_0 = 1$. Suppose that it holds for $n = k$, i.e. $x_k < L$, we have

$$x_{k+1} = \sqrt{x_k + 4} < \sqrt{L + 4} = L.$$

Thus $x_n < L$ for all n. We also have

$$x_{n+1} - x_n = \sqrt{x_n + 4} - x_n = \frac{x_n + 4 - x_n^2}{\sqrt{x_n + 4} + x_n} > 0 \text{ for all } n,$$

because $x^2 - x - 4 < 0$ for $0 \leq x < L$. Then $\{x_n\}$ is increasing and bounded from above, so its limit exists. We denote $\lim x_n = a$, we have $a = \sqrt{a+4}$ which gives $a = \frac{1+\sqrt{17}}{2}$. □

Problem 2.4. Assume a sequence $\{x_n\}_{n=1}^{\infty}$ is define recursively by $x_1 = 2$ and

$$x_{n+1} = x_n - \frac{(x_n)^3 - x_n}{3(x_n)^2 - 1} \quad \text{for all } n \geq 1$$

(a) Prove that $\{x_n\}$ is decreasing sequence for each $n, x_n \geq 1$; and
(b) Find $\lim_{n\to\infty} x_n$

Solution. We can write

$$x_{n+1} = \frac{2x_n^3}{3x_n^2 - 1}$$

To show the sequence decreasing, we find

$$\frac{x_{n+1}}{x_n} = \frac{3x_n^2 - 1 - x_n^2 + 1}{3x_n^2 - 1} = \frac{2x_n^2}{3x_n^2 - 1} = \frac{2}{3 - \frac{1}{x_n^2}} \leq 1$$

Since $x_1 = 2$, we claim that $x_n \geq 1$, suppose $x_n \geq 1$ we want to $x_{n+1} \geq 1$ notice that the function

$$f(x) = \frac{2x^3}{3x^2 - 1} \Rightarrow f'(x) = \frac{6x^2(x^2-1)}{(3x^2-1)^2} \Rightarrow f'(x) > 0 \ \ for x \geq 1$$

And $f(1) = 1$ but $x_n \geq 1$ so

$$\frac{2x_n^3}{3x_n^2 - 1} \geq 1$$

. So the sequence is decreasing and bounded so it is coverges. To find the limit,

$$\lim x_{n+1} = \lim \frac{2x_n^3}{3x_n^2 - 1} \Rightarrow l = \frac{2l^3}{3l^2 - 1} \Rightarrow \frac{2l(l^2-1)}{3l^2 - 1} = 0$$

But $l \neq 0, -1$ so $l = 1$. □

Problem 2.5. Assume that B is a positive number. Let $\{x_n\}$ be defined by $x_1 = 1$ and

$$x_{n+1} = \frac{x_n(3B + x_n^2)}{3x_n^2 + B}.$$

Prove that $x_n \to \sqrt{B}$.

Solution. Note that

$$\begin{aligned} x_{n+1} - x_n &= \frac{x_n(3B + x_n^2)}{3x_n^2 + B} - x_n \\ &= \frac{3Bx_n + x_n^3 - 3x_n^3 - Bx_n}{3x_n^2 + B} \\ &= \frac{2Bx_n - 2x_n^3}{3x_n^2 + B} \\ &= 2x_n \frac{B - x_n^2}{3x_n^2 + B} \end{aligned} \tag{1}$$

and

$$\begin{aligned} x_{n+1} - \sqrt{B} &= \frac{x_n(3B + x_n^2)}{3x_n^2 + B} - \sqrt{B} \\ &= \frac{3Bx_n + x_n^3 - 3x_n^2\sqrt{B} - B\sqrt{B}}{3x_n^2 + B} \\ &= \frac{(x_n - \sqrt{B})^3}{3x_n^2 + B}. \end{aligned} \tag{2}$$

From (2), by induction, if $1 = x_1 < \sqrt{B}$, we have $x_n < \sqrt{B}$ for all n and if $1 = x_1 > \sqrt{B}$ then $x_n > \sqrt{B}$ for all n. We consider 3 cases.

Case 1. If $B = 1$ then by (2), $x_n = 1$ for all n, thus $x_n \to 1 = \sqrt{B}$.

Case 2. If $B > 1$ then by (2) $x_n < \sqrt{B}$ for all n. From (1), we have $x_{n+1} > x_n$ for all n. Then $\lim x_n$ exists. We suppose that $\lim x_n = a$. Then from (2), we have

$$(a - \sqrt{B})(1 - \frac{(a - \sqrt{B})^2}{3a^2 + B}) = 0$$

Then $a = \sqrt{B}$ or $(a - \sqrt{B})^2 = 3a^2 + B$. Since $a > 0$ the latter means $a = -\sqrt{B}$, which is a contradiction. Therefore, $\lim x_n = \sqrt{B}$.

Case 3. If $B < 1$ then by (2) $x_n > \sqrt{B}$ for all n. From (1), we have $x_{n+1} < x_n$ for all n. Then $\lim x_n$ exists. We suppose that $\lim x_n = a \geq \sqrt{B}$. Then from (2), we have

$$(a - \sqrt{B})(1 - \frac{(a - \sqrt{B})^2}{3a^2 + B}) = 0$$

Then $a = \sqrt{B}$ or $(a - \sqrt{B})^2 = 3a^2 + B$. Since $a > 0$ the latter means $a = -\sqrt{B}$, which is a contradiction. Therefore, $\lim x_n = \sqrt{B}$.

□

Problem 2.6.

(a) For each $n \in \mathbb{N}$, prove that

$$\frac{1}{n+1} \leq \int_n^{n+1} \frac{1}{x}\, dx \leq \frac{1}{n}$$

(b) Define

$$a_n = \sum_{k=1}^{n} \frac{1}{k} - \ln(n+1),$$

$n = 1, 2, \cdots$. Prove $a_n \to L$ for some number L with $0 \leq L \leq 1$.

Solution. a) By the definition of definite integrals using Riemann sums for any partition $P = \{x_1, \cdots, x_k, x_{k+1}\}$ of $[n, n+1]$ we have

$$\sum_{j=1}^{k} m_j \Delta x_j \leq \int_n^{n+1} \frac{1}{x}\, dx \leq \sum_{j=1}^{k} M_j \Delta x_j$$

Where $m_j = \min_{x \in [x_j, x_{j+1}]} \{f(x)\}, M_j = \max_{x \in [x_j, x_{j+1}]} \{f(x)\}, \Delta x_j = x_{j+1} - x_j$. So if we consider the partition $P = \{n, n+1\}$ we get $m_1 = \frac{1}{n+1}, M_1 = \frac{1}{n}$ hence

$$\frac{1}{n+1}(n+1-n) \leq \int_n^{n+1} \frac{1}{x}\, dx \leq \frac{1}{n}(n+1-n)$$

b) Summing the left inequality of a) for $k = 1, \cdots, n$ and the right inequality of a) for $k = 1, \cdots, n-1$, we get

$$\begin{aligned}
\left(\int_1^2 \frac{1}{x} + \cdots + \int_n^{n+1} \frac{1}{x} - \ln(n+1)\right) &\leq \sum_{k=1}^{n} \frac{1}{k} - \ln(n+1) \leq \left(1 + \int_1^2 \frac{1}{x} + \cdots + \int_{n-1}^{n} \frac{1}{x} - \ln(n+1)\right) \\
(\ln(n+1) - \ln(1) - \ln(n+1)) &\leq \sum_{k=1}^{n} \frac{1}{k} - \ln(n+1) \leq (1 + \ln(n) - \ln(1) - \ln(n+1)) \\
0 &\leq \sum_{k=1}^{n} \frac{1}{k} - \ln(n+1) \leq 1
\end{aligned}$$

so a_n is bounded. If we take $a_{n+1} - a_n = \frac{1}{n+1} - \ln(n+2) + \ln(n+1) = \frac{1}{n+1} - \int_{n+1}^{n+2} \frac{1}{x}\, dx \geq 0$ by a), so it is increasing bounded sequence so it converges to $L \in [0, 1]$ □

3 Series of numbers

Problem 3.1. Suppose $\{a_n\}$ is a sequence such that $\sum_{n=1}^{\infty} a_n^2$ converges. Show that $\sum_{n=1}^{\infty} \frac{a_n}{n}$ converges.

Solution. Using the arithmetic and geometric means inequality $a_n^2 + \frac{1}{n^2} \geq 2(\sqrt{a_n^2 \cdot \frac{1}{n^2}}) =$

$2\frac{|a_n|}{n}$, but $\sum a_n^2$ converges and $\sum 1/n^2$ converges it is a p-series or we can use the integral test. Thus $\sum \frac{|a_n|}{n}$ converges absolutely by comparison test so it converges. □

Problem 3.2. Suppose that $\{a_n\}$ is a positive sequence and $\sum_{n=1}^{\infty} a_n$ converges. Show that $\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$ converges.

Solution. Note that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. By the AM-GM inequality, we have

$$a_n + \frac{1}{n^2} \geq 2\frac{\sqrt{a_n}}{n}, \text{ for all } n \in \mathbb{N}.$$

This means that

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n} \leq \frac{1}{2}\sum_{n=1}^{\infty} a_n + \frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

□

Problem 3.3.

(a) Suppose $a_n \geq 0$ and the series $\sum a_n$ converges. Prove the series

$$\sum \frac{\sqrt{a_n}}{n^{2/3}}$$

also converges.

(b) Give an example of a divergent series $\sum a_n$ such that the series

$$\sum \frac{\sqrt{a_n}}{n^{2/3}}$$

converges.

Solution.

(a) Let $N \in \mathbb{N}$. By the Cauchy-Schwarz inequality with $u_n = \sqrt{a_n}$, $v_n = \frac{1}{n^{2/3}}$, we have that

$$\left(\sum_{n=1}^{N} \frac{\sqrt{a_n}}{n^{2/3}}\right)^2 \leq \left(\sum_{n=1}^{N} a_n\right)\left(\sum_{n=1}^{N} \frac{1}{n^{4/3}}\right).$$

Using that $\sum a_n$ converges by assumption and that $\sum \frac{1}{n^{4/3}}$ converges because $4/3 > 1$, we find

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^{2/3}} \leq \sqrt{\left(\sum_{n=1}^{\infty} a_n\right)\left(\sum_{n=1}^{\infty} \frac{1}{n^{4/3}}\right)} < \infty.$$

(b) Let $a_n = \frac{1}{n}$. Then $\sum a_n$ diverges, but

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^{2/3}} = \sum_{n=1}^{\infty} \frac{\frac{1}{n^{1/2}}}{n^{2/3}} = \sum_{n=1}^{\infty} \frac{1}{n^{7/6}}$$

converges because $7/6 > 1$.

□

Problem 3.4. Suppose $a_n > 0, s_n = a_1 + \ldots + a_n$ and $\sum a_n$ diverges.

(a) Prove that $\frac{a_{n+1}}{s_{n+1}} + \ldots + \frac{a_{n+k}}{s_{n+k}} \geq 1 - \frac{s_n}{s_{n+k}}$ for all positive integers n and k and deduce that $\sum \frac{a_n}{s_n}$ diverges.

(b) Prove $\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$ for $n \geq 2$, and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

Solution.

(a) Since $a_n > 0$ for all n, we have $\{s_n\}$ is a increasing sequence. This gives

$$\begin{aligned}\frac{a_{n+1}}{s_{n+1}} + \ldots + \frac{a_{n+k}}{s_{n+k}} &\geq \frac{a_{n+1} + \ldots + a_{n+k}}{s_{n+k}} \\ &= \frac{s_{n+k} - s_n}{s_{n+k}} \\ &= 1 - \frac{s_n}{s_{n+k}}.\end{aligned}$$

Suppose by contradiction that $\sum \frac{a_n}{s_n}$ converges. Then by Cauchy criterion, with $\varepsilon = 1/2$, there is some $N \in \mathbb{N}$ such that

$$\frac{a_{n+1}}{s_{n+1}} + \ldots + \frac{a_{n+k}}{s_{n+k}} \leq \frac{1}{2}, \text{ for all } n \geq N, k \in \mathbb{N}.$$

Then we have

$$1 - \frac{s_N}{s_{N+k}} \leq \frac{1}{2}, \text{ for all } k \in \mathbb{N}.$$

Thus, $2s_N \geq s_{N+k}$ for all $k \in \mathbb{N}$. Therefore

$$s_N \geq s_{N+k} - s_N = a_{N+1} + \ldots + a_{N+k}, \text{ for all } k.$$

Let $k \to \infty$, this gives

$$s_N \geq \sum_{n=N+1}^{\infty} a_n,$$

and hence

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{N} a_n + \sum_{n=N+1}^{\infty} a_n \leq s_N + s_N = 2s_N < \infty,$$

which is a contradiction.

(b) Since $s_n \geq s_{n-1}$ we have

$$a_n \leq \frac{a_n s_n}{s_{n-1}} = \frac{s_n - s_{n-1}}{s_{n-1}s_n} s_n^2,$$

or equivalently,

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}.$$

Notice that since $\sum a_n$ diverges, $s_n \to \infty$, or equivalently, $1/s_n \to 0$, we have

$$\sum_{n=1}^{\infty} \frac{a_n}{s_n^2} \leq \frac{a_1}{s_1^2} + \sum_{n=2}^{\infty} \left(\frac{1}{s_{n-1}} - \frac{1}{s_n}\right) = \frac{a_1}{s_1^2} + \frac{1}{s_1} - \lim \frac{1}{s_n} = \frac{1}{a_1} + \frac{1}{a_1} = \frac{2}{a_1} < \infty.$$

□

Problem 3.5. Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and the series $\sum a_n$ diverges.

(a) Prove that $\sum \frac{a_n}{1+a_n}$ diverges.

(b) Prove that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$$

and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

Solution.

(a) We consider two cases.

Case 1. $a_n \to 0$. Then there is some $N \in \mathbb{N}$ such that $a_n < 1$ for all $n \geq N$. Then we have

$$\frac{a_n}{1+a_n} \geq \frac{a_n}{2}, \text{ for all } n \geq N.$$

Since $\sum a_n$ diverges, we have $\sum \dfrac{a_n}{1+a_n} = \infty$.

Case 2. $a_n \not\to 0$. Then there is a subsequence $\{a_{n_k}\}$ such that $a_{n_k} \geq \varepsilon > 0$ for all $k \in \mathbb{N}$. Since the function $\dfrac{x}{1+x}$ is increasing on $(0, +\infty)$, we have

$$\sum \frac{a_n}{1+a_n} \geq \sum \frac{a_{n_k}}{1+a_{n_k}} \geq \sum \frac{\varepsilon}{1+\varepsilon} = \infty.$$

(b) We have that

$$\frac{a_n}{s_n^2} = \frac{s_n - s_{n-1}}{s_n^2} \leq \frac{s_n - s_{n-1}}{s_n s_{n-1}} = \frac{1}{s_{n-1}} - \frac{1}{s_n}.$$

Since $\sum a_n$ diverges, $s_n \to \infty$, and we have

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{a_n}{s_n^2} &\leq \frac{a_1}{s_1^2} + \sum_{n=2}^{\infty} \left(\frac{1}{s_{n-1}} - \frac{1}{s_n} \right) \\ &= \frac{1}{a_1} + \frac{1}{s_1} - \lim_{n\to\infty} \frac{1}{s_n} \\ &= \frac{2}{a_1} < \infty.\end{aligned}$$

□

Problem 3.6. Suppose $a_n > 0$ and the series $\sum a_n$ converges. Put

$$r_n = \sum_{m=n}^{\infty} a_m.$$

(a) Prove that

$$\frac{a_m}{r_m} + \ldots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$$

if $m < n$, and deduce that $\sum \frac{a_n}{r_n}$ diverges.

(b) Prove that

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

and deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

Solution.

(a) Take $m < n$, since $a_n > 0$ we have

$$r_m = \sum_{k=m}^{\infty} a_k = \sum_{k=n}^{\infty} a_k + \sum_{k=m}^{n-1} a_k = r_n + \sum_{k=m}^{n-1} a_k > r_n.$$

Then

$$\begin{aligned}\frac{a_m}{r_m} + \ldots + \frac{a_n}{r_n} &> \frac{a_m}{r_m} + \frac{a_{m+1}}{r_m} + \ldots + \frac{a_n}{r_m} \\ &= \frac{a_m + \ldots + a_n}{r_m} \\ &= \frac{r_m - r_{n+1}}{r_m} \\ &> \frac{r_m - r_n}{r_m} = 1 - \frac{r_n}{r_m}.\end{aligned}$$

We we will show that $\sum \frac{a_n}{r_n}$ diverges. Assume by contradiction that $\sum \frac{a_n}{r_n}$ converges. Then by Cauchy criterion, there is some $N \geq 0$ such that

$$\frac{a_m}{r_m} + \ldots + \frac{a_n}{r_n} < \frac{1}{2}, \text{ for all } n > m > N.$$

This means that

$$1 - \frac{r_n}{r_m} < \frac{1}{2} \Longleftrightarrow \frac{r_n}{r_m} > \frac{1}{2} \Longleftrightarrow r_m < 2r_n, \text{ for all } n > m > N.$$

We will prove next that $r_n \to 0$ as $n \to \infty$. If it does not hold, there is some $\varepsilon > 0$ such that $r_n \geq \varepsilon$ for all $n \in \mathbb{N}$. This means

$$\sum_{k=n}^{\infty} a_k \geq \varepsilon \text{ for all } n \in \mathbb{N}.$$

Then

$$\sum_{k=1}^{\infty} a_k = \sum_{k=n}^{\infty} a_k + \sum_{k=1}^{n-1} a_k \geq \varepsilon + \sum_{k=1}^{n-1} a_k.$$

Let $n \to \infty$ and since $\sum a_k$ converges, we have $\varepsilon = 0$, which is a contradiction. However, fixing $m > N$ and letting $n \to \infty$ in $r_m < 2r_n$, $r_n \to 0$ gives $r_m \leq 0$, which is also a contradiction since $r_m > 0$. Therefore, $\sum \frac{a_n}{r_n}$ diverges.

(b) We have

$$\begin{aligned}
&\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}}) \\
\Longleftrightarrow &\frac{a_n}{\sqrt{r_n}} < 2\frac{r_n - r_{n+1}}{\sqrt{r_n} + \sqrt{r_{n+1}}} = 2\frac{a_n}{\sqrt{r_n} + \sqrt{r_{n+1}}} \\
\Longleftrightarrow &\sqrt{r_n} + \sqrt{r_{n+1}} < 2\sqrt{r_n} \\
\Longleftrightarrow &\sqrt{r_{n+1}} < \sqrt{r_n}.
\end{aligned}$$

Since the latter is true, we have

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}}).$$

Then we have

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{a_n}{\sqrt{r_n}} &< 2\sum_{n=1}^{\infty} (\sqrt{r_n} - \sqrt{r_{n+1}}) \\
&= 2\lim_{n\to\infty} (\sqrt{r_1} - \sqrt{r_{n+1}}) = 2\sqrt{r_1},
\end{aligned}$$

since $r_n \to 0$ as $n \to \infty$.

□

Problem 3.7. Prove that the series

$$\sum_{n=1}^{\infty} \frac{\cos(nt)}{n}$$

converges if $0 < t < 2\pi$.

Solution. We will use the Abel's identity which states that

$$\sum_{k=0}^{n} a_k b_k = a_n B_n - \sum_{k=0}^{n-1} (a_{k+1} - a_k) B_k, \quad B_n = \sum_{k=0}^{n} b_k$$

Let $b_k = \cos(kt)$ and $a_k = \frac{1}{k}$ then

$$\sum_{k=1}^{n} \frac{\cos(kt)}{k} = \frac{B_n}{n} - \sum_{k=1}^{n-1} \left(\frac{1}{k+1} - \frac{1}{k} \right) B_k$$

We have the following identity $2\cos(rt)\sin(t/2) = \sin(rt + t/2) - \sin(rt - t/2)$, then we get telescoping sum

$$\begin{aligned} B_k &= \cos(t) + \cos(2t) + \cdots + \cos(kt) \\ &= (\cos(t) + \cos(2t) + \cdots + \cos(kt)) \frac{2\sin(t/2)}{2\sin(t/2)} \\ &= \frac{1}{2\sin(t/2)} \sum_{r=1}^{k} \sin\left(rt + \frac{t}{2}\right) - \sin\left(rt - \frac{t}{2}\right) = \frac{1}{2\sin(t/2)} \left(\sin\left(kt + \frac{t}{2}\right) - \sin\left(t - \frac{t}{2}\right) \right) \\ |B_k| &\leq \frac{1}{|\sin(t/2)|} \end{aligned}$$

Thus $\frac{B_n}{n} \to 0$, and since $\frac{1}{k} - \frac{1}{k+1} > 0$,

$$\sum_{k=1}^{n-1} \left| \left(\frac{1}{k+1} - \frac{1}{k} \right) B_k \right| \leq \frac{1}{|\sin(t/2)|} \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = \frac{1}{|\sin(t/2)|} \left(1 - \frac{1}{n} \right) \leq \frac{1}{|\sin(t/2)|},$$

so the series $\sum_k \left(\frac{1}{k+1} - \frac{1}{k} \right) B_k$ converges absolutely. Hence the partial sums $\sum_{k=1}^{n} \frac{\cos(kt)}{k}$ converge as $n \to \infty$. □

4 Continuity, derivatives and the mean value theorem

Problem 4.1. Find all postive continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x+y) = f(x)f(y), \quad \forall x, y \in \mathbb{R}$$

Prove your conclusion.

Solution. First,

$$f(0+x) = f(x)f(0) \Rightarrow f(x)(1 - f(0)) = 0 \quad (1)$$

if $f(x) = 0$ for some $x \in \mathbb{R}$ then $f(y) = f(y - x + x) = f(y - x)f(x) = 0$ for any

$y \in \mathbb{R}$. So assume $f(x) \neq 0$ for all $x \in \mathbb{R}$. By (1) we get $f(0) = 1$. By induction, we get $f(nx) = (f(x))^n$ for $n \in \mathbb{N}$ but

$$f(x-x) = f(x)f(-x) = 1 \Rightarrow f(x) = \frac{1}{f(-x)}$$

extends $f(nx) = (f(x))^n, n \in \mathbb{Z}$, and

$$f(x) = f(n(1/nx)) = (f(1/nx))^n \Rightarrow (f(x))^{1/n} = f(1/nx)$$

extends $f(nx) = (f(x))^n, n \in \mathbb{Q}$, and it extends to the real numbers since for any real number $r \in \mathbb{R}$ there is a Cauchy sequence of rational numbers $b_n \to r$ but then by the continuity of f

$$f(b_n x) = (f(x))^{b_n} \Rightarrow \lim f(b_n x) = \lim (f(x))^{b_n} \Rightarrow f(rx) = (f(x))^r \quad r \in \mathbb{R}$$

if we set $f(1) = a$ then for any $x \in \mathbb{R}$ we have $f(x \cdot 1) = (f(1))^x = a^x$ and $a > 0$ since

$$a = f(1) = f(1/2 + 1/2) = (f(1/2))^2 > 0$$

The function should be in the form $f(x) = a^x, \;\; a \in (0, \infty)$

□

Problem 4.2. Suppose f is continuous on $\mathbb{R}$ and $f(x) = f(x + \frac{1}{n})$ for all $n = 1, 2, \ldots$ and $x \in \mathbb{R}$. Show that f is constant.

Solution. Firstly, for $m \in \mathbb{N}$, by induction, we have

$$f(x) = f(x + \frac{1}{n}) = f(x + \frac{1}{n} + \frac{1}{n}) = \ldots = f(x + \frac{m}{n}), \text{ for all } x \in \mathbb{R}, n \in \mathbb{N}.$$

Therefore, $f(x) = f(x + r)$ for all $x \in \mathbb{R}, r \in \mathbb{Q}^+$. By substituting $x - r$ for x, we can prove that $f(x) = f(x + r)$ for all $r \in \mathbb{Q}$. In particular, we have

$$f(0) = f(r), \text{ for all } r \in \mathbb{Q}.$$

Let $x \in \mathbb{R}$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, there exists a sequence $\{x_n\}_{n \in \mathbb{N}} \subset \mathbb{Q}$ such that $x_n \to x$. By the continuity of f, we have

$$f(x) = \lim_n f(x_n) = \lim_n f(0) = f(0).$$

Therefore, f is constant on $\mathbb{R}$. □

Problem 4.3.

(a) Give the precise ε, δ definition of *uniform continuity* of a function f defined on a subset of $\mathbb{R}$.

(b) Give an example of a continuous function defined on a bounded interval which is not uniformly continuous. Justify your conclusion.

(c) Prove that if $f : [a, b] \to \mathbb{R}$ is continuous, then f is uniformly continuous, where a, b are finite numbers.

Solution.

(a) Let D be a subset of $\mathbb{R}$. A function $f : D \to \mathbb{R}$ is called *uniformly continuous* on D if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x, x' \in D, |x - x'| \leq \delta \implies |f(x) - f(x')| \leq \varepsilon.$$

(b) Consider $f : (0, 1) \to \mathbb{R}$ given by

$$f(x) = \frac{1}{x}, \text{ for all } x \in (0, 1).$$

Firstly, we prove that f is continuous on $(0, 1)$. Indeed, let $x_0 \in (0, 1)$. Then

$$\lim_{x \to x_0} \frac{1}{x} = \frac{1}{x_0}.$$

Therefore, f is continuous at $x_0 \in (0, 1)$ for any x_0, thus continuous on all of $(0, 1)$. However, take $\varepsilon = 1$ and let $\delta > 0$. Because $\lim_n \frac{\delta}{n} = 0$, there exists $n_0 \in \mathbb{N}$ such that

$$0 < \frac{\delta}{n_0} < 1.$$

Let $m_0 \in \mathbb{N}$ such that $m_0 \geq [\delta] + 1$ and $x := \frac{\delta}{n_0}, x' := \frac{\delta}{n_0 + m_0}$. Then $x, x' \in (0, 1)$ and

$$|x - x'| = \frac{m_0}{n_0(n_0 + m_0)} \delta \leq \delta.$$

However,

$$\begin{aligned} |f(x) - f(x')| &= \left| \frac{n_0}{\delta} - \frac{n_0 + m_0}{\delta} \right| \\ &= \frac{m_0}{\delta} \geq \frac{[\delta] + 1}{\delta} > 1. \end{aligned}$$

Therefore, f is not uniformly continous on $(0, 1)$.

(c) Take $\varepsilon > 0$ abitrarily. Because f is continuous on $[a, b]$, for each $p \in [a, b]$ there exists $\delta_p > 0$ such that

$$x \in [a, b], |x - p| < \delta_p \implies |f(x) - f(p)| < \varepsilon/2.$$

Because $[a, b] \subset \bigcup_{p \in [a,b]} B(p, \delta_p/2)$, there exists finite numbers $p_1, p_2, \ldots, p_n$ such that

$$[a, b] \subset \bigcup_{k=1}^{n} B(p_k, \delta_{p_k}/2).$$

Put $\delta = \frac{1}{2}\min\{\delta_1, \delta_2, \ldots, \delta_n\} > 0$. Let $x, y \in [a,b]$ such that $|x-y| < \delta$. Because $[a,b] \subset \bigcup_{k=1}^{n} B(p_k, \delta_{p_k}/2)$, there exists $k \in \{1,\ldots,n\}$ such that

$$x \in B(p_k, \delta_{p_k}/2).$$

We then have

$$\begin{aligned} |y-p_k| &\leq |y-x| + |x-p_k| \\ &\leq \delta + \frac{\delta_{p_k}}{2} \\ &\leq \frac{\delta_{p_k}}{2} + \frac{\delta_{p_k}}{2} \\ &= \delta_{p_k}. \end{aligned}$$

Thus

$$|f(x)-f(y)| \leq |f(x)-f(p_k)| + |f(p_k)-f(y)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, f is uniformly continuous on $[a,b]$.

□

Problem 4.4.

(a) Give the precise $\varepsilon - \delta$ definition of a *uniformly continuous mapping* f from a metric space X to a metric space Y.

(b) Use only the definitions of uniform continuity and compactness to prove that if f is a continuous mapping of a compact metric space X into a metric space Y, then f is uniformly continuous on X. You CANNOT simply quote the theorem that states this result.

Solution.

(a) A function $f : X \to Y$ is uniformly continuous if for any $\varepsilon > 0$ there exists a $\delta > 0$ such that, for any $x, y \in X$,

$$d_X(x,y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

(b) Let $\varepsilon > 0$. For any $x \in X$, continuity gives us a $\delta_x > 0$ such that $d_Y(f(x), f(y)) < \varepsilon/2$ for any $y \in X$ such that $d_X(x,y) < \delta_x$. Consider the open cover $\{B(x, \delta_x/2)\}_{x \in X}$. By compactness, there exists a finite subcover $B(x_0, \delta_0/2), \ldots, B(x_n, \delta_n/2)$. Let $\delta = \min(\delta_0/2, \ldots, \delta_n/2)$.
Let $x \in X$. Since we have an open cover, there exists an x_i such that $d_X(x, x_i) < \delta_i/2$. Then, for any $y \in X$ such that $d_X(x,y) < \delta$,

$$d_X(x_i, y) \leq d_X(x_i, x) + d_X(x,y) < \delta_i/2 + \delta \leq \delta_i,$$

which implies that

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(x_i)) + d_Y(f(x_i), f(y)) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Problem 4.5. Let $f : [0,1] \to \mathbb{R}$ be continuous on $[0,1]$ and differentiable in $(0,1)$ such that
- $f(0) = 0$
- and there exists $M > 0$ such that $|f'(x)| \leq M|f(x)|$ for all $x \in (0,1)$

Show that $f(x) = 0$ for all $x \in [0,1]$.

Solution. Let $a_1 \in (0,1)$ then by the mean value theorem there exists $a_2 \in (0, a_1)$ such that

$$\frac{f(a_1) - f(0)}{a_1 - 0} = f'(a_2) \Rightarrow |f(a_1)| = |a_1||f'(a_2)| \leq Ma_1|f(a_2)|$$

the last term due to the assumption that $|f'(x)| \leq M|f(x)|$. Keep that by induction we get a sequence $a_1 > a_2 > a_3 > \cdots$ such that

$$|f(a_n)| \leq M(a_n)|f(a_{n+1})|$$

Hence we get

$$|f(a_1)| \leq \prod_{i=1}^{n-1}(a_i M)|f(a_n)|$$

Now take $a_1 < \frac{1}{2M}$. Then every factor satisfies $a_i M < \frac{1}{2}$, and $|f(a_n)| \leq \max_{[0,1]} |f|$, so $|f(a_1)| \leq \frac{1}{2^{n-1}} \max_{[0,1]} |f| \to 0$ as $n \to \infty$. Thus $f(a_1) = 0$, and since a_1 was arbitrary, $f = 0$ on $[0, \frac{1}{2M})$. (Without the restriction on a_1 the factors $a_i M$ need not tend to 0: the decreasing sequence a_n need not tend to 0.) Finally, if $f = 0$ on $[0, c]$, the same argument with c in place of 0 (using $f(c) = 0$) gives $f = 0$ on $[c, c + \frac{1}{2M}) \cap [0,1]$; after finitely many steps, $f = 0$ on $[0,1]$. □

Problem 4.6. Suppose that f is differentiable function defined on reals, $f(0) = 0$ and $|f'(x)| \leq |f(x)|$ for all x. Prove that f is identically 0.

Solution. Firstly, we prove that f vanishes on $[-1/2, 1/2]$. Since $|f|$ is continuous on $[-1/2, 1/2]$, $|f|$ attains it maximum M_n on $[-1/2, 1/2]$. Then we have

$$|f'(x)| \leq |f(x)| \leq M_n, \text{ for all } x \in [-1/2, 1/2].$$

By Lagrange theorem for all $x \in [-1/2, 1/2]$ there is some c between x and 0 such that

$$\frac{f(x) - f(0)}{x - 0} = f'(c).$$

This means $|f(x)| \leq |f'(c)||x| \leq M_n|x|$, for all $x \in [-1/2, 1/2]$. Then

$$M_n \leq \frac{M_n}{2},$$

which shows that $M_n = 0$, or $f = 0$ on $[-1/2, 1/2]$. Next we will prove that $f(x) = 0$ on

$\left[\frac{k-1}{2}, \frac{k+1}{2}\right]$ for all $k \in \mathbb{N}$. Defining $g(x) = f(x + \frac{k}{2})$, where $x \in [-1/2, 1/2]$ gives us

$$|g'(x)| = |f'(x + \frac{k}{2})| \leq |f(x + \frac{k}{2})| = |g(x)|, \text{ for all } x \in [-1/2, 1/2]\,.$$

Moreover, $g(0) = f(k/2) = 0$. By the same arguments, we get $g(x) = 0$ for all $x \in [-1/2, 1/2]$, or equivalently, $f(x) = 0$ on $\left[\frac{k-1}{2}, \frac{k+1}{2}\right]$. Therefore, $f(x) = 0$ on $[-1/2, +\infty)$. The same argument gives $f(x) = 0$ on $(-\infty, 1/2]$. Therefore, $f(x) = 0$ for all x. □

Problem 4.7. Suppose $f : [1, \infty) \to \mathbb{R}$ is differentiable with continuous derivative and
(a) $f(1) = 1$.
(b) $xf'(x) \leq \frac{1}{2} f(x)$.
Show that $f(x) \leq x^{1/2}$ for all $x \geq 1$.

Solution. Let $g(x) = f(x)/\sqrt{x}$ for all $x \geq 1$. By the second condition, we have

$$\begin{aligned} x\left(g(x)\sqrt{x}\right)' &\leq \frac{1}{2} g(x)\sqrt{x}, \text{ for all } x \geq 1 \\ xg'(x)\sqrt{x} + xg(x)\frac{1}{2\sqrt{x}} &\leq \frac{1}{2} g(x)\sqrt{x}, \text{ for all } x \geq 1 \\ xg'(x)\sqrt{x} &\leq 0, \text{ for all } x \geq 1 \\ g'(x) &\leq 0, \text{ for all } x \geq 1. \end{aligned}$$

Therefore, $g(x) \leq g(1) = 1$ for all $x \geq 1$, or equivalently,

$$f(x) \leq x^{\frac{1}{2}}, \text{ for all } x \geq 1.$$

□

Problem 4.8. If f is a bounded continuously differentiable function defined on the reals satisfying

$$|f(x) + f'(x)| \leq 1$$

for all x, prove that $|f(x)| \leq 1$ for all x.

Solution. Define $g(x) = e^x f(x)$, we have $g'(x) = e^x(f(x) + f'(x))$. Therefore,

$$|g'(x)| \leq e^x \iff -e^x \leq g'(x) \leq e^x, \text{ for all } x \in \mathbb{R}.$$

Integrating three sides gives

$$\begin{aligned} &\int_a^b -e^x dx \leq \int_a^b g'(x)dx \leq \int_a^b e^x dx \\ \iff &e^a - e^b \leq g(b) - g(a) \leq e^b - e^a. \end{aligned}$$

Since f is bounded,

$$\lim_{a\to-\infty} g(a) = \lim_{a\to-\infty} (e^a f(a)) = 0.$$

Therefore,

$$-e^b \le g(b) \le e^b, \text{ for all } b \in \mathbb{R}.$$

This gives $|g(x)| \le e^x$ or equivalently, $|f(x)| \le 1$ for all $x \in \mathbb{R}$. □

Problem 4.9. Suppose the real constants $C_0, \cdots, C_n$ satisfy

$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$$

Prove that the equation

$$C_0 + C_1 x + C_2 x^2 + \cdots + C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real root between 0 and 1.

Solution. I will show that $f(x) = \sum_{i=0}^{n} C_i x^i$ will change the sign on the interval $[0,1]$ then by intermediate value theorem we get a zero of the function. Note that if $f(x) \ge 0$ on the interval $[0,1]$ then by the linearity of integration we get $\int_0^1 f(x)\, dx \ge 0$, and if $f(x) \le 0$ on $[0,1]$ we get $\int_0^1 f(x)\, dx \le 0$.

We have

$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0 = \sum_{i=0}^{n} \frac{C_i}{i+1}$$

Consider

$$\int_0^1 \sum_{i=0}^{n} C_i x^i\, dx = \sum_{i=0}^{n} \frac{C_i x^{i+1}}{i+1}\bigg|_0^1 = \sum_{i=0}^{n} \frac{C_i}{i+1} = 0$$

Since the integral is zero, either $f \equiv 0$ on $[0,1]$ (and then every point is a root), or f takes both positive and negative values on $[0,1]$ (a continuous $f \ge 0$ which is not identically zero has a positive integral, and similarly for $f \le 0$). In the second case the intermediate value theorem gives a root strictly between a point where $f > 0$ and a point where $f < 0$, hence in $(0,1)$. □

Problem 4.10. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function that satisfies $f \circ f(x) = x$ for all $x \in \mathbb{R}$.

(a) Prove f is either strictly increasing or strictly decreasing.
(b) In the case f is strictly increasing prove that $f(x) = x$ for all $x \in \mathbb{R}$.
(c) In the case f is strictly decreasing prove that there exists $x_0 \in \mathbb{R}$ such that $f(x_0) = x_0$

and $f'(x_0) = -1$.

Solution.

(a) By differentiating the relation $f(f(x)) = x$ we find

$$f'(f(x))f'(x) = 1,$$

which implies that $f'(x) \neq 0$ for all $x \in \mathbb{R}$. Since a derivative has the intermediate value property (Darboux's theorem), the derivative never changes sign, so f is strictly monotone.

(b) Fix $x \in \mathbb{R}$ and let $f(x) = y$. Notice that we also have $f(y) = x$ by assumption. If $x \neq y$, then by the Mean Value Theorem, there exists a $c \in [x, y]$ such that

$$f'(c) = \frac{x - y}{y - x} = -1,$$

so f is not strictly increasing.

(c) Let $f(0) = a$. If $a \geq 0$, then since f is strictly decreasing there must exist a point $x_0 \in [0, a]$ such that the graph of f intersects the line $x = y$, i.e. such that $f(x_0) = x_0$. Similarly, if $a \leq 0$ then there exists a point $x_0 \in [a, 0]$ such that $f(x_0) = x_0$.
By the relation we derived in (a), we have that

$$1 = f'(f(x_0))f'(x_0) = (f'(x_0))^2.$$

It follows that $f'(x_0) = \pm 1$; since f is strictly decreasing, it must be the case that $f'(x_0) = -1$.

□

Problem 4.11. Suppose f is a twice-differentiable real function defined on $(0, \infty)$. Let $M_0 = \sup_{(0,\infty)} |f(x)|$, $M_1 = \sup_{(0,\infty)} |f'(x)|$, and $M_2 = \sup_{(0,\infty)} |f''(x)|$. Prove that

$$M_1^2 \leq 4M_0M_2.$$

Hint: Apply the Taylor's expansion.

Solution. Fix $x \in (0, \infty)$. As f is twice-differentiable on $(0, \infty)$ which is a neighborhood of x, the Taylor's expansion theorem says that with $h > 0$, there is $\xi \in (x, x + 2h)$ such that

$$f(x + 2h) = f(x) + f'(x)2h + \frac{f''(\xi)(2h)^2}{2}.$$

Thus one can write:

$$f'(x) = \frac{1}{2h}\left[f(x + 2h) - f(x)\right] - hf''(\xi).$$

And then

$$\left|f'(x)\right| \leq \frac{1}{2h}(M_0 + M_0) + hM_2 = \frac{M_0}{h} + hM_2.$$

Since x does not depend on h, taking the infimum of both sides with respect to all $h > 0$ gives us:

$$|f'(x)| \leq \inf_{h>0} \left(\frac{M_0}{h} + hM_2 \right) = 2\sqrt{M_0 M_2}.$$

As the above inequality is true for x taken arbitrarily in $(0, \infty)$, we deduce that $M_1 = \sup_{(0,\infty)} |f'(x)| \leq 2\sqrt{M_0 M_2}$ and thus $M_1^2 \leq 4M_0 M_2$. □

Problem 4.12.

(a) Assume f and g are two differentiable functions defined in an open neighborhood of a compact subset K of $\mathbb{R}$ such that

$$[f(x) - g(x)]^2 + [f'(x) - g'(x)]^2 \neq 0 \text{ for all } x \in K.$$

Prove that either $f(x) \neq g(x)$ for all $x \in K$ or $f(x) = g(x)$ only at finitely many $x \in K$.

(b) Is the following generalization of part (a) to higher dimensions true: Assume f and g are two differentiable functions defined in an open neighborhood of a compact subset K of $\mathbb{R}^n$ such that

$$[f(x) - g(x)]^2 + |\nabla f(x) - \nabla g(x)|^2 \neq 0 \text{ for all } x \in K,$$

where ∇f is the gradient vector of f. Then $f(x) \neq g(x)$ for all $x \in K$ or $f(x) = g(x)$ only at finitely only many $x \in K$.

Solution.

(a) Notice that $[f(x) - g(x)]^2 \geq 0$ and $[f'(x) - g'(x)]^2 \geq 0$ for all $x \in K$ because every square of a real number is non-negative. Furthermore, at any $x \in K$ it must be the case that either $[f(x) - g(x)]^2 > 0$ or $[f'(x) - g'(x)]^2 > 0$ (or both).
Suppose that the set $E = \{x \in K \mid f(x) = g(x)\}$ is infinite. Since E is an infinite subset of a compact set, it must have a limit point $x_0 \in K$. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence of elements of E which witnesses the fact that x_0 is a limit point. We must have $x_0 \in E$, because, since f and g are continuous,

$$f(x_0) = f(\lim_{n\to\infty} x_n) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} g(x_n) = g(\lim_{n\to\infty} x_n) = g(x_0).$$

Our assumption implies that $[f'(x_0) - g'(x_0)]^2 > 0$, so that $y := (f-g)'(x_0) = f'(x_0) - g'(x_0) \neq 0$.
By the definition of derivative, there exists an $h > 0$ such that for any $x \in (x_0 - h, x_0 + h)$, $x \neq x_0$,

$$\left| \frac{(f-g)(x) - (f-g)(x_0)}{x - x_0} - y \right| < \left| \frac{y}{2} \right|.$$

Hence $\frac{(f-g)(x)}{x-x_0} \neq 0$, that is, $(f-g)(x) \neq 0$, for $0 < |x - x_0| < h$: this interval contains no root other than x_0. This contradicts the fact that x_0 is a limit point. Therefore, E is at most finite.

(b) This is false. Take any pair of f and g that satisfy the hypothesis in (a), such that $f(x) = g(x)$ at at least one point $x = x_0 \in K$. Define two new functions with domain $K \times [0,1]$ as

$$\tilde{f}(x,t) = f(x)$$
$$\tilde{g}(x,t) = g(x).$$

Then $K \times [0,1]$ is also compact, and

$$\begin{aligned}[\tilde{f}(x,t) - \tilde{g}(x,t)]^2 + |\nabla\tilde{f}(x,t) - \nabla\tilde{g}(x,t)|^2 &= [f(x) - g(x)]^2 + \left|\begin{bmatrix} f'(x) \\ 0 \end{bmatrix} - \begin{bmatrix} g'(x) \\ 0 \end{bmatrix}\right|^2 \\ &= [f(x) - g(x)]^2 + [f'(x) - g'(x)]^2 \\ &\neq 0 \text{ for all } (x,t) \in K \times [0,1].\end{aligned}$$

However, $\tilde{f}(x_0,t) - \tilde{g}(x_0,t) = 0$ for all (uncountably many) $t \in [0,1]$.

□

Problem 4.13.

(a) Give an example of a differentiable function which is not continuously differentiable.

(b) Give an example of a continuously differentiable function which is not twice differentiable.

(c) Give an example of an infinitely continuously differentiable function which is is not analytic.

Solution.

(a) Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin\dfrac{1}{x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Then we have

$$\lim_{x\to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{x^2 \sin\frac{1}{x}}{x} = \lim_{x\to 0^+} (x \sin\frac{1}{x}) = 0,$$
$$\lim_{x\to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^-} \frac{0}{x} = 0.$$

Thus, $f'(0) = 0$, and moreover,

$$f'(x) = \begin{cases} 2x \sin\dfrac{1}{x} - \cos\dfrac{1}{x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

However, take $x_n = \dfrac{1}{2n\pi} \to 0$, we have

$$2x_n \sin\frac{1}{x_n} - \cos\frac{1}{x_n} = 2x_n \sin(2n\pi) - \cos(2n\pi) = -1, \text{ for all } n \in \mathbb{N}.$$

Therefore, the limit of $f'(x)$ at 0 will not be $f'(0) = 0$, i.e. $f'(x)$ does not continuous at 0.

(b) Consider function $g : \mathbb{R} \to \mathbb{R}$ given by

$$g(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

Then we have

$$\lim_{x\to 0^+} \frac{g(x) - g(0)}{x - 0} = \lim_{x\to 0^+} \frac{x^2}{x} = \lim_{x\to 0^+} x = 0,$$
$$\lim_{x\to 0^-} \frac{g(x) - g(0)}{x - 0} = \lim_{x\to 0^-} \frac{-x^2}{x} = \lim_{x\to 0^-} (-x) = 0.$$

Thus we have $g'(0) = 0$. Moreover,

$$g'(x) = \begin{cases} 2x, & x > 0, \\ 0, & x = 0, \\ -2x, & x < 0 \end{cases} = 2|x|$$

which is continuous but not differentiable. Therefore, g is a continuously differentiable function which is not twice differentiable.

(c) Consider the function $h : \mathbb{R} \to \mathbb{R}$ given by

$$h(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

Then h is C^∞ but not analytic at 0, because its Taylor expansion at $x = 0$ is the 0 series.

□

Problem 4.14. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^{-\frac{1}{x}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Prove that f is infinitely continuously differentiable (C^∞) but not real analytic.

Solution. We claim that

$$\frac{d^n}{dx^n}e^{-\frac{1}{x}} = p_n\left(\frac{1}{x}\right)e^{-\frac{1}{x}}$$

where p_n is some polynomial, for any $n \geq 0$. This is clearly the case for $n = 0$, with $p_0 = 1$. Suppose this holds for n. Then

$$\begin{aligned}\frac{d^{n+1}}{dx^{n+1}}e^{-\frac{1}{x}} &= \frac{d}{dx}\left(\frac{d^n}{dx^n}e^{-\frac{1}{x}}\right)\\ &= \frac{d}{dx}\left(p_n\left(\frac{1}{x}\right)e^{-\frac{1}{x}}\right)\\ &= \left(-\frac{1}{x^2}p_n'\left(\frac{1}{x}\right)+\frac{1}{x^2}p_n\left(\frac{1}{x}\right)\right)e^{-\frac{1}{x}}\end{aligned}$$

and the bracketed expression is a polynomial since the derivative of a polynomial stays a polynomial. By induction, we have proved our claim.
Away from $x = 0$, f is either identically 0 or a composite of two C^∞ functions, so it is C^∞. It remains to be shown that f is C^∞ at $x = 0$. It's easy to see that $\lim_{x\to 0^-}\frac{d^n}{dx^n}f(x) = \lim_{x\to 0^-} 0 = 0$. On the other hand,

$$\begin{aligned}\lim_{x\to 0^+}\frac{d^n}{dx^n}f(x) &= \lim_{x\to 0^+}\frac{d^n}{dx^n}e^{-\frac{1}{x}}\\ &= \lim_{x\to 0^+} p_n\left(\frac{1}{x}\right)e^{-\frac{1}{x}}\\ &= \lim_{x\to\infty} p_n(x)e^{-x}\\ &= \lim_{x\to\infty}\frac{p_n(x)}{e^x}\\ &= 0\end{aligned}$$

by repeated application of L'Hôpital's rule ($\deg p_n$ times). Therefore f is C^∞ at $x = 0$ with all derivatives being 0.
It follows that f is not real-analytic, because the series expansion of f at 0 is the 0 series, but $f(x) > 0$ for all $x > 0$. □

5 The Riemann integral and integral inequalities

Problem 5.1. Let f be a bounded function on $[0,1]$. For any partition $P = \{x_0 < x_1 < \cdots < x_n\}$ of $[0,1]$, let $\mu(P)$ denote the mesh of P and

$$R(P) = \sum_{i=1}^{n}(M_i - m_i)(x_i - x_{i-1}),$$

where $m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}$ and $M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\}$. Prove that the following two assertions are equivalent (i.e., (i) $\implies$ (ii) and (ii) $\implies$ (i)).

(i) Given $\varepsilon > 0$ there exists $\delta > 0$ such that $R(P) < \varepsilon$ whenever $\mu(P) < \delta$.
(ii) There exists a real number A for which given $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\left| \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) - A \right| < \varepsilon$$

for every partition P and $x_{i-1} \leq t_i \leq x_i$, $i = 1, \ldots, n$ with $\mu(P) < \delta$.

Solution. For any partition $P = \{x_0 < x_1 < \cdots < x_n\}$ of $[0, 1]$, let

$$L_P = \sum_{i=1}^{n} m_i(x_i - x_{i-1}),$$
$$U_P = \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

(These are called the lower and upper Darboux sums.) Notice that

$$R(P) = U_P - L_P$$

by definition of $R(P)$. Since f is bounded, there is an $M \geq 0$ such that $|f(x)| \leq M$ for all $x \in [0, 1]$; we have that $L_P \leq M$ and $U_P \geq -M$ for any partition P, so we can define

$$L = \sup\{L_P \mid P \text{ is a partition of } [0, 1]\},$$
$$U = \inf\{U_P \mid P \text{ is a partition of } [0, 1]\}.$$

[(i) $\implies$ (ii)] For any $\varepsilon > 0$, (i) implies that

$$U - L \leq U_P - L_P = R(P) < \varepsilon$$

for any fine enough partition P. Since ε was arbitrary, we must have $L = U$. Set $A = L = U$.
Let $\varepsilon > 0$. For any partition P, by definition of m_i and M_i, we must have

$$L_P = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) \leq \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) \leq \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = U_P.$$

Let δ be such that $R(P) < \varepsilon$ whenever $\mu(P) < \delta$, and let P be such a partition. Then we find that

$$A - \varepsilon < L_P \leq \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) \leq U_P < A + \varepsilon,$$

which implies that

$$\left| \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) - A \right| < \varepsilon.$$

[(ii) $\implies$ (i)] Let $\varepsilon > 0$ and let δ be such that

$$\left| \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) - A \right| < \frac{\varepsilon}{4}$$

for every partition P and $x_{i-1} \le t_i \le x_i$, $i = 1, \dots, n$ with $\mu(P) < \delta$, and let P be such a partition. The infimum m_i and the supremum M_i need not be attained (f is not assumed to be continuous), but by their definition there exist $\ell_i, u_i \in [x_{i-1}, x_i]$ such that $f(\ell_i) < m_i + \frac{\varepsilon}{4}$ and $f(u_i) > M_i - \frac{\varepsilon}{4}$. Since $\sum_{i=1}^{n}(x_i - x_{i-1}) = 1$, our assumption implies that

$$L_P > \sum_{i=1}^{n} f(\ell_i)(x_i - x_{i-1}) - \frac{\varepsilon}{4} > A - \frac{\varepsilon}{2},$$
$$U_P < \sum_{i=1}^{n} f(u_i)(x_i - x_{i-1}) + \frac{\varepsilon}{4} < A + \frac{\varepsilon}{2}.$$

It follows that

$$R(P) = U_P - L_P < \left(A + \frac{\varepsilon}{2}\right) - \left(A - \frac{\varepsilon}{2}\right) = \varepsilon.$$

□

Problem 5.2. Prove: if f is a monotonic function on the interval $[a, b]$, then f is integrable on $[a, b]$. (You may use any definition of the integral you wish.)

Solution. By setting $f := -f$ if need be, we may assume without loss of generality that f is increasing. Notice that f is bounded, because $f(a) \le f(x) \le f(b)$ for any $x \in [a, b]$. If $f(a) = f(b)$ then f is constant and there is nothing to prove; therefore, we can assume that $f(a) \ne f(b)$. Given a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$, let

$$L_P = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) = \sum_{i=1}^{n} f(x_{i-1})(x_i - x_{i-1}),$$
$$U_P = \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = \sum_{i=1}^{n} f(x_i)(x_i - x_{i-1}),$$

where $m_i = \inf\{f(x) \mid x_{i-1} \le x \le x_i\}$, $M_i = \sup\{f(x) \mid x_{i-1} \le x \le x_i\}$, and where we used the fact that f is increasing for the rightmost equalities. Recall that a function f is said to be *Darboux-integrable* if for any $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$U_P - L_P < \varepsilon$$

for any partition P such that $\mu(P) < \delta$, where $\mu(P) = \max\{x_i - x_{i-1} \mid i = 1, \dots, n\}$ is the

mesh of P. Let $\varepsilon > 0$, and $\delta = (f(b) - f(a))^{-1}\varepsilon$.

$$\begin{aligned} U_P - L_p &= \sum_{i=1}^{n} f(x_i)(x_i - x_{i-1}) - \sum_{i=1}^{n} f(x_{i-1})(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))\mu(P) \\ &= (f(b) - f(a))\mu(P) \\ &< \varepsilon. \end{aligned}$$

□

Problem 5.3. Define

$$f(x) = \int_0^{\pi} \frac{\sin(xt)}{t}\, dt,\ x \in \mathbb{R}$$

(a) Prove f is a continuous function on $\mathbb{R}$
(b) Prove f is differentiable on $\mathbb{R}$ and compute $f'(x)$.

Solution. a) Let $c \in \mathbb{R}$ and given $\varepsilon > 0$, for any x by the mean value theorem there exists $d_x \in B(c, |x-c|)$ such that $\sin(tx) - \sin(tc) = \left(\frac{d}{dx}\sin(tx)|_{x=d_x}\right)(x-c) = t\cos(td_x)(x-c)$ hence,

$$\begin{aligned} |f(x) - f(c)| &= \left| \int_0^{\pi} \frac{\sin(xt)}{t}\, dt - \int_0^{\pi} \frac{\sin(ct)}{t}\, dt \right| \\ &= \left| \int_0^{\pi} \frac{\sin(xt) - \sin(ct)}{t}\, dt \right| \\ &\leq \int_0^{\pi} \left| \frac{\sin(xt) - \sin(ct)}{t} \right| dt \\ &\leq \int_0^{\pi} \left| \frac{t\cos(td_x)(x-c)}{t} \right| dt \leq \pi|x-c| \end{aligned}$$

Thus for $\delta < \frac{\varepsilon}{\pi}$ we get $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$

b) For any $x \in \mathbb{R}$ function $g(t) = \begin{cases} \frac{\sin(xt)}{t} & t \neq 0 \\ x & t = 0 \end{cases}$ is continuous for $t \in [0, \pi]$ continuous on compact set is uniform continuous so limit of the integral is the integral of the limit.

Then

$$\begin{aligned}
\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} &= \lim_{h\to 0}\frac{1}{h}\left[\int_0^\pi \frac{\sin(xt+ht)}{t}\,dt - \int_0^\pi \frac{\sin(xt)}{t}\,dt\right] \\
&= \lim_{h\to 0}\frac{1}{h}\left[\int_0^\pi \frac{\sin(xt+ht)-\sin(xt)}{t}\,dt\right] \\
&= \lim_{h\to 0}\frac{1}{h}\left[\int_0^\pi \frac{2\cos\left(\frac{xt+ht+xt}{2}\right)\sin\left(\frac{ht}{2}\right)}{t}\,dt\right] \\
&= \left[\int_0^\pi \lim_{h\to 0}\frac{1}{h}\frac{2\cos\left(\frac{xt+ht+xt}{2}\right)\sin\left(\frac{ht}{2}\right)}{t}\,dt\right] \\
&= \int_0^\pi \cos(xt)\,dt = \frac{\sin(xt)}{x}\Big|_0^\pi = \begin{cases}\frac{\sin(\pi x)}{x} & x\neq 0\\ \pi & x=0\end{cases}
\end{aligned}$$

□

Problem 5.4. For every integer $n \geq 0$, let $I_n = \int_0^{\frac{\pi}{2}} \cos^n(t)\,dt$.

(a) Show that $(n+2)I_{n+2} = (n+1)I_n$. Then use it to find explicitly I_{2n} and I_{2n+1}

(b) Show that $\lim_{n\to\infty}\frac{I_{n+1}}{I_n} = 1$

(c) Show that $\{(n+1)I_nI_{n+1}\}$ is a constant sequence. Then conclude that $\lim_{n\to\infty} I_n\sqrt{2n} = \sqrt{\pi}$

Solution. a) Using integration by parts

$$\begin{aligned}
I_{n+2} &= \int_0^{\pi/2} \underbrace{\cos(t)}_{dv}\underbrace{\cos^{n+1}(t)}_{u}\,dt \\
&= \sin(t)\cos^{n+1}(t)\Big|_0^{\pi/2} - \int_0^{\pi/2}\sin(t)((n+1)\cos^n(t)(-\sin t))\,dt \\
&= (n+1)\int_0^{\pi/2}\underbrace{\sin^2(t)}_{1-\cos^2(t)}\cos^n(t)\,dt \\
&= (n+1)[I_n - I_{n+2}]
\end{aligned}$$

So we get $(n+2)I_{n+2} = (n+1)I_n$

By the previous we have $I_n = \frac{n-1}{n}I_{n-2}$ and we have $I_0 = \int_0^{\pi/2} 1\,dt = \frac{\pi}{2}$ and $I_1 =$

$\int_0^{\pi/2} \cos t \, dt = \sin t|_0^{\pi/2} = 1$ thus for $n = 2k+1$ odd then

$$I_{2k+1} = \frac{2k}{2k+1} I_{2k-1} = \frac{2k}{2k+1} \cdot \frac{2k-2}{2k-1} \cdots \frac{2}{3} = \frac{2k}{2k+1} \cdot \frac{2k}{2k} \cdot \frac{2k-2}{2k-1} \cdot \frac{2k-2}{2k-2} \cdots \frac{2}{3} = \frac{4^k (k!)^2}{(2k+1)!}$$

for $n = 2k$ even

$$I_{2k} = \frac{2k-1}{2k} I_{2k-2} = \frac{2k-1}{2k} \cdot \frac{2k-3}{2k-2} \cdots \frac{1}{2} \cdot I_0 = \frac{2k}{2k} \cdot \frac{2k-1}{2k} \cdot \frac{2k-2}{2k-2} \cdot \frac{2k-3}{2k-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2} = \frac{(2k)!\pi}{4^k (k!)^2 (2)}$$

b) Note that

$$I_{n+1} - I_n = \int_0^{\pi/2} \cos^{n+1}(t) - \cos^n(t) \, dt = \int_0^{\pi/2} \cos^n(t)(\cos t - 1) \, dt \leq 0$$

Since $1 \geq \cos t, \forall t$ hence $I_{n+1} \leq I_n$ for all n so the sequence I_n is decreasing. Thus

$$\begin{array}{rcccl} I_{n-1} & \geq & I_n & \geq & I_{n+1} \\ 1 & \geq & \dfrac{I_n}{I_{n-1}} & \geq & \dfrac{I_{n+1}}{I_{n-1}} \\ 1 & \geq & \frac{I_n}{I_{n-1}} & \geq & \frac{n}{n+1} \end{array}$$

Taking the limit by squeeze theorem we get $\lim \frac{I_n}{I_{n-1}} \to 1$

c) Multiplying $(n+2)I_{n+2} = (n+1)I_n$ by I_{n+1} gives $(n+2)I_{n+1}I_{n+2} = (n+1)I_n I_{n+1}$, so the sequence is constant. Its value can be read from the odd and even I_n:

$$(2k+1)I_{2k+1}I_{2k} = (2k+1)\frac{4^k (k!)^2}{(2k+1)!} \frac{(2k)!\pi}{4^k (k!)^2 (2)} = \frac{\pi}{2}$$

Thus

$$\sqrt{2(n)} I_n \sqrt{2(n+1)} I_{n+1} = \frac{\pi}{2(n+1)} 2\sqrt{n(n+1)} = \pi \sqrt{\frac{n}{n+1}}$$

By b), $\dfrac{\sqrt{2n} I_n}{\sqrt{2(n+1)} I_{n+1}} = \sqrt{\dfrac{n}{n+1}} \dfrac{I_n}{I_{n+1}} \to 1$, so

$$\left(\sqrt{2n} I_n\right)^2 = \sqrt{2n} I_n \sqrt{2(n+1)} I_{n+1} \cdot \frac{\sqrt{2n} I_n}{\sqrt{2(n+1)} I_{n+1}} \to \pi \cdot 1 = \pi,$$

and since $I_n > 0$, $\lim \sqrt{2n} I_n = \sqrt{\pi}$. □

Problem 5.5. Suppose $f : [0,1] \to [0,\infty)$ is a continuous function. Let $A = \int_0^1 f(x) \, dx$. Prove that

$$\sqrt{1 + A^2} \leq \int_0^1 \sqrt{1 + [f(x)]^2} \, dx \leq 1 + A.$$

Solution. For the second inequality, notice that, for any $x \geq 0$,

$$1 + x^2 \leq 1 + 2x + x^2 = (1+x)^2 \implies \sqrt{1+x^2} \leq 1 + x.$$

Therefore,

$$\int_0^1 \sqrt{1 + [f(x)]^2}\, dx \leq \int_0^1 1 + f(x)\, dx = 1 + A.$$

On the other hand, the function $\phi(x) = \sqrt{1+x^2}$ is convex, because

$$\phi''(x) = \left(\frac{x}{\sqrt{1+x^2}}\right)' = \frac{\sqrt{1+x^2} - \frac{x^2}{\sqrt{1+x^2}}}{1+x^2} = \frac{1}{(1+x^2)^{3/2}} > 0$$

for all $x \in \mathbb{R}$. By Jensen's inequality,

$$\sqrt{1+A^2} = \phi\left(\int_0^1 f(x)\, dx\right) \leq \int_0^1 \phi(f(x))\, dx = \int_0^1 \sqrt{1 + [f(x)]^2}\, dx.$$

Alternative proof for the first inequality: Notice that the graph of $F(x) = \int_0^x f(t)\, dt$ passes through both $(0,0)$ and $(1,A)$. The left-hand side is the straight line distance between the two points, while the middle part is the arc length of the graph of F (which then has to be at least as long.) □

Problem 5.6. Suppose f is a real, continuously differentiable function on $[0,1]$, $f(0) = f(1) = 0$ and

$$\int_0^1 f(x)e^x\, dx = 1$$

Prove

(a) $\int_0^1 f'(x)e^x\, dx = -1$ and

(b) $\int_0^1 (f'(x))^2\, dx \geq \frac{2}{e^2}$.

Solution. Integrating by parts we get

$$\int_0^1 f(x)e^x\, dx = e^x f(x)|_0^1 - \int_0^1 f'(x)e^x\, dx \Rightarrow 1 = -\int_0^1 f'(x)e^x\, dx \Rightarrow \int_0^1 f'(x)e^x\, dx = -1$$

For the second part, apply the Cauchy Schwartz integral inequality

$$\begin{aligned}
\int_0^1 (f'(x))^2\, dx \int_0^1 (e^x)^2\, dx &\geq \left(\int_0^1 f'(x)e^x\, dx\right)^2 \\
\int_0^1 (f'(x))^2 \cdot \left(\frac{e^2}{2} - \frac{1}{2}\right) &\geq 1 \\
\int_0^1 (f'(x))^2\, dx &\geq \frac{2}{e^2 - 1} \\
\int_0^1 (f'(x))^2\, dx &\geq \frac{2}{e^2}
\end{aligned}$$

□

Remark. In the version of this problem we worked from, part (a) asked to show $\int_0^1 f'(x)e^x\,dx = 1$. Integration by parts gives -1, as the solution shows; part (b) is not affected, since the integral is squared.

Problem 5.7. Suppose that f is a real, continuously differentiable function on $[a,b]$ with $b > a \geq 0$, $f(a) = f(b) = 0$ and

$$\int_a^b f(x)^2\,dx = 1.$$

Prove that

$$\int_a^b xf(x)f'(x)\,dx = -\frac{1}{2}$$

and that

$$\int_a^b x(f'(x))^2\,dx \cdot \int_a^b x(f(x))^2\,dx > \frac{1}{4}.$$

Be aware that the second inequality is a strict inequality.

Solution. Use integral by parts and note that $f(a) = f(b) = 0$, we have

$$\begin{aligned}\int_a^b xf(x)f'(x)dx &= xf(x)f(x)|_a^b - \int_a^b (xf(x))'f(x)dx \\ &= -\int_a^b (f(x) + xf'(x))f(x)dx \\ &= -\int_a^b xf(x)f'(x)dx - \int_a^b f(x)^2dx\end{aligned}$$

and thus

$$\int_a^b xf(x)f'(x)dx = -\frac{1}{2}\int_a^b f(x)^2dx = -\frac{1}{2}.$$

Note that

$$\left(\int_a^b g(x)h(x)dx\right)^2 \leq \int_a^b g(x)^2dx \int_a^b h(x)^2dx,$$

we have

$$\frac{1}{4} = \left(\int_a^b xf(x)f'(x)dx\right)^2 \leq \int_a^b x(f(x))^2dx \int_a^b x(f'(x))^2dx.$$

The equality holds only if there is some $\lambda \neq 0$ such that

$$\sqrt{x}f(x) = \lambda\sqrt{x}f'(x), \text{ for all } x \in [a,b],$$

and thus

$$f(x) = \lambda f'(x), \text{ for all } x \in (a, b).$$

Then there is some $C \in \mathbb{R}$ such that

$$f(x) = Ce^{x/\lambda}, \text{ for all } x \in (a, b).$$

Clearly, $C \neq 0$ since the integral is non-zero. Then this function is never 0, which contradicts the assumption that $f(a) = f(b) = 0$. Therefore we must have a strict inequality. □

Remark. In the version of this problem we worked from, the hypothesis was $\int_a^b f(x)\,dx = 1$. The first identity needs $\int_a^b f(x)^2\,dx = 1$, as stated above.

6 Sequences and series of functions

Problem 6.1. Let $\{f_n\}$ be a sequence of continuous functions which converges pointwise to a function f on a set D in $\mathbb{R}^d$.

(a) If $\{f_n\}$ converges uniformly to f, prove that

$$\lim_{n\to\infty} f_n(x_n) = f(x)$$

for any sequence of points $x_n \in D$ and $x \in D$ such that $x_n \to x$.

(b) Suppose D is compact and f is continuous on D. Prove that the convergence condition above implies that $\{f_n\}$ converges uniformly to f on D.

Solution.

(a) Take any sequence of points $x_n \in D$ and $x \in D$ such that $x_n \to x$. Take $\varepsilon > 0$, there is some $N \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2}, \text{ for all } x \in D, n \geq N.$$

Therefore,

$$|f_n(x_n) - f(x_n)| < \frac{\varepsilon}{2}, \text{ for all } n \geq N.$$

Note that f_n converges uniformly to f and f_n is continuous imply that f is continuous. Then $x_n \to x$ gives $f(x_n) \to f(x)$. Thus, there is some $N_1 \geq N$ such that

$$|f(x_n) - f(x)| < \frac{\varepsilon}{2}, \text{ for all } n \geq N_1.$$

Thus,

$$|f_n(x_n) - f(x)| < \varepsilon, \text{ for all } n \geq N_1.$$

Therefore, $f_n(x_n) \to f(x)$ as $n \to \infty$.

(b) Assume the opposite. Then f_n does not uniformly converge to f, so there is some $\varepsilon > 0$ such that

$$\sup_{x \in D} |f_n(x) - f(x)| > \varepsilon, \text{ for infinitely many } n \in \mathbb{N}.$$

Then for each such n, we can find some $x_n \in D$ such that $|f_n(x_n) - f(x_n)| > \varepsilon/2$. Since D is compact, by taking a subsequence $n_1 < n_2 < \cdots$ of these n we can assume that $x_{n_k} \to x \in D$. For the indices n which are not of the form n_k, put $x_n := x$; then the whole sequence $x_n \to x$. Then the hypothesis gives

$$\lim_{n \to \infty} |f_n(x_n) - f(x)| = 0,$$

and thus, since f is continuous,

$$\lim_{n \to \infty} |f_n(x_n) - f(x_n)| = 0,$$

which contradicts $|f_n(x_n) - f(x_n)| > \varepsilon/2$ for all $n = n_k$.

□

Problem 6.2.

(a) Let $\{f_n\}$ be a sequence of continuous functions which converges uniformly to a function f on a set $E \subset \mathbb{R}^d$. Prove that

$$\lim_{n \to \infty} f_n(x_n) = f(x)$$

for every sequence of points $x_n \in E$ and $x \in E$ such that $x_n \to x$.

(b) Let $\{f_n\}$ be a sequence of continuous function defined on $[0, \infty)$ as

$$f_n(x) = \begin{cases} \sin^2(\pi x), & n \le x \le n+1 \\ 0, & \text{otherwise.} \end{cases}$$

For any sequence $0 \le x_n < \infty$ and $x_n \to x < \infty$, show that $\lim f_n(x_n) = 0$ but f_n does not converges uniformly to 0.

Solution.

(a) This is part (a) of Problem 6.1, which is solved there.

(b) Take any sequence $0 \le x_n < \infty$ and $x_n \to x < \infty$. Then there exists some integer $N \ge -1$ such that $N + 1 \le x < N + 2$. Thus $N < x < N + 3$. Since $x_n \to x$, there is some $n_0 \in \mathbb{N}$, $n_0 > N + 3$, such that

$$N < x_n < N + 3, \text{ for all } n \ge n_0.$$

Then $n_0 > N + 3$ also implies that

$$x_n \notin [n, n+1], \text{ for all } n \ge n_0.$$

Therefore, $f_n(x_n) = 0$ for all $n \geq n_0$ and thus $f_n(x_n) \to 0$. We next show that f_n does not converge uniformly to 0. We will show that for $\varepsilon = 1/2$, for each $N \in \mathbb{N}$, there is some $n \geq N$ $x \in [0, +\infty)$ such that

$$|f_n(x)| > \frac{1}{2}.$$

Indeed, take $n = 2N + 1, x = n + \frac{1}{2}$, we have

$$|f_n(x)| = \sin^2(\pi x) = \sin^2\left(\pi(2N+1) + \frac{\pi}{2}\right) = 1 > 1/2.$$

Therefore, f_n does not uniformly converge to 0.

□

Problem 6.3.

(a) Let $\{f_n\}$ be a sequence of complex-valued functions defined on a metric space E. Suppose that each supremum norm $||f_n||$ (defined as the sup $\{|f_n(x)| \mid x \in E\}$) is finite. Prove that if the series of numbers $\sum ||f_n||$ converges, then the series of functions $\sum f_n$ converges uniformly on E.

(b) Prove that the converse is false. For that purpose consider the series of real-valued functions $\sum f_n$ defined on $[0, 1]$ for $n = 1, 2, \ldots$ by

$$f_n(x) = \begin{cases} 1/n & 1/2^n < x \leq 1/2^{n-1}, \\ 0 & \text{otherwise.} \end{cases}$$

Solution.

(a) This is literally the Weierstrass M-test. Here's a proof anyway. Let $\varepsilon > 0$. Since the series $\sum ||f_n||$ converges, it is in particular Cauchy, so there exists an $N \in \mathbb{N}$ such that for any $m > n > N$, we have

$$\sum_{i=n+1}^{m} ||f_i|| < \varepsilon.$$

Let $x \in E$, $m > n > N$, and consider the partial sums $s_n = \sum_{i=0}^{n} f_i$. Then

$$\begin{aligned} |s_m(x) - s_n(x)| &= \left| \sum_{i=n+1}^{m} f_i(x) \right| \\ &\leq \sum_{i=n+1}^{m} |f_i(x)| \\ &\leq \sum_{i=n+1}^{m} ||f_i|| < \varepsilon, \end{aligned}$$

so the sequence $\{s_n(x)\}$ is Cauchy. By completeness of $\mathbb{C}$, it converges to some value $s(x)$. It remains to be shown that the sequence $\{s_n\}$ converges uniformly to

this function s. Let $n > N$. Then

$$\begin{aligned} |s(x) - s_n(x)| &= |\lim_{m\to\infty} s_m(x) - s_n(x)| \\ &= \lim_{m\to\infty} |s_m(x) - s_n(x)| \\ &< \varepsilon \text{ when } m > n. \end{aligned}$$

Since N did not depend on x, the convergence is uniform.

(b) For each $n \in \mathbb{N}$ we have

$$\|f_n\| = \sup\{|f_n(x)| : x \in [0,1]\} = \frac{1}{n}.$$

Thus $\sum \|f_n\|$ does not converge. We next show that $\sum f_n$ converges uniformly, i.e.

$$\lim_{n\to\infty} \sup_{x\in[0,1]} |\sum_{k=n+1}^{\infty} f_k(x)| = 0$$

Note that $(1/2^i, 1/2^{i-1}]$ for all $i \in \mathbb{N}$ does not overlap, we have

$$\sup_{x\in[0,1]} |\sum_{k=n+1}^{\infty} f_k(x)| = \frac{1}{n+1},$$

attained when $x \in (1/2^{n+1}, 1/2^n]$. Thus

$$\lim_{n\to\infty} \sup_{x\in[0,1]} |\sum_{k=n+1}^{\infty} f_k(x)| = 0$$

and hence, $\sum f_n$ converges uniformly.

□

Problem 6.4. Let $\{f_n\}$ be a sequence of continuously differentiable functions defined on $[a,b]$. Suppose $\{f_n\}$ converges to some function f pointwise on $[a,b]$ and $\{f_n'\}$ converges uniformly to some function g on $[a,b]$. Prove that f is differentiable in (a,b) and $f'(x) = g(x)$ for all $x \in (a,b)$.

Solution. Let $\varepsilon > 0$ and fix $x_0 \in [a,b]$. Choose an N such that for all $m, n \geq N$, we have

$$|f_n(x_0) - f_m(x_0)| < \frac{\varepsilon}{2}$$

and

$$|f_n'(t) - f_m'(t)| < \frac{\varepsilon}{2(b-a)} \quad (a \leq t \leq b).$$

Applying the Mean Value Theorem to the function $f_n - f_m$ gives

$$|f_n(x) - f_m(x) - f_n(t) + f_m(t)| \leq \frac{|x-t|\varepsilon}{2(b-a)} \leq \frac{\varepsilon}{2},$$

for any $x, t \in [a,b]$ and $m, n \geq N$. Notice that

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)|,$$

we have

$$|f_n(x) - f_m(x)| < \varepsilon \quad (a \leq x \leq b, m, n \geq N).$$

This means that f_n uniformly converges to f. Now fix a point $x \in [a,b]$, and define

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}, \quad \phi(t) = \frac{f(t) - f(x)}{t - x}, a \leq t \leq b, t \neq x.$$

Then

$$\lim_{t \to x} \phi_n(t) = f_n'(x), \text{ for all } n.$$

Then

$$|\phi_n(t) - \phi_m(t)| \leq \frac{\varepsilon}{2(b-a)}, \text{ for all } m, n \geq N.$$

This means ϕ_n converges uniformly for $t \neq x$. Since $f_n \to f$ we have $\phi_n \to \phi$ when $t \neq x$. Thus we have

$$\lim_{t \to x} \phi(t) = \lim_{n \to \infty} f_n'(x).$$

This means

$$f'(x) = g(x) \quad (a \leq x \leq b).$$

□

Problem 6.5. Suppose $f_1 \in C([0,1])$ and c is a constant. Define, for all $n \geq 1$,

$$f_{n+1}(x) = c + \int_0^x f_n(t)\, dt$$

for all $x \in [0,1]$. Prove that

(a) $\{f_n\}$ converges uniformly to some function $f \in C([0,b])$ for all $b \in (0,1)$;
(b) $f \in C((0,1))$; and
(c) $f(x) = ce^x$ for all $x \in [0,1]$.

Solution. The following proof shows directly the result *(c)*, and thus it implies also *(a)*

and *(b)*. Iteratively, we write:

$$\begin{aligned}
f_2(x) &= c + \int_0^x f_1(t)dt, \\
f_3(x) &= c + \int_0^x f_2(t)dt \\
&= c + \int_0^x \left(c + \int_0^t f_1(t_1)dt_1 \right) dt \\
&= c + cx + \int_0^x \int_0^t f_1(t_1)dt_1 \, dt, \\
f_4(x) &= c + \int_0^x f_3(t)dt \\
&= c + \int_0^x \left(c + ct + \int_0^t \int_0^{t_1} f_1(t_2)dt_2 \, dt_1 \right) dt \\
&= c + cx + \frac{cx^2}{2} + \int_0^x \int_0^t \int_0^{t_1} f_1(t_2)dt_2 \, dt_1 \, dt, \\
&\ldots, \\
f_{n+1}(x) &= c \sum_{k=0}^{n-1} \frac{x^k}{k!} + \int_0^x \int_0^t \int_0^{t_1} \ldots \int_0^{t_{n-2}} f_1(t_{n-1})dt_{n-1} \, dt_{n-2} \ldots dt_1 \, dt \\
&= g_{n+1}(x) + h_{n+1}(x).
\end{aligned}$$

We see from the Taylor expansion theorem that g_{n+1} converges pointwise to g, with $g(x) := ce^x$ for all $x \in [0,1]$. Moreover,

$$\left| \frac{x^k}{k!} \right| \leq \frac{1}{k!} \quad \text{for all } k \in \mathbb{N}, x \in [0,1],$$

and $\sum_{k=0}^{\infty} \frac{1}{k!}$ converges by the D'Alembert's ratio test. That implies from the Weierstrass M-test that g_{n+1} converges uniformly to g in $[0,1]$.

Consider $M > 0$ such that $|f_1(x)| \leq M$ for all $x \in [0,1]$ (such M exists as f_1 is continuous). Then we have the following inequalities:

$$\begin{aligned}
|h_{n+1}(x)| &\leq \int_0^x \int_0^t \int_0^{t_1} \ldots \int_0^{t_{n-2}} |f_1(t_{n-1})| \, dt_{n-1} \, dt_{n-2} \ldots dt_1 \, dt \\
&\leq \int_0^x \int_0^t \int_0^{t_1} \ldots \int_0^{t_{n-2}} M dt_{n-1} \, dt_{n-2} \ldots dt_1 \, dt \\
&= \int_0^x \int_0^t \int_0^{t_1} \ldots \int_0^{t_{n-3}} M t_{n-2} dt_{n-2} \ldots dt_1 \, dt \\
&= \int_0^x \int_0^t \int_0^{t_1} \ldots \int_0^{t_{n-4}} M \frac{t_{n-3}^2}{2} dt_{n-3} \ldots dt_1 \, dt \\
&\ldots \\
&= M \frac{x^n}{n!} \leq \frac{M}{n!} \quad \text{for all } n \in \mathbb{N} \text{ and } x \in [0,1].
\end{aligned}$$

Thus $\sup_{[0,1]} |h_{n+1}(x)| \xrightarrow{n\to\infty} 0$ and therefore h_{n+1} converges uniformly to 0 in $[0,1]$. In conclusion, $f_{n+1} = g_{n+1} + h_{n+1}$ converges uniformly to f in $[0,1]$, with $f(x) := ce^x$. □

Problem 6.6.

(a) Prove the series $\frac{x}{1!} + \frac{x^3}{3!} + \ldots + \frac{x^{2n-1}}{(2n-1)!} + \ldots$ and $1 + \frac{x^2}{2!} + \ldots + \frac{x^{2n}}{(2n)!} + \ldots$ both converge uniformly on any bounded interval of $\mathbb{R}$.

(b) Define $f(x) = \frac{x}{1!} + \frac{x^3}{3!} + \ldots + \frac{x^{2n-1}}{(2n-1)!} + \ldots, g(x) = 1 + \frac{x^2}{2!} + \ldots + \frac{x^{2n}}{(2n)!} + \ldots$ on $\mathbb{R}$. Prove that

$$\begin{cases} 1 + \int_0^x f(t)dt & = g(x), \\ \int_0^x g(t)dt & = f(x), \end{cases} \forall x \in \mathbb{R}.$$

Solution.

(a) There exists $M > 0$ such that for all $x \in (a,b) \subset \mathbb{R}$, $|x| < M$. Therefore

$$\left| \frac{x^{2n-1}}{(2n-1)!} \right| \leq \frac{M^{2n-1}}{(2n-1)!} := a_n, \text{ for all } n \in \mathbb{N}.$$

Then we have

$$\frac{a_{n+1}}{a_n} = \frac{M^{2n+1}}{(2n+1)!} \cdot \frac{(2n-1)!}{M^{2n-1}} = \frac{M^2}{2n(2n+1)} \to 0 \text{ as } n \to \infty.$$

By D'Alembert's criterion for series convergence, we have

$$\sum_{n=1}^{\infty} a_n \text{ converges .}$$

By Weierstrass criterion,

$$\sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)!} \text{ converges uniformly on } (a,b).$$

Similarly, we have

$$\sum_{n=1}^{\infty} \frac{x^{2n}}{(2n)!} \text{ converges uniformly on } (a,b).$$

(b) Claims are obviously true for $x = 0$. If $x > 0$, we have

$$\sum_{n=1}^{\infty} \frac{t^{2n-1}}{(2n-1)!} \text{ and } \sum_{n=1}^{\infty} \frac{t^{2n}}{(2n)!} \text{ converge uniformly on } [0,x].$$

Aditionally, $\frac{t^{2n-1}}{(2n-1)!}$ and $\frac{t^{2n}}{(2n)!}$ are integrable for all $n \in \mathbb{N}$ then

$$\begin{aligned}\int_0^x f(t)dt &= \int_0^x \sum_{n=1}^{\infty} \frac{t^{2n-1}}{(2n-1)!}dt \\ &= \sum_{n=1}^{\infty} \int_0^x \frac{t^{2n-1}}{(2n-1)!}dt \\ &= \sum_{n=1}^{\infty} \frac{x^{2n}}{(2n)!} \\ &= g(x) - 1,\end{aligned}$$

and

$$\begin{aligned}\int_0^x g(t)dt &= \int_0^x \sum_{n=0}^{\infty} \frac{t^{2n}}{(2n)!}dt \\ &= \sum_{n=0}^{\infty} \int_0^x \frac{t^{2n}}{(2n)!}dt \\ &= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \\ &= f(x).\end{aligned}$$

The case $x < 0$ is the same by substitution $[0, x]$ by $[x, 0]$.

□

Problem 6.7. Let $\mathcal{X}$ be the metric space $C([0,1])$ with the metric

$$d(f,g) = \sup_{x\in[0,1]} |f(x) - g(x)|, \quad f,g \in \mathcal{X}.$$

Let $\mathcal{S} := \{f \in \mathcal{X} : f(0) = 0, f \text{ is differentiable, and } |f'(x)| \leq 1, \forall x \in [0,1]\}$. Show that S is a pre-compact subset of $\mathcal{X}$.

Solution. First, notice that this metric d is the metric induced by the uniform norm

$$||f||_\infty = \sup_{x\in[0,1]} |f(x)|.$$

Unrolling the definitions, convergence in this norm is equivalent to uniform convergence. We want to show that every sequence of functions in $\mathcal{S}$ has a subsequence which uniformly converges to some element of $\mathcal{X}$. By the Arzelà-Ascoli theorem, it suffices to show that every sequence is uniformly bounded and uniformly equicontinuous.
For any $f \in \mathcal{S}$, we have that $f(0) = 0$ and $|f'(x)| \leq 1$ for all $x \in [0,1]$; in particular, it is bounded above by the function which satisfies $g(0) = 0$ and $g'(x) \equiv 1$, i.e. $g(x) = x$. Similarly, it is bounded below by the function which satisfies $h(0) = 0$ and $h'(x) \equiv -1$,

i.e. $h(x) = -x$. Therefore, $|f(x)| \leq 1$, and so

$$||f||_\infty = \sup_{x\in[0,1]} |f(x)| \leq 1.$$

In other words, every sequence is uniformly bounded.
Let $\varepsilon > 0$ and $\delta = \varepsilon$. Let $f \in \mathcal{S}$, $x \in [0,1]$ and $y \in B(x,\delta)$, the δ-ball around x, and furthermore suppose that $x \neq y$. By the Mean Value Theorem, there exists a $c \in [x,y]$ (if $x < y$, and $[y,x]$ if $x > y$) such that $f'(c) = \frac{f(x)-f(y)}{x-y}$. Therefore,

$$\begin{aligned} |f(x) - f(y)| &= |f'(c)||x-y| \\ &\leq |x-y| \\ &< \varepsilon. \end{aligned}$$

Since δ does not depend on f or x, every sequence is uniformly equicontinuous. □

Problem 6.8. Let $f \in C([0,1])$ such that $f(0) = 0$ and

$$||f||_{[0,1]} := \sup_{x\in[0,1]} |f(x)| \leq 1.$$

Prove that the sequence of powers $f, f^2, f^3, \ldots$ is equicontinuous if and only if $||f||_{[0,1]} < 1$.

Solution. Let $X_0 = \{x \in [0,1] \mid |f(x)| < 1\}$ and $X_1 = \{x \in [0,1] \mid |f(x)| = 1\}$. Clearly, $X_0 \sqcup X_1 = [0,1]$ by assumption, and $f(0) = 0$ implies that $X_0 \neq \emptyset$. For any $x \in X_0$,

$$|(f(x))^n| = |f(x)|^n \to 0$$

because $|f(x)| < 1$. On the other hand, for any $x \in X_1$,

$$|(f(x))^n| = |f(x)|^n = 1^n = 1.$$

Therefore, the sequence $\{|f|^n\}$ converges (pointwise!) to the function $\bar{f} = \chi_{X_1}$, the indicator function for X_1. (The sequence $\{f^n\}$ itself need not converge at points where $f(x) = -1$, which is why we pass to absolute values.) Notice that, since $X_0 \neq \emptyset$ and $[0,1]$ is connected, $\bar{f}$ is continuous if and only if $X_1 = \emptyset$.
Moreover, the sup norm is submultiplicative, because for any two $g, h \in C([0,1])$,

$$||gh||_{[0,1]} = \sup_{x\in[0,1]} |g(x)h(x)| \leq (\sup_{x\in[0,1]} |g(x)|)(\sup_{x\in[0,1]} |h(x)|) = (||g||_{[0,1]})(||h||_{[0,1]}).$$

It follows that

$$||f^n||_{[0,1]} \leq (||f||_{[0,1]})^n \leq 1,$$

so $\{f^n\}$ is bounded by 1.

[$\Longrightarrow$] If the sequence $\{f^n\}$ is equicontinuous, then so is $\{|f|^n\}$, because $\big||a|-|b|\big| \leq |a-b|$. By the Arzelà-Ascoli theorem, the sequence $\{|f|^n\}$, which is bounded by 1, has a convergent subsequence. This subsequence must converge to $\bar{f}$. Convergence in the sup norm is equivalent to uniform convergence, and the uniform limit of continuous functions is continuous. Therefore, $\bar{f}$ is continuous, and so $X_1 = \emptyset$. But since $[0,1]$ is compact, $|f(x)|$ reaches its maximum, so we have

$$||f||_{[0,1]} = \max_{x\in[0,1]} |f(x)| < 1.$$

[$\Longleftarrow$] If $X_1 = \emptyset$, then

$$||f^n||_{[0,1]} \to 0,$$

so $\{f^n\}$ converges to $\bar{f} = 0$ in the sup norm; that is, it converges uniformly. By the Arzelà-Ascoli theorem, $\{f^n\}$ is bounded and equicontinuous.

□

Problem 6.9. Construct sequences $\{f_n\} \subset C_c([0,1])$ (the space of continuous functions with compact support in $[0,1]$) that satisfy statements (a), (b), or (c) below (one sequence for each of the statements).

(a) $||f_n||_\infty \to \infty$ and $||f_n||_1 \to 0$.
(b) $f_n(x) \to 0$ for all x and $||f_n||_1 \to \infty$.
(c) $||f_n||_1 \to 0$ and $f_n(x)$ does not converge for any x.

Solution.

(a) Let

$$f_n(x) = \begin{cases} n - n^3 x & 0 \leq x \leq \frac{1}{n^2}, \\ 0 & \frac{1}{n^2} < x. \end{cases}$$

Then

$$||f_n||_\infty = \max\{|f_n(x)| \mid x \in [0,1]\} = n \to \infty$$

and

$$||f_n||_1 = \int_0^1 |f_n(x)|\,dx = \left[nx - n^3\frac{x^2}{2}\right]_0^{\frac{1}{n^2}} = \frac{1}{n} - \frac{1}{2n} \to 0.$$

(b) Let

$$f_n(x) = \begin{cases} 2n^3 x & 0 \leq x \leq \frac{1}{2n}, \\ 2n^2 - 2n^3 x & \frac{1}{2n} < x \leq \frac{1}{n}, \\ 0 & \frac{1}{n} < x. \end{cases}$$

Then, for any $0 \neq x \in [0,1]$, we have $f_n(x) = 0$ for all $n > \frac{1}{x}$. Moreover, $f_n(0) = 0$

for all n, so $f_n(x) \to 0$ for all $x \in [0,1]$. On the other hand,

$$\begin{aligned} ||f_n||_1 &= \int_0^1 |f_n(x)|\,dx \\ &= \left[n^3x^2\right]_0^{\frac{1}{2n}} + \left[2n^2x - n^3x^2\right]_{\frac{1}{2n}}^{\frac{1}{n}} \\ &= \frac{n}{4} + \left((2n-n) - \left(n - \frac{n}{4}\right)\right) \\ &= \frac{n}{2} \to \infty. \end{aligned}$$

(c) Use moving tents. For $k \geq 1$ and $j = 0, 1, \ldots, k$ let

$$T_{k,j}(x) = \max\left(0,\ 1 - k\left|x - \frac{j}{k}\right|\right),$$

a tent of height 1 centred at j/k, of half-width $\frac{1}{k}$. List them in the order $T_{1,0}, T_{1,1}, T_{2,0}, T_{2,1}, T_{2,2}, T_{3,0}, \ldots$ and let f_n be the n-th one. Then:

- $||T_{k,j}||_1 \leq \frac{1}{k}$, and $k \to \infty$ along the list, so $||f_n||_1 \to 0$.
- Fix $x \in [0,1]$. For each k some centre j/k is within $\frac{1}{2k}$ of x, and then $T_{k,j}(x) \geq \frac{1}{2}$. So $f_n(x) \geq \frac{1}{2}$ for infinitely many n.
- For $k \geq 3$ at most two of the $k+1 \geq 4$ centres j/k are within $\frac{1}{k}$ of x, so $T_{k,j}(x) = 0$ for some j. So $f_n(x) = 0$ for infinitely many n.

Hence $\{f_n(x)\}_n$ does not converge, for any $x \in [0,1]$.

□

7 Metric spaces and topology

Problem 7.1. Show that a convergent sequence in a metric space is bounded.

Solution. Let (X,d) be a metric space and a_n a convergent sequence which converges to a. Then for $\varepsilon = 1$ there exists N such that

$$d(a_n, a) < 1\ , \forall n > N$$

Then let $M = \max_{i\in\{1,\cdots N\}}\{d(a_i,a)\}$ let $M' = \max\{M,1\}$ clearly for any n we have $d(a_n,a) \leq M'$, so $a_n \in B(a, M'+1)$. □

Problem 7.2.

(a) Define a compact metric space

(b) Suppose $f : X \to Y$ is a continuous function from a compact metric space X to a metric space Y. Show that f is bounded.

Solution.

(a) A topological space (X,d) is said to be compact metric if (X,d) is a metric space

and X is a compact set. Note that (X, d) is a metric space if
- For any $x, y \in X$, $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
- For any $x, y \in X, d(x, y) = d(y, x)$.
- For any $x, y, z \in X, d(x, y) \leq d(x, z) + d(y, z)$.

The set X is said to be compact if for any sequence $\{x_n\} \subset X$, there is some subsequence $\{x_{n_k}\}$ converging to some $\bar{x} \in X$.

(b) We will prove that $f(X)$ is a compact set. Firstly, we prove the closednees. Take any convergent sequence $\{y_n\} \subset f(X)$, then there is a sequence $\{x_n\} \subset X$ such that

$$y_n = f(x_n), \text{ for all } n \in \mathbb{N}.$$

Since X is compact, there is a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that $x_{n_k} \to \bar{x} \in X$. The continuity of f implies that $y_{n_k} = f(x_{n_k}) \to f(\bar{x}) \in f(X)$. Since $\{y_n\}$ converges, we deduce that $y_n \to f(\bar{x}) \in f(X)$.
We next show the boundedness of $f(X)$. Assume by contradiction that there is some sequence $y_n \subset f(X)$ which is unbounded. Fixing $y_0 \in Y$, this means that we can choose the sequence so that $d(y_0, y_n) \to \infty$. Note that there is some sequence $\{x_n\} \subset X$ such that

$$y_n = f(x_n), \text{ for all } n \in \mathbb{N}.$$

Since $\{x_n\} \subset X$, by taking subsequence, we can also assume that x_n converges to some $\bar{x} \in X$. The continuity of f implies that $y_n = f(x_n) \to f(\bar{x}) \in f(X)$. This contradicts the fact that $d(y_0, y_n) \to \infty$, since a convergent sequence is bounded.

□

Problem 7.3.

(a) State the definition of a compact set in a metric space.

(b) Let X be a metric space in which every infinite subset has a limit point. Prove X is compact.

Solution.

(a) Given a metric space X, a subset $K \subseteq X$ is compact if for any sequence a_i in K there exists a subsequence a_{n_i} which converges to some $a \in K$.

(b) Let a_i be a sequence in X. The set $A = \{x \in X \mid x = a_n \text{ for some } n \geq 0\}$ is either finite or infinite.
If A is finite, by the pigeonhole principle there exists an $a \in A$ such that $a = a_n$ for infinitely many n. This gives a constant subsequence, which clearly converges to a.
If A is infinite, by hypothesis there exists an $a \in X$ which is a limit point of A, i.e. such that any open set containing a contains infinitely many elements of A. In particular, there exists an n_1 such that $a_{n_1} \in B(a, 1)$. Now suppose we have an n_i such that $a_{n_i} \in B(a, \frac{1}{i})$. Since a is a limit point, there are infinitely many n such that $d(a_n, a) < \frac{1}{i+1}$; in particular, there exists an $n_{i+1} > n_i$ such that $d(a_{n_{i+1}}, a) < \frac{1}{i+1}$. By induction, this gives a subsequence $\{a_{n_i}\}_{i \geq 1}$ which converges to a, so X is compact.

□

Problem 7.4.

(a) State the Heine-Borel Theorem for a Euclidean space.
(b) Prove the Heine-Borel Theorem.
(c) Give a counter-example to the Heine-Borel Theorem in a general metric space. Justify your example.

Solution.

(a) For a subset of Euclidean space $\mathbb{R}^m$, the following two statements are equivalent
  (i) S is closed, and bounded;
  (ii) S is compact, i.e. every open cover of S has a finite subcover.

(b) (i)⇒(ii) Suppose that S is closed and bounded. Then there is some $a > 0$ such that $S \subset B(0,a)$. Since $B(0,a) \subset [-a,a]^m$, we have

$$S \subset T := [-a,a]^m.$$

Note that closed subset of a compact set is compact, it suffices to show that T is compact. Since $[-a,a]$ is a closed interval in $\mathbb{R}$, it is compact. We show that T is compact using sequences (in a metric space, compactness is equivalent to sequential compactness). Let $\{t_k\}$ be a sequence in T. Its first coordinates lie in $[-a,a]$, so by the Bolzano-Weierstrass theorem a subsequence has convergent first coordinates; a further subsequence has convergent second coordinates, and so on. After m steps we get a subsequence which converges to a point of T, because $[-a,a]$ is closed. So T is compact. Since closed subset of compact set is compact, we have S is compact.
(ii)⇒(i) Suppose that S is compact. Then take the open cover $\{B(x,1)|x \in S\}$ of S. Then there are $x_1, \ldots, x_n \in S$ such that

$$S \subset \bigcup_{i=1}^{n} B(x_i, 1).$$

Define

$$r = 2 + \max_{i \neq j} \left\| x_i - x_j \right\|,$$

we will prove that $S \subset B(x_1, r)$. Indeed, for any $x \in S$, there is some i such that $x \in B(x_i, 1)$. Then

$$\|x - x_1\| \leq \|x - x_i\| + \|x_i - x_1\| \leq 1 + \max_{i \neq j} \left\| x_i - x_j \right\| = r - 1 < r.$$

Therefore, $x \in B(x_1, r)$. This showed that S is bounded.
Next we will show that S is closed. Assume by contradiction that S is not closed, i.e. there is some $a \in \mathbb{R}^m$ which is a limit point of S but does not belong to S. Then for any $x \in S$, define

$$r_x = \frac{1}{2} \|x - a\|.$$

Then we have

$$S \subset \bigcup_{x \in S} B(x, r_x) := \mathcal{O}$$

and $a \notin \mathcal{O}$. Since S is compact, there are $x_1, \ldots, x_n \in S$ such that

$$S \subset \bigcup_{i=1}^{n} B(x_i, r_{x_i}).$$

Write $r_i = r_{x_i}$. For $y \in B(x_i, r_i)$ we have

$$\|y - a\| \geq \|x_i - a\| - \|y - x_i\| > 2r_i - r_i = r_i.$$

Then define $r = \min_{1 \leq i \leq n} r_i$, we have $B(a, r)$ does not intersect $\bigcup_{i=1}^{n} B(x_i, r_i)$, and thus S, which contradicts with the fact that a is a limit point of S.

(c) Take the discrete metric on an infinite set. Then every subset is bounded (by 1) as well as closed, but only the finite subsets are compact.

□

Problem 7.5. Suppose (X, d) and (Y, δ) are metric spaces. Assume K is a compact subset of X. Recall the graph of function $f : K \to Y$ is the set

$$\text{gph}(f) = \{(x, f(x)) : x \in K\}.$$

Prove that if f is continuous is and only if $\text{gph}(f)$ is compact.

Solution.

($\Rightarrow$) Suppose that f is continuous. Let $(x_n, f(x_n))$ be an arbitrary sequence in $\text{gph}(f)$. Because $\{x_n\}_{n \in \mathbb{N}} \subset K$, there exists a subsequence $\{x_{n_k}\}_{k \in \mathbb{N}}$ such that

$$x_{n_k} \to x \in K.$$

Since f is continuous on K, $f(x_{n_k}) \to f(x)$, thus $(x_{n_k}, f(x_{n_k})) \to (x, f(x)) \in \text{gph}(f)$. Therefore, $\text{gph}(f)$ is sequentially compact. Since $\text{gph}(f)$ is a metric space, it is compact.

Alternatively, we can remark that, if we set $g = (1 \times f) \circ \Delta : K \to K \times K \to K \times Y$, then $g(K) = \text{gph}(f)$. Since f is continuous, g is continuous. Since the continuous image of a compact set is compact, this implies that $\text{gph}(f)$ is compact.

($\Leftarrow$) Assume that $\text{gph}(f)$ is compact. Let x is a point in K and $\{x_n\}_{n \in \mathbb{N}} \subset K, x_n \to x$. It suffices to prove that $f(x_n) \to f(x)$. Since $\text{gph}(f)$ is compact and $\{(x_n, f(x_n))\}_{n \in \mathbb{N}} \subset \text{gph}(f)$, there exists a subsequence $\{(x_{n_k}, f(x_{n_k}))\}_{k \in \mathbb{N}}$ such that

$$(x_{n_k}, f(x_{n_k})) \to (x, f(x)) \in \text{gph}(f).$$

If $f(x_n) \not\to f(x)$, there exist $\varepsilon > 0$ and a subsequence $\{x_{n_l}\}$ such that $x_{n_l} \to x$ but

$$|f(x_{n_l}) - f(x)| > \varepsilon, \forall l \in \mathbb{N}.$$

However, $\{(x_{n_l}, f(x_{n_l})\} \subset \text{gph}(f)$, so the latter being compact implies that there exists a subsequence $\left\{(x_{n_{l_i}}, f(x_{n_{l_i}})\right\}$ such that

$$(x_{n_{l_i}}, f(x_{n_{l_i}})) \to (x, f(x)) \text{ as } i \to \infty.$$

For this reason,

$$|f(x_{n_{l_i}}) - f(x)| \to 0 \text{ as } i \to \infty.$$

This contradicts the fact that $|f(x_{n_l}) - f(x)| > \varepsilon, \forall l \in \mathbb{N}$. Therefore, $f(x_n) \to f(x)$. □

Problem 7.6. Suppose that K is a compact subset of $\mathbb{R}$ and $f : K \to \mathbb{R}$ is a continuous function which is one-to-one. Prove that inverse of f is continuous.

Solution. Notice that $f : K \to f(K)$ is bijective and we need to prove that $f^{-1} : f(K) \to K$ is continuous. Take some closed $A \subset K$, we will prove $f(A)$ is closed. Since K is compact and any closed subset of compact set is compact, we have A is compact. Take any sequence $\{y_n\} \subset f(A)$ converging to some y_0. Then there is a sequence $\{x_n\} \subset A$ such that $y_n = f(x_n)$. Since A is compact, there is a subsequence $\{x_{n_k}\}$ converging to some x_0. Since f is continuous

$$y_{n_k} = f(x_{n_k}) \to f(x_0),$$

or equivalently, $y_0 = f(x_0)$. Therefore, $f(A)$ is closed. Finally, take $A \subset K$ is open, we will show that $f(A)$ is open. Note that $K \setminus A$ is closed in K, hence compact, and f is one to one and onto $f(K)$, so by the first part $f(K \setminus A) = f(K) \setminus f(A)$ is closed, that is, $f(A)$ is open in $f(K)$. Therefore, f^{-1} is continuous.
Alternative proof: Notice that f^{-1} is continuous iff the preimages of open sets by f^{-1} are open, iff the preimages of closed sets by f^{-1} are closed, iff the images of closed sets by f are closed. If $C \subseteq K$ is closed, then it is compact, because all closed subsets of compact spaces are compact. Then the image $f(C)$ is compact, because continuous functions map compact sets to compact sets. Finally, $f(C)$ is closed because $\mathbb{R}$ is Hausdorff and compact sets are closed in Hausdorff spaces. □

Problem 7.7. Let $X = \{f : [0,1] \to \mathbb{R} \mid f \text{ is continuous}\}$.

(a) Prove that $d(f,g) = \int_0^1 |f(t) - g(t)|\, dt$ is a metric on X.

(b) Give an example, without proof, of a Cauchy sequence in X which does not converge under this metric.

Solution.

(a) For any $f, g, h \in X$, we have

- If $f = g$, it is clear that $d(f,g) = \int_0^1 |f(t) - f(t)|\, dt = 0$.
  Conversely, if $f \neq g$, there is some $x_0 \in [0,1]$ such that $f(x_0) \neq g(x_0)$. We can

assume that $f(x_0) > g(x_0)$ and put $\varepsilon := \frac{f(x_0)-g(x_0)}{2} > 0$. By the continuity of $f - g$, there is some $\delta > 0$ such that if $x \in A := (x_0 - \delta, x_0 + \delta) \cap [0,1]$, we always have

$$f(x) - g(x) > f(x_0) - g(x_0) - \varepsilon = \varepsilon.$$

Notice that $x_0 \in [0,1]$, we always have $A := (x_0 - \delta, x_0 + \delta) \cap [0,1]$ is a nonempty interval. Therefore,

$$\int_0^1 |f(t) - g(t)|\, dt \geq \int_A |f(t) - g(t)|\, dt \geq \int_A \varepsilon\, dt > 0$$

Thus we have $d(f,g) = 0$ if and only if $f = g$.

- $d(f,g) = \int_0^1 |f(t) - g(t)|\, dt = d(g,f)$.
-

$$\begin{aligned} d(f,h) &= \int_0^1 |f(t) - h(t)|\, dt \\ &= \int_0^1 |f(t) - g(t) + g(t) - h(t)|\, dt \\ &\leq \int_0^1 (|f(t) - g(t)| + |g(t) - h(t)|)\, dt \\ &= d(f,g) + d(g,h). \end{aligned}$$

(b) Let $f_n : [0,1] \to \mathbb{R}$ be given by

$$f_n(x) = \begin{cases} 0, & x \in [0, 1/2], \\ n(x - 1/2), & x \in (1/2, 1/2 + 1/n], \\ 1, & x \in (1/2 + 1/n, 1]. \end{cases}$$

□

Problem 7.8. If $\{\mathcal{O}_n : n \in \mathbb{N}\}$ is a countable collection of open dense subsets of $\mathbb{R}$. Show that $\bigcap_{n=1}^{\infty} \mathcal{O}_n$ is dense in $\mathbb{R}$.

Solution. A subset is dense iff it intersects every nonempty open set non-trivially. Let $U \subseteq \mathbb{R}$ be open and nonempty. It suffices to show that there exists $x \in U$ such that $x \in \mathcal{O}_n$ for all n.

Since $\mathcal{O}_1$ is dense, there exists $x_1 \in U \cap \mathcal{O}_1$. Moreover, the latter set is open, so we can find $0 < r_1 < 1$ such that $\overline{B}(x_1, r_1) \subseteq U \cap \mathcal{O}_1$. By induction, for any n we can find x_n and $0 < r_n < \frac{1}{n}$ such that $\overline{B}(x_n, r_n) \subseteq B(x_{n-1}, r_{n-1}) \cap \mathcal{O}_n$. Since the sequence x_i is Cauchy, it converges to a point $x \in \mathbb{R}$. By closedness, $x \in \overline{B}(x_n, r_n)$ for all n, so $x \in \mathcal{O}_n$ for all n. □

Problem 7.9. Prove each of the following assertion using the definition of connectedness

(a) If A and B are connected sets in a metric space (X, d) such that $A \cap B \neq \emptyset$ then $A \cup B$ is connected.

(b) Every interval I in $\mathbb{R}$ is connected.

(c) If X is a connected metric space and $f : X \to \mathbb{R}^d$ is continuous, then the range $f(X)$ of f is connected.

Solution.

(a) Suppose by contradiction that $M := A \cup B$ is not connected. Then there are two disjoint open sets C, D such that C and D intersect M and $M \subset C \cup D$. We will show that M is contained entirely in C or D. Firstly, since

$$A \subset M \subset C \cup D \implies C \cap D = \emptyset$$

and A is connected, A must be contained entirely in C or D. Similarly, B must be contained entirely in C or D. However, $A \cap B \neq \emptyset$, $C \cap D = \emptyset$ implies that A and B are contained entirely in the same set C or D. Without loss of generality, we can assume that $A, B \subset C$. Then $M = A \cup B \subset C$. This contradicts the fact that D intersects M but does not intersect C. Therefore, $A \cup B$ is connected.

(b) We will prove that $I \subset \mathbb{R}$ is connected if and only if $x, y \in I$, $x < z < y$ implies $z \in I$.

Suppose the property fails for some x, y, z, i.e. $x, y \in I, x < z < y$ but $z \notin I$. Then define $A = (-\infty, z)$, $B = (z, +\infty)$, we have A, B intersect I but $A \cap B = \emptyset$, which means I is not connected.

Conversely, suppose that I is not connected, then there are two numbers $x < y$, and two disjoint open subsets $A, B \subset \mathbb{R}$ both intersect I such that $I \subset A \cup B$, $x \in A, y \in B$. Then consider

$$S := A \cap [x, y] \neq \emptyset.$$

Define $z = \sup S$. Since $y \in B$ which does not intersect A and B is open, we have $z < y$. If we have $z \in A$, the fact A is open implies that z is not $\sup S$. Therefore, $z \notin A$.

Since $x \in A$ and A is open, $x < z$ Thus if we have $z \in B$, the fact that B is open implies that z is not $\sup S$. Thus $z \notin B$. But $z \in [x, y] \subset I \subset A \cup B$ by the property, a contradiction. We complete the statement.

Next, for an interval I, since the property holds, we have I is connected.

(c) Suppose that $f(X)$ is not connected. Then there are two disjoint open sets C, D intersect $f(X)$ such that $f(X) \subset C \cup D$. This means that $X \subset f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$. Since C, D are open and does not intersect, the continuity of f implies $f^{-1}(C), f^{-1}(D)$ are open and moreover,

$$f^{-1}(C) \cap f^{-1}(D) = f^{-1}(C \cap D) = \emptyset.$$

Therefore, X is not connected.

□

Problem 7.10.

(a) Give the definition of a perfect set in a metric space.

(b) Let P be a nonempty perfect set in $\mathbb{R}^k$. Prove P is uncountable.

Solution.

(a) A set S of a metric space X is said to be perfect if S is closed and it does not have any isolated point, i.e. any element of S is a limit point of S.

(b) If P is finite, any point of P is isolated, which is a contradiction. Suppose P is countable infinite, and let $P = \{p_1, p_2, \ldots\}$. Let U_0 be any bounded neighborhood of p_1, then obviously, $U_0 \cap P \neq \emptyset$.
Suppose we have an open set U_i such that $U_i \cap P \neq \emptyset$. Then since P is perfect, there exists an open subset U_{i+1} of $\mathbb{R}^k$ such that $\overline{U_{i+1}} \subset U_i$, $p_{i+1} \notin \overline{U_{i+1}}$, and $U_{i+1} \cap P \neq \emptyset$. By induction with the existence of U_0, we have such an U_i for all $i \in \mathbb{N}$.
Notice that $\overline{U_i} \cap P$ is closed because every perfect set is closed, and bounded because U_0 is bounded, so it is compact. Then

$$\overline{U_0} \cap P \supset \overline{U_1} \cap P \supset \cdots$$

is a descending chain of compact subsets of $\mathbb{R}^k$. Moreover, they are all non-empty, so by Cantor's intersection theorem,

$$\bigcap_{i \in \mathbb{N}} (\overline{U_i} \cap P) \neq \emptyset.$$

But by construction, no p_i can be contained in this intersection. This is a contradiction.

□

Problem 7.11.

(a) Prove every open subset $\mathcal{O}$ of $\mathbb{R}^1$ is an at most countable union of disjoint open intervals $\{I_k\}_k$. Then one can define the *measure* of $\mathcal{O}$ to be

$$|\mathcal{O}| = \sum_k |I_k|$$

where $|I_k|$ is the length of the interval I_k, if the right-hand sum is finite.

(b) Construct a closed subset F of the interval $[0,1]$ so that $|\mathcal{O} \cap (0,1)| = \frac{1}{2}$ where $\mathcal{O} = \mathbb{R} \setminus F$ is the complement of F in $\mathbb{R}$.
Hint: Modify the construction of the Cantor set properly.

Solution.

(a) For each $x \in \mathcal{O}$, let I_x be the largest interval containing x and contained in $\mathcal{O}$. More precisely, since $\mathcal{O}$ is open, x is contained in some small interval, and therefore, if we define

$$a_x = \inf\{a < x : (a, x) \subset \mathcal{O}\} \quad b_x = \sup\{b > x : (x, b) \subset \mathcal{O}\}$$

then $I_x = (a_x, b_x)$. Then we must have

$$\mathcal{O} = \bigcup_{x \in \mathcal{O}} I_x.$$

Now suppose that two intervals I_x and I_y intersect. Then their union, which is also an interval, is contained in $\mathcal{O}$ and contains x. Since I_x is maximal, we must have $I_x \cup I_y \subset I_x$, and similarly $I_x \cup I_y \subset I_y$. This only happens when $I_x = I_y$. Therefore, two distinct intervals in $\mathcal{I} = \{I_x\}_{x \in \mathcal{O}}$ must be disjoint. It remains to show that there are only countable intervals in $\mathcal{I}$. Indeed, since every open interval I_x contain a rational number. Since different intervals are disjoint, they must contain different rationals, and therefore $\mathcal{I}$ is countable.

(b) Let $F = [0, \frac{1}{2}]$. It is closed, and $|(\mathbb{R} \setminus [0, \frac{1}{2}]) \cap (0,1)| = |(\frac{1}{2}, 1)| = \frac{1}{2}$.

□

8 Functions of several variables

Problem 8.1. Let $f : \mathbb{R}^2 \to \mathbb{R}$.

(a) Define what it means for f to be differentiable at the point $(1,1)$.

(b) Prove that if $f(x,y) = x^2 - y^2$ then f is differentiable at the point $(1,1)$.

Solution.

(a) f is called differentiable at the point $(1,1)$ if there exists a 1×2 matrix A such that

$$\lim_{(h_1,h_2)\to(0,0)} \frac{|f(1+h_1, 1+h_2) - f(1,1) - A(h_1,h_2)|}{\sqrt{h_1^2 + h_2^2}} = 0.$$

(b) Let $A = \begin{bmatrix} 2 & -2 \end{bmatrix}$. We have that

$$\begin{aligned} \frac{|f(1+h_1, 1+h_2) - f(1,1) - A(h_1,h_2)|}{\sqrt{h_1^2 + h_2^2}} &= \frac{|(1+h_1)^2 - (1+h_2)^2 - (2h_1 - 2h_2)|}{\sqrt{h_1^2 + h_2^2}} \\ &= \frac{|h_1^2 - h_2^2|}{\sqrt{h_1^2 + h_2^2}} \\ &\leq \frac{|(h_1 - h_2)(h_1 + h_2)|}{\sqrt{\frac{(h_1+h_2)^2}{2}}} \\ &= \sqrt{2} \cdot |h_1 - h_2| \to 0 \text{ as } (h_1, h_2) \to (0,0). \end{aligned}$$

Therefore, f is differentiable at $(1,1)$. In the inequality above (where $h_1 + h_2 \neq 0$; when $h_1 + h_2 = 0$ the quotient is 0) we used that, by the AM-GM inequality, $2h_1h_2 \leq h_1^2 + h_2^2$, and therefore $(h_1 + h_2)^2 = h_1^2 + 2h_1h_2 + h_2^2 \leq 2(h_1^2 + h_2^2)$. □

Problem 8.2. A careless student claims to have a proof of the following incorrect statement:

If $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous in each variable separately, then f is continuous.

The "proof" given is as follows. Let (x_0, y_0) in $\mathbb{R}^2$. For any (x, y) in $\mathbb{R}^2$,

$$|f(x,y) - f(x_0,y_0)| \leq |f(x,y) - f(x,y_0)| + |f(x,y_0) - f(x_0,y_0)|.$$

There exist $\delta_1 > 0$ and $\delta_2 > 0$ such that, if $|y - y_0| < \delta_1$, then $|f(x,y) - f(x,y_0)| < \varepsilon/2$, and if $|x - x_0| < \delta_2$, then $|f(x,y_0) - f(x_0,y_0)| < \varepsilon/2$. Thus if $\delta = \min\{\delta_1, \delta_2\}$, $|x - x_0| < \delta$, and $|y - y_0| < \delta$, then $|f(x,y) - f(x_0,y_0)| < \varepsilon$.

(a) Show that the function defined by $f(x,y) = \frac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$ is a counterexample to the student's statement.

(b) What is the error in the students's "proof"?

(c) Find an added hypothesis that would make the student's statement correct and the outline of his "proof" valid.

Solution.

(a) We will show that

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0) \end{cases}$$

is separately continuous but not continuous. Fix $\bar{x}$. We have

$$\lim_{y \to \bar{y}} \frac{\bar{x}y}{\bar{x}^2 + y^2} = \begin{cases} 0 & \bar{x} = 0, \\ \dfrac{\bar{x}\bar{y}}{\bar{x}^2 + \bar{y}^2} & \bar{x} \neq 0. \end{cases} = f(\bar{x}, \bar{y}).$$

Thus we see that f is continuous in terms of y. Similarly, we can prove that f is continuous in terms of x. But f is not continuous, since $f(0,0) = 0$ and for $(x_n, y_n) = (1/n, 1/n) \to (0,0)$, we have

$$f(x_n, y_n) = \frac{1}{2}, \text{ for all } n \in \mathbb{N}.$$

(b) The error is that the student does not fix x and y in the proof. That means, δ_1 is not invariant for y and δ_2 is not invariant for x.

(c) Suppose that for (x_0, y_0) fixed, for any $\varepsilon > 0$, there is some $\delta_1, \delta_2 > 0$ such that

$$|y - y_0| < \delta_1 \implies |f(x,y) - f(x,y_0)| < \varepsilon, \text{ for all } x \in \mathbb{R}.$$
$$|x - x_0| < \delta_2 \implies |f(x,y) - f(x_0,y)| < \varepsilon, \text{ for all } y \in \mathbb{R}.$$

Then f is continuous.

□

Problem 8.3. Let $f(x,y)$ be a continuous function on the rectangle $R = [a_1, b_1] \times [a_2, b_2]$ in $\mathbb{R}^2$.

(a) Let $F(y) = \int_{a_1}^{b_1} f(x,y)\,dx$. Prove that $F(y)$ is continuous on $[a_2, b_2]$.

(b) Give the definition of the area integral $\iint_R f(x,y)\,dA$ of f over R.

(c) Using the notation above, prove iterated integrals formula; that is, prove that

$$\iint_R f(x,y)\,dA = \int_{a_2}^{b_2} F(y)\,dy.$$

Solution.

(a) Consider $\overline{y} \in [a_2, b_2]$ and a sequence $y_n \xrightarrow{n\to\infty} \overline{y}$ in $[a_2, b_2]$. We aim to show that

$$\int_{a_1}^{b_1} f(x, y_n)dx \xrightarrow{n\to\infty} \int_{a_1}^{b_1} f(x, \overline{y})dx$$

with the help of the Lebesgue's Dominated Convergence Theorem. Notice here that f is continuous on R compact, hence f is Riemann-integrable, Lebesgue-integrable on R and those two integrals are equal.
First we check the convergence condition. For $x \in [a_1, b_1]$, since $y_n \to \overline{y}$ in $[a_2, b_2]$ we also have $(x, y_n) \xrightarrow{n\to\infty} (x, \overline{y})$ in $R = [a_1, b_1] \times [a_2, b_2]$. And from the fact that f is continuous on R, we deduce that $f(x, y_n) \xrightarrow{n\to\infty} f(x, \overline{y})$ for any $x \in [a_1, b_1]$.

Next, we check the dominated condition. As f is continuous on $[a_1, b_1] \times [a_2, b_2]$, from the Extreme Value Theorem one can find $M > 0$ such that

$$|f(x,y)| \leq M,\ \forall (x,y) \in [a_1, b_1] \times [a_2, b_2].$$

Thus $|f(x, y_n)| \leq M,\ \forall x \in [a_1, b_1]$ and $\forall n \in \mathbb{N}$. Moreover, $\int_{a_1}^{b_1} M dx = M(b_1 - a_1) < \infty$. Therefore, from the Lebesgue's Dominated Convergence Theorem we deduce that $F(y_n) \xrightarrow{n\to\infty} F(\overline{y})$. That implies F is continuous on $[a_2, b_2]$.

(b) Consider P_i a partition of the interval $[a_i, b_i]$, $i = 1, 2$. Then $P = P_1 \times P_2$ is a partition of $R = [a_1, b_1] \times [a_2, b_2]$. The partition P splits R into finite sub-rectangles A_k, $k = 1, 2, ..., n_P$. S_k denotes the area of A_k. For any such P, we set

$$U(P,f) := \sum_{k=1}^{n_P} S_k \max_{x\in A_k} f(x),$$
$$L(P,f) := \sum_{k=1}^{n_P} S_k \min_{x\in A_k} f(x).$$

Let $\mathcal{P}$ be the set of all partitions of R and

$$U(f) := \inf_{P\in\mathcal{P}} U(P,f),$$
$$L(f) := \sup_{P\in\mathcal{P}} L(P,f).$$

As f is continuous on R, we notice that:

(i) If $P, P' \in \mathcal{P}$ and $P \subset P'$ then

$$L(P,f) \le L(P',f) \le U(P',f) \le U(P,f).$$

As $P \subset P'$, a rectangle A_k corresponding to P is the union of some finite rectangles A_k^j corresponding to P'. This implies

$$L(P,f) = \sum_{k=1}^{n_P} S_k \min_{x\in A_k} f(x) \le \sum_{k=1}^{n_P} \left(\sum_j S_k^j \min_{x\in A_k^j} f(x) \right) = L(P',f).$$

Similarly, we get the third inequality. The second inequality is directly implied from the definitions.

(ii) $L(P,f) \le U(P',f) \ \ \forall P, P' \in \mathcal{P}$. Indeed, consider $P = P_1 \times P_2, P' = P_1' \times P_2' \in \mathcal{P}$ and $P'' := (P_1 \cup P_1') \times (P_2 \cup P_2')$. Thus $P'' \supset P' \cup P$. From (i) we obtain that

$$L(P,f) \le L(P'',f) \le U(P'',f) \le U(P',f).$$

(iii) $U(f) = L(f)$. Indeed, we will show that

$$\inf_{P\in\mathcal{P}} U(P,f) = \sup_{P\in\mathcal{P}} L(P,f).$$

First, for every $P, P' \in \mathcal{P}$ we always have $U(P,f) \ge L(P',f)$. Hence $\inf\limits_{P\in\mathcal{P}} U(P,f) \ge L(P',f)$ for every $P' \in \mathcal{P}$.

Next, for $\varepsilon > 0$ we find $P' \in \mathcal{P}$ such that

$$\inf_{P\in\mathcal{P}} U(P,f) < L(P',f) + \varepsilon,$$

equivalent to the existence of some $P \in \mathcal{P}$ such that

$$U(P,f) < L(P',f) + \varepsilon.$$

Since f is continuous on the compact set $R = [a_1,b_1] \times [a_2,b_2]$, the Heine-Cantor theorem implies that f is also uniformly continuous on R. Thus we find $\delta > 0$ such that for every points $x, x' \in R$, we have the implication:

$$\|x - x'\| < \delta \Longrightarrow |f(x) - f(x')| < \frac{\varepsilon}{\text{area}(R)}.$$

By choosing $P = P'$ a partition of R such that each sub-rectangle A_k has sides of length at most $\dfrac{\delta}{2}$, we see that

$$\forall A_k \text{ and } \forall x, x' \in A_k : \|x - x'\| \le \frac{\delta}{\sqrt{2}} < \delta.$$

Therefore,

$$U(P,f) - L(P,f) = \sum_{k=1}^{n_P} S_k \big[\max_{x\in A_k} f(x) - \min_{x\in A_k} f(x) \big] < \sum_{k=1}^{n_P} S_k \frac{\varepsilon}{\text{area}(R)} = \varepsilon.$$

Thus we have found $P, P' \in \mathcal{P}$ such that $U(P,f) < L(P',f) + \varepsilon$.

Then, we define

$$\iint_R f(x,y)dA := U(f) = L(f).$$

(c) Take arbitrary partitions $P_1 := \{a_1 = x_0, x_1, ..., x_n = b_1\}$ of $[a_1, b_1]$ and $P_2 := \{a_2 = y_0, y_1, ..., y_m = b_2\}$ of $[a_2, b_2]$. Then $P := P_1 \times P_2$ is a partition of R into $n \times m$ sub-rectangles A_{ij}. For simplicity, we use the notations

$$\Delta x_i := x_i - x_{i-1} \quad \Delta y_j := y_j - y_{j-1} \quad m_{ij} := \min_{x \in A_{ij}} f(x) \quad M_{ij} := \max_{x \in A_{ij}} f(x).$$

Since $m_{ij} \leq f(x,y) \leq M_{ij}$ for every $(x,y) \in A_{ij}$, we deduce that

$$m_{ij}\Delta x_i \leq \int_{x_{i-1}}^{x_i} f(x,y)dx \leq M_{ij}\Delta x_i, \ \ \forall y \in [y_{j-1}, y_j], \ \ \forall j = 1, ..., m, \ \ \forall i = 1, ..., n.$$

Then taking the sum as i runs from 1 to n, it shows:

$$\sum_{i=1}^{n} m_{ij}\Delta x_i \leq \int_{a_1}^{b_1} f(x,y)dx \leq \sum_{i=1}^{n} M_{ij}\Delta x_i, \ \ \forall y \in [y_{j-1}, y_j], \ \ \forall j = 1, ..., m.$$

Therefore, again we get:

$$\left(\sum_{i=1}^{n} m_{ij}\Delta x_i\right)\Delta y_j \leq \int_{y_{j-1}}^{y_j}\int_{a_1}^{b_1} f(x,y)dxdy \leq \left(\sum_{i=1}^{n} M_{ij}\Delta x_i\right)\Delta y_j, \ \ \forall j = 1, ..., m$$

and thus

$$\sum_{j=1}^{m}\sum_{i=1}^{n} m_{ij}\Delta x_i \Delta y_j \leq \int_{a_2}^{b_2}\int_{a_1}^{b_1} f(x,y)dxdy \leq \sum_{j=1}^{m}\sum_{i=1}^{n} M_{ij}\Delta x_i \Delta y_j.$$

Recalling the definitions of $U(P,f)$ and $L(P,f)$, we deduce that

$$L(P,f) \leq \int_{a_2}^{b_2}\int_{a_1}^{b_1} f(x,y)dxdy \leq U(P,f), \ \ \forall P \in \mathcal{P},$$

and moreover, $L(f) \leq \int_{a_2}^{b_2}\int_{a_1}^{b_1} f(x,y)dxdy \leq U(f)$. As $\iint_R f(x,y)dA := U(f) = L(f)$, we conclude that

$$\iint_R f(x,y)dA = \int_{a_2}^{b_2}\int_{a_1}^{b_1} f(x,y)dxdy = \int_{a_2}^{b_2} F(y)dy.$$

□

Problem 8.4. Let $\mathbf{F} : \mathbb{R}^2 \to \mathbb{R}^2$ be a continuous differentiable vector field. We write $\mathbf{F}(x,y) = \big(f(x,y), g(x,y)\big)$ and define the **divergence** of $\mathbf{F}$ as $\mathrm{div}\mathbf{F} = f_x(x,y) + g_y(x,y)$. For a bounded piecewise smooth domain Ω in $\mathbb{R}^2$, let $\partial\Omega$ denote its boundary.

(a) If $\Omega = [a,b] \times [c,d]$, prove that

$$\iint_\Omega \mathrm{div}\mathbf{F}(x,y)\, dxdy = \int_{\partial\Omega} \mathbf{F} \cdot \nu \, dS(x,y), \tag{3}$$

where ν is the exterior normal to the boundary and dS denotes the arc length element.

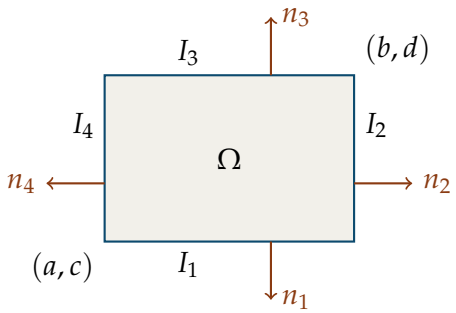
(b) If Ω is the unit disk in $\mathbb{R}^2$, prove that

$$\iint_\Omega \mathrm{div}\mathbf{F}(x,y)\, dxdy = \int_{\partial\Omega} \mathbf{F} \cdot \nu \, dS(x,y), \tag{4}$$

where ν is the exterior normal to $\partial\Omega$.

Solution.

(a)



Recall. When calculating $\int_C f(x,y)\, dS$ and C is parameterized by

$$x = x(t),\ y = y(t),\ a \le t \le b,$$

we have that:

$$\int_C f(x,y)\, dS = \int_a^b f\big(x(t), y(t)\big)\sqrt{[x'(t)]^2 + [y'(t)]^2}dt.$$

As $\Omega = [a,b] \times [c,d]$, the RHS of (3) is computed through 4 integrals:

$$\begin{aligned}
\int_{\partial\Omega} \mathbf{F} \cdot \nu \, dS(x,y) &= \int_{I_1} \mathbf{F} \cdot \nu \, dS + \int_{I_2} \mathbf{F} \cdot \nu \, dS + \int_{I_3} \mathbf{F} \cdot \nu \, dS + \int_{I_4} \mathbf{F} \cdot \nu \, dS \\
&= \int_{I_1} \mathbf{F} \cdot n_1 \, dS + \int_{I_2} \mathbf{F} \cdot n_2 \, dS + \int_{I_3} \mathbf{F} \cdot n_3 \, dS + \int_{I_4} \mathbf{F} \cdot n_4 \, dS \\
&= -\int_a^b g(x,c)\, dx + \int_c^d f(b,y)\, dy + \int_a^b g(x,d)\, dx - \int_c^d f(a,y)dy.
\end{aligned}$$

Besides, we also have

$$\begin{aligned}
&\iint_\Omega \operatorname{div}\mathbf{F}(x,y)\,dxdy = \int_a^b\int_c^d \operatorname{div}\mathbf{F}(x,y)\,dxdy\\
&= \int_a^b\int_c^d f_x(x,y)\,dxdy + \int_a^b\int_c^d g_y(x,y)\,dxdy\\
&= \int_c^d \left(\int_a^b f_x(x,y)\,dx\right)dy + \int_a^b\left(\int_c^d g_y(x,y)\,dy\right)dx\\
&= \int_c^d \big[f(b,y)-f(a,y)\big]dy + \int_a^b \big[g(x,d)-g(x,c)\big]dx\\
&= \int_{\partial\Omega} \mathbf{F}\cdot\nu\,dS(x,y).
\end{aligned}$$

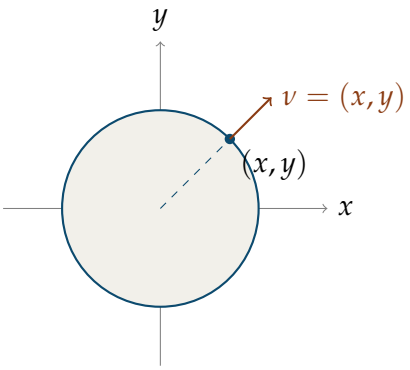
(b) Suppose now that $\Omega = B_1$ which is the unit disk in $\mathbb{R}^2$. We can look at Ω as

$$\Omega = \left\{(x,y)\in\mathbb{R}^2 \mid -1\le y\le 1, -\sqrt{1-y^2}\le x\le\sqrt{1-y^2}\right\},$$
$$\Omega = \left\{(x,y)\in\mathbb{R}^2 \mid -1\le x\le 1, -\sqrt{1-x^2}\le y\le\sqrt{1-x^2}\right\}.$$

First, we compute the LHS of (4):

$$\begin{aligned}
&\iint_\Omega \operatorname{div}\mathbf{F}(x,y)\,dxdy\\
&= \iint_\Omega f_x(x,y)\,dxdy + \iint_\Omega g_y(x,y)\,dxdy\\
&= \int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f_x dxdy + \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} g_y dydx\\
&= \int_{-1}^{1}\left[f\left(\sqrt{1-y^2},y\right) - f\left(-\sqrt{1-y^2},y\right)\right]dy + \int_{-1}^{1}\left[g\left(x,\sqrt{1-x^2}\right) - g\left(x,-\sqrt{1-x^2}\right)\right]dx.
\end{aligned}$$

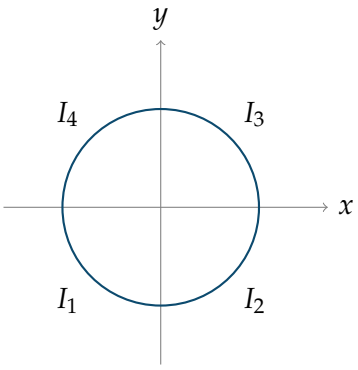
To compute the RHS of (4), first remark that the exterior unit vector to $\partial\Omega$ at (x,y) is exactly (x,y).



Thus

$$\begin{aligned}\int_{\partial\Omega} \mathbf{F}\cdot\nu\, dS &= \int_{\partial\Omega} [xf(x,y)+yg(x,y)]\, dS \\ &= M_1 + M_2 + M_3 + M_4,\end{aligned}$$

where M_1, M_2, M_3, M_4 respectively are the line integrals on I_1, I_2, I_3, I_4:



+ Calculate M_1:

$$\begin{aligned}M_1 &= \int_{I_1} [xf(x,y)+yg(x,y)]\, dS \\ &= \int_{-1}^{0} -\sqrt{1-y^2}\, f\Big(-\sqrt{1-y^2}, y\Big)\frac{1}{\sqrt{1-y^2}}dy \\ &\qquad + \int_{-1}^{0} -\sqrt{1-x^2}\, g\Big(x, -\sqrt{1-x^2}\Big)\frac{1}{\sqrt{1-x^2}}dx. \\ &= \int_{-1}^{0} -f\Big(-\sqrt{1-y^2}, y\Big)dy + \int_{-1}^{0} -g\Big(x, -\sqrt{1-x^2}\Big)dx.\end{aligned}$$

+ Similarly, we have:

$$\begin{aligned}M_2 &= \int_{I_2} [xf(x,y)+yg(x,y)]\, dS \\ &= \int_{-1}^{0} f\Big(\sqrt{1-y^2}, y\Big)dy + \int_{0}^{1} -g\Big(x, -\sqrt{1-x^2}\Big)dx,\end{aligned}$$

and

$$\begin{aligned}M_3 &= \int_{I_3} [xf(x,y)+yg(x,y)]\, dS \\ &= \int_{0}^{1} f\Big(\sqrt{1-y^2}, y\Big)dy + \int_{0}^{1} g\Big(x, \sqrt{1-x^2}\Big)dx,\end{aligned}$$

and

$$\begin{aligned}M_4 &= \int_{I_4} [xf(x,y)+yg(x,y)]\, dS \\ &= \int_{0}^{1} -f\Big(-\sqrt{1-y^2}, y\Big)dy + \int_{-1}^{0} g\Big(x, \sqrt{1-x^2}\Big)dx.\end{aligned}$$

The four equalities lead to:

$$\begin{aligned}
&\int_{\partial\Omega} \mathbf{F}\cdot\nu\, dS \\
&= M_1 + M_2 + M_3 + M_4 \\
&= \int_{-1}^{1} \left[f\Big(\sqrt{1-y^2}, y\Big) - f\Big(-\sqrt{1-y^2}, y\Big)\right] dy + \int_{-1}^{1} \left[g\Big(x, \sqrt{1-x^2}\Big) - g\Big(x, -\sqrt{1-x^2}\Big)\right] dx \\
&= \iint_{\Omega} \mathrm{div}\mathbf{F}(x,y)\, dxdy.
\end{aligned}$$

□

Problem 8.5. Let $\mathbf{F} : \mathbb{R}^2 \to \mathbb{R}^2$ be a continuous differentiable vector field. We write $\mathbf{F}(x,y) = \big(f(x,y), g(x,y)\big)$ and define the **divergence** of $\mathbf{F}$ as $\mathrm{div}\mathbf{F} = f_x(x,y) + g_y(x,y)$. For a bounded piecewise smooth domain Ω in $\mathbb{R}^2$, let $\partial\Omega$ denote its boundary.

(a) Prove the divergence theorem in a square, that is, prove that if $\Omega = [a,b]\times[c,d]$,

$$\iint_{\Omega} \mathrm{div}\mathbf{F}(x,y)\, dxdy = \int_{\partial\Omega} \mathbf{F}\cdot\nu\, dS(x,y), \tag{5}$$

where ν is the unit exterior normal to the boundary and dS denotes the arc length element.

(b) Given an $n\times n$ matrix M, use the divergence theorem in general dimensions to prove that for any $R > 0$

$$\frac{1}{|B_R(0)|}\int_{B_R(0)} Mx\cdot x\, dx = \frac{R^2}{n+2}\mathrm{trace}(M),$$

where $\mathrm{trace}(M) = M_{11} + M_{22} + \ldots + M_{nn}$ is the trace of the square matrix M.

Solution.

(a) This is part (a) of Problem 8.4, with the same proof.

(b) Let $g(x) := x\cdot x - R^2 \in \mathbb{R}$, $f(x) := Mx \in \mathbb{R}^n$ and $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ with $\mathbf{F}(x) := g(x)f(x)$ for all $x\in\mathbb{R}^n$. It is clear that $\mathbf{F}$ is continuous differentiable on $\mathbb{R}^n$, and

$$\begin{aligned}
\mathrm{div}\mathbf{F}(x,y) = \sum_{i=1}^{n} \frac{\partial\,[g(x)f(x)]_i}{\partial x_i} &= \sum_{i=1}^{n}\left[\frac{\partial g(x)}{\partial x_i} f_i(x) + \frac{\partial f_i(x)}{\partial x_i} g(x)\right] \\
&= \nabla g(x)\cdot f(x) + g(x)\mathrm{div} f(x) \\
&= 2x\cdot Mx + \big(x\cdot x - R^2\big)\mathrm{trace}(M).
\end{aligned}$$

By using the divergence theorem in general dimensions for the function $\mathbf{F}$ and observing that g (and thus $\mathbf{F}$) vanishes on the boundary of $B_R(0)$, one has:

$$\iint_{B_R(0)} \mathrm{div}\mathbf{F}(x,y)\, dxdy = 0,$$

or

$$2\int_{B_R(0)} Mx\cdot x\,dx = \operatorname{trace}(M)\int_{B_R(0)}\left(R^2 - x\cdot x\right)dx$$
$$= \operatorname{trace}(M)\left(R^2\,|B_R(0)| - \int_{B_R(0)} x\cdot x\,dx\right). \tag{6}$$

• Now we proceed to calculate $I = \int_{B_R(0)} x\cdot x\,dx$. As usual when $n = 2$, we use spherical coordinates $(r,\theta)\in[0,R]\times[0,2\pi]$ so that

$$\begin{pmatrix} x_1\\ x_2\end{pmatrix} = \begin{pmatrix} r\cos\theta\\ r\sin\theta\end{pmatrix} =: \psi_2(\theta, r).$$

For $n = 3$ we let

$$\begin{pmatrix} x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}\psi_2(\theta, r\sin\varphi_1)\\ r\cos\varphi_1\end{pmatrix} = \begin{pmatrix} r\sin\varphi_1\cos\theta\\ r\sin\varphi_1\sin\theta\\ r\cos\varphi_1\end{pmatrix} =: \psi_3(\theta,\varphi_1, r).$$

For $n = 4$:

$$\begin{pmatrix} x_1\\ x_2\\ x_3\\ x_4\end{pmatrix} = \begin{pmatrix}\psi_3(\theta,\varphi_1, r\sin\varphi_2)\\ r\cos\varphi_2\end{pmatrix} = \begin{pmatrix} r\sin\varphi_2\sin\varphi_1\cos\theta\\ r\sin\varphi_2\sin\varphi_1\sin\theta\\ r\sin\varphi_2\cos\varphi_1\\ r\cos\varphi_2\end{pmatrix} =: \psi_4(\theta,\varphi_1,\varphi_2, r).$$

We continue to work inductively this way to define:

$$\begin{pmatrix} x_1\\ \vdots\\ x_n\\ x_{n+1}\end{pmatrix} = \begin{pmatrix}\psi_n(\theta,\varphi_1,...,\varphi_{n-2}, r\sin\varphi_{n-1})\\ r\cos\varphi_{n-1}\end{pmatrix} =: \psi_{n+1}(\theta,\varphi_1,...,\varphi_{n-2},\varphi_{n-1}, r).$$

So, generally we have:

$$\begin{aligned} x_1 &= r\sin\varphi_{n-2}...\sin\varphi_2\sin\varphi_1\sin\theta\\ x_2 &= r\sin\varphi_{n-2}...\sin\varphi_2\sin\varphi_1\cos\theta\\ x_3 &= r\sin\varphi_{n-2}...\sin\varphi_2\cos\varphi_1\\ &\vdots\\ x_{n-2} &= r\sin\varphi_{n-2}\sin\varphi_{n-3}\cos\varphi_{n-4}\\ x_{n-1} &= r\sin\varphi_{n-2}\cos\varphi_{n-3}\\ x_n &= r\cos\varphi_{n-2},\end{aligned}$$

with $0\le r\le R$, $0\le\varphi_i\le\pi$ and $0\le\theta\le 2\pi$. The Jacobian determinant is known as $\Delta_{n+1}(\theta,\varphi_1,...,\varphi_{n-1}, r) = \det\psi'_{n+1}(\theta,\varphi_1,...,\varphi_{n-1}, r)$. Inductively, we will show that:

$$\Delta_n(\theta,\varphi_1,...,\varphi_{n-2}, r) = r^{n-1}\sin^{n-2}\varphi_{n-2}...\sin^2\varphi_2\sin^1\varphi_1\sin^0\theta. \tag{7}$$

Indeed, let $\rho := r\sin\varphi_{n-1}$ and write $\frac{\partial\psi_n}{\partial x}$ for $\frac{\partial\psi_n}{\partial x}(\theta,\varphi_1,...,\varphi_{n-2},\rho)$, we have:

$$\begin{aligned}\Delta_{n+1}(\theta,\varphi_1,...,\varphi_{n-1},r) &= \det\psi'_{n+1}(\theta,\varphi_1,...,\varphi_{n-1},r)\\ &= \begin{vmatrix} \frac{\partial\psi_n}{\partial\theta} & \frac{\partial\psi_n}{\partial\varphi_1} & \dots & \frac{\partial\psi_n}{\partial\varphi_{n-2}} & \frac{\partial\psi_n}{\partial\rho}r\cos\varphi_{n-1} & \frac{\partial\psi_n}{\partial\rho}\sin\varphi_{n-1} \\ 0 & 0 & \dots & 0 & -r\sin\varphi_{n-1} & \cos\varphi_{n-1}\end{vmatrix}\\ &= r\cos^2\varphi_{n-1}\Delta_n(\theta,\varphi_1,...,\varphi_{n-2},\rho) + r\sin^2\varphi_{n-1}\Delta_n(\theta,\varphi_1,...,\varphi_{n-2},\rho)\\ &= r\Delta_n(\theta,\varphi_1,...,\varphi_{n-2},\rho),\end{aligned}$$

with the determinant computed by expanding the last row. Starting with $\Delta_2(\theta,r) = r$, it implies that:

$$\begin{aligned}\Delta_3(\theta,\varphi_1,r) &= r\Delta_2(\theta,r\sin\varphi_1) = r^2\sin\varphi_1\\ \Delta_4(\theta,\varphi_1,\varphi_2,r) &= r\Delta_3(\theta,\varphi_1,r\sin\varphi_2) = r^3\sin^2\varphi_2\sin\varphi_1\\ &\vdots\\ \Delta_n(\theta,\varphi_1,...,\varphi_{n-2},r) &= r^{n-1}\sin^{n-2}\varphi_{n-2}...\sin^2\varphi_2\sin\varphi_1,\end{aligned}$$

and thus (7) is true. Then

$$I = \int_{B_R(0)} x\cdot x\,dx = \left(\int_0^R r^{n+1}\,dr\right)\left(\int_0^{2\pi} d\theta\right)\prod_{i=1}^{n-2}\int_0^\pi \sin^i x\,dx.$$

On the other hand, one also has

$$|B_R(0)| = \int_{B_R(0)} dx = \left(\int_0^R r^{n-1}\,dr\right)\left(\int_0^{2\pi} d\theta\right)\prod_{i=1}^{n-2}\int_0^\pi \sin^i x\,dx.$$

And we deduce that:

$$I = |B_R(0)|\frac{\int_0^R r^{n+1}\,dr}{\int_0^R r^{n-1}\,dr} = \frac{n}{n+2}R^2|B_R(0)|.$$

Plugging this into (6), one receives:

$$\begin{aligned}2\int_{B_R(0)} Mx\cdot x\,dx &= \text{trace}(M)\left(R^2|B_R(0)| - \frac{n}{n+2}R^2|B_R(0)|\right)\\ &= \frac{2}{n+2}R^2|B_R(0)|\text{trace}(M),\end{aligned}$$

or

$$\frac{1}{|B_R(0)|}\int_{B_R(0)} Mx\cdot x\,dx = \frac{R^2}{n+2}\text{trace}(M).$$

□

Problem 8.6.

(a) State the Implicit Function Theorem for a continuously differentiable function $f : \mathbb{R}^{m+n} \to \mathbb{R}^n$.

(b) State the Inverse Function Theorem for a continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}^n$.

(c) Use the Implicit Function Theorem to prove the Inverse Function Theorem.

(d) Let $g : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function satisfying $g(2) = 1$ and $g'(2) \neq 0$. Prove that there exists a neighborhood V of 1 in $\mathbb{R}$ and a unique continuously differentiable function $f : V \to \mathbb{R}$ such that

$$f(1) = 2 \quad \text{and} \quad 2xf(x) = 2f(x) - g(f(x)) + 1.$$

Solution.

(a) **The Implicit Function Theorem:** Let $f : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable function, and $(a, b) \in \mathbb{R}^m \times \mathbb{R}^n$. If $f(a, b) = 0$ and $\det Df(a, b) \neq 0$, where

$$Df(x,y) := \begin{bmatrix} \frac{\partial f_1}{\partial y_1}(x,y) & \frac{\partial f_1}{\partial y_2}(x,y) & \dots & \frac{\partial f_1}{\partial y_n}(x,y) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial y_1}(x,y) & \frac{\partial f_n}{\partial y_2}(x,y) & \dots & \frac{\partial f_n}{\partial y_n}(x,y) \end{bmatrix},$$

then there exist open subsets $A \subset \mathbb{R}^m$ containing a, $B \subset \mathbb{R}^n$ containing b and a unique continuously differentiable function $g : A \to B$ such that

$$g(a) = b \quad \text{and } f(x, g(x)) = 0_{\mathbb{R}^n},\ \forall x \in A.$$

(b) **The Inverse Function Theorem:** Let $f : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable and $a \in \mathbb{R}^n$. If $\det f'(a) \neq 0$, there exists open sets $U \subset \mathbb{R}^n$ containing a, V containing $f(a)$ such that $f : U \to V$ is bijective and $f^{-1} : V \to U$ is also continuously differentiable, with derivatives given by the inverse matrices

$$(f^{-1})'(y) = \left[f'(f^{-1}(y))\right]^{-1}, \forall y \in V,$$

(c) Let $f = (f_1, f_2, \dots, f_n) : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable function and $a \in \mathbb{R}^n$ with $\det f'(a) \neq 0$.

First, we prove that there exists an $r > 0$ such that f is injective on $B(a, r)$, the ball with radius r centered at a, by noticing that:

- Each $\dfrac{\partial f_i}{\partial x_j}$ $(1 \leq i, j \leq n)$ is continuous on $\mathbb{R}^n$. Indeed, since $f = (f_1, f_2, \dots, f_n)$ is continuously differentiable on $\mathbb{R}^n$, each f_i' $(i = 1, 2, \dots, n)$ is continuous on $\mathbb{R}^n$. Besides, for $x, y \in \mathbb{R}^n$ we get:

$$\left\| f_i'(x) - f_i'(y) \right\| = \left[\sum_{j=1}^{n} \left(\frac{\partial f_i}{\partial x_j}(x) - \frac{\partial f_i}{\partial x_j}(y) \right)^2 \right]^{1/2}.$$

Thus, for every $i = 1, 2, ..., n$ the continuity of $f_i^{'}$ implies continuities of $\dfrac{\partial f_i}{\partial x_j}$, $j = 1, 2, ..., n$.

- The map $d_1 : (\mathbb{R}^n)^{n^2} \to M_n(\mathbb{R})$, with

$$d_1\left(a_{11}, ..., a_{1n}, ..., a_{nn}\right) = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1}(a_{11}) & \dfrac{\partial f_1}{\partial x_2}(a_{12}) & \dots & \dfrac{\partial f_1}{\partial x_n}(a_{1n}) \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial f_n}{\partial x_1}(a_{n1}) & \dfrac{\partial f_n}{\partial x_2}(a_{n2}) & \dots & \dfrac{\partial f_n}{\partial x_n}(a_{nn}) \end{bmatrix},$$

is continuous. Indeed, from the previous point we see that for every $1 \leq i, j \leq n$:

$$a_{ij} \mapsto \frac{\partial f_i}{\partial x_j}(a_{ij}) \text{ is continuous.}$$

It deduces that $(a_{11}, ..., a_{1n}, ..., a_{nn}) \mapsto \dfrac{\partial f_i}{\partial x_j}(a_{ij})$ is also continuous. In conclusion, d_1 is continuous.

- The determinant map $d_2 : M_n(\mathbb{R}) \to \mathbb{R}$, with $d_2(B) = \det B$ is continuous. From the Leibniz formula, for $B = (b_{i,j}) \in M_n(\mathbb{R})$ we have

$$\det B = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} b_{i,\sigma(i)}.$$

Thus $d_2(B) = \det B$ can be seen as a polynomial in n^2 variables $b_{1,1}, b_{1,2}, ..., b_{n,n}$. Therefore d_2 is continuous on $M_n(\mathbb{R})$.

By setting $d := d_2 \circ d_1$, we have $d : (\mathbb{R}^n)^{n^2} \to \mathbb{R}$ is continuous with

$$d\left(a_{11}, ..., a_{1n}, ..., a_{nn}\right) = \det \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1}(a_{11}) & \dfrac{\partial f_1}{\partial x_2}(a_{12}) & \dots & \dfrac{\partial f_1}{\partial x_n}(a_{1n}) \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial f_n}{\partial x_1}(a_{n1}) & \dfrac{\partial f_n}{\partial x_2}(a_{n2}) & \dots & \dfrac{\partial f_n}{\partial x_n}(a_{nn}) \end{bmatrix}.$$

Seeing that $d(a, ..., a, ..., a) = \det f'(a) \neq 0$ and d is continuous, we find $\rho > 0$ such that

$$d(A) \neq 0 \text{ for any } A \in B\big((a), \rho\big) \subset (\mathbb{R}^n)^{n^2}.$$

Besides, notice that $A \in B\big((a), \rho\big)$ if each of its component (a vector in $\mathbb{R}^n$) belongs to $B\left(a, \dfrac{\rho}{n}\right)$. With $r := \rho/n$, we conclude that

$$d(A) \neq 0, \ \forall A \in (\mathbb{R}^n)^{n^2} \text{ s.t. } a_{ij} \in B(a, r). \tag{8}$$

Let u, v be distinct in $B(a,r)$. By employing the mean-value theorem in several variables to each component f_i of f, we find c_i in the open segment connecting u and v (thus, in $B(a,r)$) such that $f_i(v) - f_i(u) = f_i'(c_i) \cdot (v-u)$. Hence

$$\begin{bmatrix} f_1(v) - f_1(u) \\ f_2(v) - f_2(u) \\ \cdots \\ f_n(v) - f_n(u) \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(c_1) & \cdots & \frac{\partial f_1}{\partial x_n}(c_1) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(c_n) & \cdots & \frac{\partial f_n}{\partial x_n}(c_n) \end{bmatrix} \begin{bmatrix} v_1 - u_1 \\ v_2 - u_2 \\ \cdots \\ v_n - u_n \end{bmatrix}.$$

Using (8) and the fact that all c_i, $i = 1, 2, ..., n$ belong to $B(a,r)$, the matrix is invertible, and since $v - u \neq 0$ we deduce that $f(v) \neq f(u)$. Thus f is injective on $B(a,r)$.

Secondly, we utilize the Implicit Function Theorem to prove the Inverse Function Theorem. Consider $F : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, given by $F(x,y) = f(y) - x$. Now we check the assumptions of the Implicit Function Theorem:

i) $F : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ *is continuously differentiable?* Set $F^1, F^2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ with

$$F^1(x,y) = f(y) \quad \text{and } F^2(x,y) = x.$$

Assume $f = (f_1, f_2, ..., f_n)$ and $F^1 = \left(F_1^1, F_2^1, ..., F_n^1\right)$. Thus $F_i^1(x,y) = f_i(y)$ for all $i = 1, 2, ..., n$. Since F^1 is continuously differentiable if and only if each $F_i^1, i = 1, 2, ..., n$ is continuously differentiable, we just need to check F_1^1. Consider $(\overline{x}, \overline{y}) \in \mathbb{R}^n \times \mathbb{R}^n$. As f_1 is differentiable at $\overline{y}$, for any $h \in \mathbb{R}^n$ we have

$$f_1(\overline{y} + h) - f_1(\overline{y}) = f_1'(\overline{y}) \cdot h + \|h\|_{\mathbb{R}^n} \varphi(h), \tag{9}$$

where $\lim\limits_{h \to 0_{\mathbb{R}^n}} \varphi(h) = 0_{\mathbb{R}}$. For $k = (k_1, k_2) \in \mathbb{R}^n \times \mathbb{R}^n$ and choosing $h = k_2$, from (9) we get

$$F_1^1(\overline{x} + k_1, \overline{y} + k_2) - F_1^1(\overline{x}, \overline{y}) = f_1'(\overline{y}) \cdot k_2 + \|k_2\|_{\mathbb{R}^n} \varphi(k_2),$$

or

$$F_1^1\left((\overline{x}, \overline{y}) + k\right) - F_1^1(\overline{x}, \overline{y}) = \left(0_{\mathbb{R}^n}, f_1'(\overline{y})\right) \cdot k + \|k\|_{\mathbb{R}^n \times \mathbb{R}^n} \frac{\|k_2\|_{\mathbb{R}^n}}{\|k\|_{\mathbb{R}^n \times \mathbb{R}^n}} \varphi(k_2).$$

As $\dfrac{\|k_2\|_{\mathbb{R}^n}}{\|k\|_{\mathbb{R}^n \times \mathbb{R}^n}} \varphi(k_2) \leq \varphi(k_2) \to 0$ when $k \to 0_{\mathbb{R}^n \times \mathbb{R}^n}$, we see that F_1^1 is differentiable at $(\overline{x}, \overline{y})$ and

$$F_1^{1'}(\overline{x}, \overline{y}) = \left(0_{\mathbb{R}^n}, f_1'(\overline{y})\right).$$

For any sequence $(x_n, y_n) \to (\overline{x}, \overline{y})$, we have $y_n \to \overline{y}$ and thus $f_1'(y_n) \to f_1'(\overline{y})$. Hence

$$F_1^{1'}(x_n, y_n) = \left(0_{\mathbb{R}^n}, f_1'(y_n)\right) \to (0_{\mathbb{R}^n}, f_1'(\overline{y}) = F_1^{1'}(\overline{x}, \overline{y}).$$

That deduces F_1^1 is continuously differentiable on $\mathbb{R}^n \times \mathbb{R}^n$. Similarly, we have

$$F_1^{2'}(x,y) = \big(e_1, 0_{\mathbb{R}^n}\big),$$

which is continuous on $\mathbb{R}^n \times \mathbb{R}^n$.

ii) $F\big(f(a),a\big) = 0$ *and* $\det D_2F\big(f(a),a\big) \neq 0$? Since $F(x,y) = f(y) - x$, there holds

$$F\big(f(a),a\big) = f(a) - f(a) = 0$$

and

$$\det D_2F\big(f(a),a\big) = \det f'(a) \neq 0.$$

From the Implicit Function Theorem, there are open subsets $V \subset \mathbb{R}^n$ containing $f(a)$, $a \in U \subset \mathbb{R}^n$ and a unique continuously differentiable $g : V \to U$ such that

$$g(f(a)) = a \quad \text{and } F\big(x, g(x)\big) = 0_{\mathbb{R}^n},\ \forall x \in V,$$

or

$$g(f(a)) = a \quad \text{and } f(g(x)) = x,\ \forall x \in V. \tag{10}$$

With $\mathcal{U} := U \cap f^{-1}(V) \cap B(a,r)$ an open set containing a, it is clear that f is injective on $\mathcal{U}$. Putting $\mathcal{V} := f(\mathcal{U}) \subset V$, we see that $f : \mathcal{U} \to \mathcal{V}$ is bijective. Besides, (10) implies $f(g(x)) = x$, $\forall x \in \mathcal{V}$. Combining with the bijectivity of f on $\mathcal{U}$, one gets

$$g(x) = f^{-1}(x),\ \forall x \in \mathcal{V}.$$

Hence $g = f^{-1}$ on $\mathcal{V}$. Moreover, $\mathcal{V}$ is open: by the uniqueness in the Implicit Function Theorem, for $x \in V$ the point $g(x)$ is the only point $y \in U$ with $f(y) = x$, so $\mathcal{V} = V \cap g^{-1}(\mathcal{U})$, which is open because g is continuous. Taking derivatives of both sides of (10) with respect to x in $\mathcal{V}$, we arrive at

$$f'(g(x))g'(x) = I,\ \forall x \in \mathcal{V}.$$

Therefore $g'(x) = \big[f'(g(x))\big]^{-1}$, $\forall x \in \mathcal{V}$.

(d) Consider $h : \mathbb{R}^2 \to \mathbb{R}$, given by

$$h(x,y) = g(y) + (2x-2)y - 1.$$

We see that $h(1,2) = g(2) - 1 = 0$, $\dfrac{\partial h}{\partial y} = g'(y) + 2x - 2$ and $\dfrac{\partial h}{\partial y}(1,2) = g'(2) \neq 0$. Choosing $m = n = 1$, $(a,b) = (1,2)$, we have got all the assumptions needed to use the Implicit Function Theorem. Utilizing it, we find an open neighborhood V of 1 and a unique continuously differentiable function $f : V \to \mathbb{R}$ such that

$$f(1) = 2 \text{ and } h\big(x, f(x)\big) = 0,\ \forall x \in V.$$

Notice that for every x in V,

$$\begin{aligned} h\big(x,f(x)\big) = 0 &\iff g\big(f(x)\big) + (2x-2)f(x) - 1 = 0 \\ &\iff 2xf(x) = 2f(x) - g\big(f(x)\big) + 1. \end{aligned}$$

□

9 Real integrals by residues

Problem 9.1. By applying the Residue theorem, evaluate the definite integral

$$\int_0^{2\pi} \frac{4}{5-\cos t}\, dt,$$

and sketch the contour you use.

Solution. Take the contour as the unit circle $C : |z| = 1$ of the following integral

$$\int_C \frac{4}{iz(5-\frac{1}{2}(z+z^{-1}))}\, dz = 2\pi i \operatorname{Res}_{z=5-\sqrt{24}} \frac{8i}{(z-5+\sqrt{24})(z-5-\sqrt{24})} = 2\sqrt{\frac{2}{3}}\pi$$

On the other hand, if you do the integral using the parametrization $z = e^{it}, t \in (0, 2\pi)$ then $dz = ie^{it}dt$ thus,

$$\int_C \frac{4}{iz(5-\frac{1}{2}(z+z^{-1}))}\, dz = \int_0^{2\pi} \frac{4}{ie^{it}(5-\frac{1}{2}(e^{it}+e^{-it}))}\, ie^{it}\, dt = \int_0^{2\pi} \frac{4}{5-\cos t}\, dt$$

□

Problem 9.2. Suppose that α is a complex number such that $|\alpha| \neq 1$. By applying the Residue theorem, evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{1-2\alpha\cos\theta+\alpha^2}.$$

Sketch the contour you use, and give explicit estimates for any integrals you claim tend to zero if there are any.

Solution. Put $z = e^{i\theta}$, we have $dz = ie^{i\theta}d\theta$ and $\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2} = \dfrac{z+z^{-1}}{2}$. Thus we have

$$\begin{aligned} I = \int_0^{2\pi} \frac{d\theta}{1-2\alpha\cos\theta+\alpha^2} &= \int_{|z|=1} \frac{dz}{iz(1-\alpha(z+z^{-1})+\alpha^2)} \\ &= \int_{|z|=1} \frac{-idz}{(1+\alpha^2)z-\alpha(z^2+1)} \\ &= \int_{|z|=1} \frac{idz}{\alpha z^2-(1+\alpha^2)z+\alpha} \\ &= \int_{|z|=1} \frac{idz}{\alpha(z-\alpha)(z-1/\alpha)} \end{aligned}$$

Define $f(z) = \dfrac{i}{\alpha(z-\alpha)(z-1/\alpha)}$. We consider two cases.

Case 1. $|\alpha| < 1$, then the residue theorem gives

$$I = 2\pi i \text{Res}_{z=\alpha} f(z) = 2\pi i \frac{i}{\alpha(\alpha - 1/\alpha)} = \frac{2\pi}{1-\alpha^2}.$$

Case 2. $|\alpha| > 1$, then the residue theorem gives

$$I = 2\pi i \text{Res}_{z=1/\alpha} f(z) = 2\pi i \frac{i}{\alpha(1/\alpha - \alpha)} = \frac{2\pi}{\alpha^2 - 1}.$$

□

Problem 9.3. Evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{1 + r\sin\theta}$$

for $0 < r < 1$.

Solution. Put $z = e^{i\theta}$, then $\sin\theta = \dfrac{z - z^{-1}}{2i}$. Thus we have

$$\begin{aligned}
\int_0^{2\pi} \frac{d\theta}{1 + r\sin\theta} &= \int_{|z|=1} \frac{dz}{iz(1 + r\frac{z - z^{-1}}{2i})} \\
&= \int_{|z|=1} \frac{2dz}{2iz(1 + r\frac{z - z^{-1}}{2i})} \\
&= \int_{|z|=1} \frac{2dz}{2iz + rz^2 - r} \\
&= \int_{|z|=1} \frac{2dz}{r(z - z_1)(z - z_2)}
\end{aligned}$$

where

$$z_1 = \frac{-i + i\sqrt{1-r^2}}{r}, z_2 = \frac{-i - i\sqrt{1-r^2}}{r}.$$

Then we have $|z_1| < 1$ and $|z_2| > 1$. Then by Residue theorem, we have

$$\int_{|z|=1} \frac{2dz}{r(z - z_1)(z - z_2)} = 2\pi i \text{Res}_{z=z_1} f(z),$$

where

$$f(z) = \frac{2}{r(z - z_1)(z - z_2)}.$$

Thus,

$$\int_{|z|=1} \frac{2dz}{r(z - z_1)(z - z_2)} = 2\pi i \frac{2}{r(z_1 - z_2)} = \frac{4\pi i}{r \cdot \frac{2i\sqrt{1-r^2}}{r}} = \frac{2\pi}{\sqrt{1-r^2}}.$$

Thus

$$\int_0^{2\pi} \frac{d\theta}{1+r\sin\theta} = \frac{2\pi}{\sqrt{1-r^2}}.$$

□

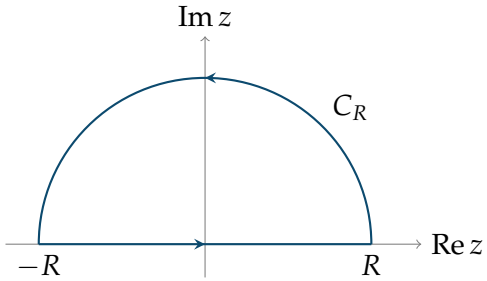
Problem 9.4. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^4+x^2+1}\,dx.$$

Solution. Consider the function

$$f(z) = \frac{e^{iz}}{z^4+z^2+1}, z \in \mathbb{C}.$$

Consider the contour $C := C_R + [-R, R]$ where C_R is the upper half part of the positively oriented circle $|z| = R$.



Consider further the equation

$$\begin{aligned} z^4+z^2+1=0 \iff z^4+2z^2+1=z^2 &\iff (z^2-z+1)(z^2+z+1)=0 \\ &\iff z=\frac{1}{2}\pm\frac{\sqrt{3}}{2}i, \text{ or } z=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i. \end{aligned}$$

On the upper half plane, f has only two poles at $z_1 = \frac{1}{2}+\frac{\sqrt{3}}{2}i$ and $z_2 = -\frac{1}{2}+\frac{\sqrt{3}}{2}i$. Then by the Residue Theorem, we have

$$\int_{C_R} f(z)\,dz + \int_{-R}^{R} f(z)\,dz = 2\pi i(\mathrm{Res}_{z=z_1} f(z) + \mathrm{Res}_{z=z_2} f(z)).$$

We have

$$f(z) = \frac{\phi_1(z)}{z-z_1}, \text{ where } \phi_1(z) = \frac{e^{iz}}{(z^2+z+1)\left(z-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)}.$$

Therefore,

$$\begin{aligned}\mathrm{Res}_{z=z_1} f(z) = \phi_1(z_1) &= \left(-\frac{1}{4} - \frac{\sqrt{3}}{12}i\right) e^{-\sqrt{3}/2}(\cos\frac{1}{2} + i\sin\frac{1}{2}) \\ &= e^{-\sqrt{3}/2}\left[(-\frac{1}{4}\cos\frac{1}{2} + \frac{\sqrt{3}}{12}\sin\frac{1}{2}) + i(-\frac{\sqrt{3}}{12}\cos\frac{1}{2} - \frac{1}{4}\sin\frac{1}{2})\right].\end{aligned}$$

Moreover,

$$\begin{aligned}\mathrm{Res}_{z=z_2} f(z) = \frac{e^{iz_2}}{4z_2^3 + 2z_2} &= \left(\frac{1}{4} - \frac{\sqrt{3}}{12}i\right) e^{-\sqrt{3}/2}(\cos\frac{1}{2} - i\sin\frac{1}{2}) \\ &= e^{-\sqrt{3}/2}\left[(\frac{1}{4}\cos\frac{1}{2} - \frac{\sqrt{3}}{12}\sin\frac{1}{2}) + i(-\frac{\sqrt{3}}{12}\cos\frac{1}{2} - \frac{1}{4}\sin\frac{1}{2})\right].\end{aligned}$$

Thus

$$\mathrm{Res}_{z=z_1} f(z) + \mathrm{Res}_{z=z_2} f(z) = ie^{-\sqrt{3}/2}(-\frac{\sqrt{3}}{6}\cos\frac{1}{2} - \frac{1}{2}\sin\frac{1}{2})$$

and hence,

$$2\pi i(\mathrm{Res}_{z=z_1} f(z) + \mathrm{Res}_{z=z_2} f(z)) = \pi e^{-\sqrt{3}/2}(\frac{\sqrt{3}}{3}\cos\frac{1}{2} + \sin\frac{1}{2}).$$

On the other hand, for any $z \in C_R$, we have

$$\left|z^4 + z^2 + 1\right| \geq |z|^4 - |z^2 + 1| \geq |z|^4 - |z|^2 - 1 = R^4 - R^2 - 1.$$

Therefore

$$\left|\mathrm{Re}\int_{C_R} f(z)dz\right| = \left|\mathrm{Re}\int_{C_R} \frac{e^{iz}}{z^4 + z^2 + 1}dz\right| \leq \frac{1}{R^4 - R^2 - 1} \cdot \pi R \to 0, \text{ as } R \to \infty.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^4 + x^2 + 1} = \pi e^{-\sqrt{3}/2}(\frac{\sqrt{3}}{3}\cos\frac{1}{2} + \sin\frac{1}{2}).$$

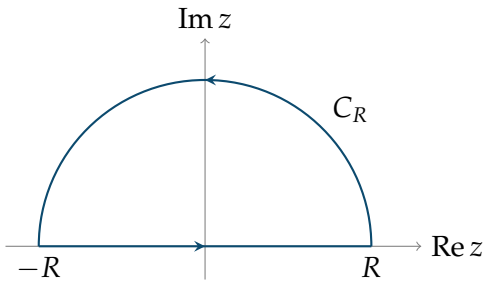
□

Problem 9.5. By applying the Residue Theorem, evaluate the following improper integral

$$\int_{-\infty}^{\infty} \frac{\cos x}{1 + x^4}dx.$$

Sketch the contour you use, and give explicit estimates for any integrals you claim tend to zero.

Solution. Consider the function $f(z) = \dfrac{e^{iz}}{1+z^4}$ and the contour $C := C_R + [-R, R], R > 1$.



Then f has two simple poles at $z_1 = \dfrac{1}{\sqrt{2}} + i\dfrac{1}{\sqrt{2}}$ and $z_2 = -\dfrac{1}{\sqrt{2}} + i\dfrac{1}{\sqrt{2}}$ with the residue

$$\operatorname{Res}_{z=z_1} f(z) = \frac{e^{iz_1}}{4z_1^3} = \frac{e^{i(\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}})}}{-2\sqrt{2}+2\sqrt{2}i} = \frac{e^{-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}}}{-2\sqrt{2}+2\sqrt{2}i}$$

$$\operatorname{Res}_{z=z_2} f(z) = \frac{e^{iz_2}}{4z_2^3} = \frac{e^{i(-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}})}}{2\sqrt{2}+2\sqrt{2}i} = \frac{e^{-\frac{1}{\sqrt{2}}-i\frac{1}{\sqrt{2}}}}{2\sqrt{2}+2\sqrt{2}i}$$

Thus,

$$2\pi i(\operatorname{Res}_{z=z_1} f(z) + \operatorname{Res}_{z=z_2} f(z)) = \frac{1}{2}\sqrt{2}\pi e^{-1/\sqrt{2}}(\cos\frac{1}{\sqrt{2}} + \sin\frac{1}{\sqrt{2}}).$$

On the other hand, for $z \in C_R$ we have $|1+z^4| \geq R^4 - 1$ and $|e^{iz}| = e^{-\operatorname{Im} z} \leq 1$, so

$$\left|\int_{C_R} f(z)dz\right| \leq \frac{\pi R}{R^4-1} \to 0, \text{ as } R \to \infty.$$

Since $\int_{C_R} f(z)dz + \int_{-R}^{R} f(x)dx = 2\pi i(\operatorname{Res}_{z=z_1} f(z) + \operatorname{Res}_{z=z_2} f(z))$ and $\operatorname{Re}\int_{-R}^{R} f(x)dx = \int_{-R}^{R} \dfrac{\cos x}{1+x^4}dx$, letting $R \to \infty$ gives the value. Therefore,

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^4}dx = \frac{1}{2}\sqrt{2}\pi e^{-1/\sqrt{2}}(\cos\frac{1}{\sqrt{2}} + \sin\frac{1}{\sqrt{2}}).$$

□

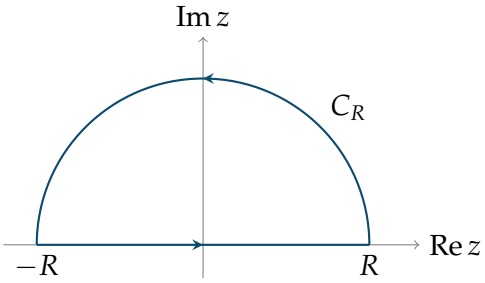
Problem 9.6. Calculate the integral

$$\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+2x+2}dx$$

Solution. Consider the function

$$f(z) = \frac{ze^{iz}}{z^2+2z+2}$$

and the contour $C := C_R + [-R, R]$, where $R > 3$.



Then f has a simple pole at $z = z_0 = -1 + i$. We also have

$$\begin{aligned}\mathrm{Res}_{z=z_0} f(z) &= \frac{z_0 e^{iz_0}}{2z_0+2} = \frac{(i+1)e^{i(-1+i)}}{2} = \frac{(i+1)e^{-i-1}}{2} \\ &= \frac{(i+1)e^{-1}(\cos 1 - i\sin 1)}{2} \\ &= \frac{(i+1)(\cos 1 - i\sin 1)}{2e} \\ &= \frac{\cos 1 + \sin 1 + i(\cos 1 - \sin 1)}{2e}.\end{aligned}$$

Thus

$$2\pi i \mathrm{Res}_{z=z_0} f(z) = \frac{\pi}{e} i(\cos 1 + \sin 1) - \frac{\pi}{e}(\cos 1 - \sin 1).$$

By Residue Theorem, we have

$$\int_{C_R} f(z)dz + \int_{-R}^{R} f(z)dz = 2\pi i \mathrm{Res}_{z=z_0} f(z) = \frac{\pi}{e} i(\cos 1 + \sin 1) - \frac{\pi}{e}(\cos 1 - \sin 1).$$

This means

$$\int_{-R}^{R} \frac{x\sin x}{x^2+2x+2} dx = \frac{\pi}{e}(\cos 1 + \sin 1) - \mathrm{Im} \int_{C_R} f(z)dz.$$

Notice that for $z \in C_R$, we have $|z^2+2z+2| \geq |z|^2 - 2|z| - 2 = R^2 - 2R - 2$ and thus

$$\left| \frac{z}{z^2+2z+2} \right| \leq \frac{R}{R^2-2R-2} \to 0, \text{ as } R \to \infty.$$

Then, by Jordan's lemma (with $a = 1$),

$$\lim_{R\to\infty} \int_{C_R} f(z)dz = \lim_{R\to\infty} \int_{C_R} \frac{ze^{iz}}{z^2+2z+2} dz = 0,$$

and hence $\mathrm{Im} \int_{C_R} f(z)dz \to 0$ as $R \to \infty$. Therefore,

$$\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+2x+2} dx = \frac{\pi}{e}(\cos 1 + \sin 1).$$

□

Problem 9.7. For constants $a, b > 0$, evaluate

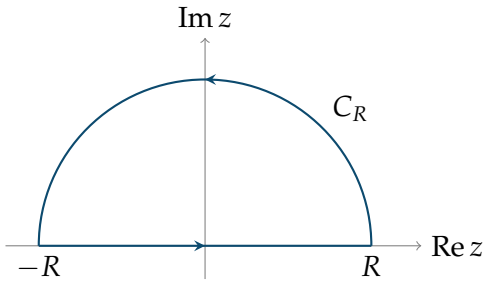
$$\int_0^\infty \frac{x \sin(bx)}{a^2 + x^2} dx$$

by use of a version of the Residue Theorem. Give a simple sketch of the type of contours used in the proof of the residue theorem you use.

Solution. Consider the function

$$f(z) = \frac{z e^{ibz}}{a^2 + z^2}$$

and the contour $C = C_R + [-R, R]$, with $R > a$.



Then f has a simple pole at $z_0 = ai$ and

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{z_0 e^{ibz_0}}{2z_0} = \frac{e^{ibz_0}}{2} = \frac{e^{ibia}}{2} = \frac{e^{-ab}}{2}.$$

Thus

$$2\pi i \operatorname{Res}_{z=z_0} f(z) = i\pi e^{-ab}.$$

The residue theorem gives

$$\int_{C_R} f(z)\, dz + \int_{-R}^{R} f(z)\, dz = 2\pi i \operatorname{Res}_{z=z_0} f(z) = i\pi e^{-ab}.$$

This means

$$\operatorname{Im} \int_{C_R} f(z)\, dz + \int_{-R}^{R} \frac{x \sin(bx)}{a^2 + x^2}\, dx = \pi e^{-ab}.$$

For $z \in C_R$, we have $|a^2 + z^2| \geq |z|^2 - a^2 = R^2 - a^2$, and thus

$$\left| \frac{z}{a^2 + z^2} \right| \leq \frac{R}{R^2 - a^2} \to 0, \text{ as } R \to \infty.$$

Then, by Jordan's lemma (with $b > 0$ in the exponent),

$$\lim_{R \to \infty} \int_{C_R} \frac{z}{a^2 + z^2} e^{ibz}\, dz = 0$$

and thus

$$\lim_{R\to\infty} \operatorname{Im} \int_{C_R} f(z)\, dz = 0.$$

Notice further that $\dfrac{x\sin(bx)}{a^2+x^2}$ is an even function, we have

$$\operatorname{Im} \int_{C_R} f(z)dz + 2\int_0^R \frac{x\sin(bx)}{a^2+x^2}dx = \pi e^{-ab}.$$

Then passing to the limit $R \to \infty$ gives

$$\int_0^\infty \frac{x\sin(bx)}{a^2+x^2}dx = \frac{\pi e^{-ab}}{2}.$$

□

Problem 9.8. By applying Residue Theorem, evaluate the following improper integral

$$\int_0^\infty \frac{x\sin x}{x^2+1}dx.$$

Sketch the contour you use, and give explicit estimates for any integrals you claim tend to zero.

Solution. Consider the function

$$f(z) = \frac{ze^{iz}}{z^2+1}.$$

Then we have

$$\int_0^\infty \frac{x\sin x}{x^2+1}dx = \operatorname{Im}\left(\frac{1}{2}2\pi i \operatorname*{Res}_{z=i} f(z)\right) = \operatorname{Im}(\pi i.\frac{e^{-1}}{2}) = \frac{\pi e^{-1}}{2}.$$

Here the factor $\frac{1}{2}$ comes from the evenness of the integrand, and the integral over the upper half-circle C_R tends to 0 by Jordan's lemma, exactly as in Problem 9.7 (the case $a = b = 1$). □

Problem 9.9.

(a) State a version of the Residue Theorem.

(b) By applying the Residue Theorem, evaluate the following improper integral

$$\int_0^\infty \frac{\sin x}{x}\, dx.$$

Sketch the contour you use, and give explicit estimates for any integrals you claim tend to zero.

Hint: Consider the principal value of the improper integral

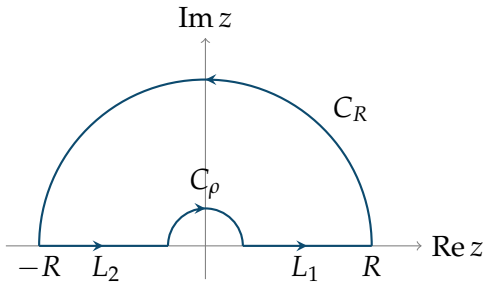
$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x}\, dx$$

Solution.

(a) Let C be a simple, closed, positive oriented contour, f be analytic on and inside C except finitely points $z_0, \ldots, z_n$. Then

$$\int_C f(z)dz = 2\pi i \sum_{k=0}^{n} \operatorname*{Res}_{z=z_k} f(z).$$

(b) Consider the function $f(z) = \dfrac{e^{iz}}{z}$ and the contour $C := L_1 + C_R + L_2 + C_\rho$, where $0 < \rho < R$.



Since f is analytic on and inside C, by Cauchy-Goursat theorem, we have

$$\int_{L_1} f(z)dz + \int_{C_R} f(z)dz + \int_{L_2} f(z)dz + \int_{C_\rho} f(z)dz = 0.$$

This means that

$$\int_{L_1} f(z)dz - \int_{-L_2} f(z)dz = -\int_{C_R} f(z)dz - \int_{C_\rho} f(z)dz.$$

Note that L_1 and $-L_2$ have parametric representations

$$\begin{aligned} L_1 &: \quad z = re^{i0} = r \quad (\rho \le r \le R) \\ -L_2 &: \quad z = re^{i\pi} = -r \quad (\rho \le r \le R). \end{aligned}$$

Then we have

$$\int_\rho^R \frac{e^{ir}}{r} dr - \int_\rho^R \frac{e^{-ir}}{r} dr = -\int_{C_R} f(z)dz - \int_{C_\rho} f(z)dz.$$

This means that

$$\int_\rho^R \frac{2i\sin r}{r} dr = -\int_{C_R} \frac{e^{iz}}{z} dz - \int_{C_\rho} \frac{e^{iz}}{z} dz.$$

We will use two following theorems.

Theorem A (Jordan's lemma). *Suppose that f is analytic in all points in the upper half plane above circle $|z| = R_0$, for $R > R_0$, for each $z \in C_R$, we have $|f(z)| \leq M_R \to 0$ as $R \to \infty$. Then*

$$\lim_{R\to\infty} \int_{C_R} e^{iaz} f(z)dz = 0 \quad (a > 0).$$

Theorem B. *Suppose that f has a simple pole at $z = x_0$ on the real line with a Laurent series representation in a punctured disk $0 < |z - x_0| < R_2$ with residue B_0. If C_ρ denote the upper half of a circle $|z - x_0| = \rho$, where $\rho < R_2$, and clockwise oriented, then*

$$\lim_{\rho\to 0} \int_{C_\rho} f(z)dz = -B_0\pi i.$$

Since for $z \in C_R$, we have

$$\left|\frac{1}{z}\right| = \frac{1}{R} \to 0, \text{ as } R \to \infty.$$

Then by Theorem A

$$\lim_{R\to\infty} \int_{C_R} e^{iz}/z dz = 0$$

We also have that

$$B_0 = \operatorname*{Res}_{z=0} f(z) = e^{i0} = 1.$$

Therefore, the Theorem B gives

$$\lim_{\rho\to 0} \int_{C_\rho} f(z)dz = -\pi i.$$

Therefore,

$$2i \int_0^\infty \frac{\sin x}{x} dx = \pi i,$$

or equivalently

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

□

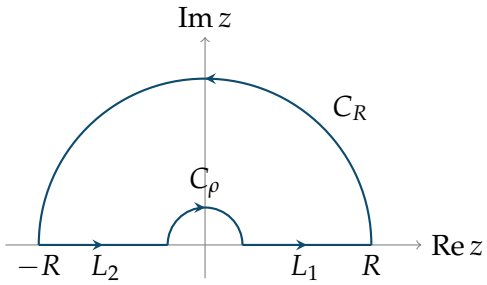
Problem 9.10. Let $-1 < a < 3$. Evaluate the integral

$$\int_0^\infty \frac{x^a}{(x^2+1)^2} dx.$$

Solution. Consider the branch

$$f(z) = \frac{\exp[a \log z]}{(z^2+1)^2}, \quad (|z| > 0,\ -\pi/2 < \arg z < 3\pi/2)$$

of the multivalued function $f(z) = \dfrac{\exp[a \log z]}{(z^2+1)^2}$ and the contour $C := L_1 + C_R + L_2 + C_\rho$, where $\rho < 1 < R$.



Since

$$f(z) = \frac{\phi(z)}{(z-i)^2}, \text{ where } \phi(z) = \frac{\exp[a \log z]}{(z+i)^2}$$

and

$$\begin{aligned}\phi'(z) &= \frac{a \exp[(a-1)\log z](z+i)^2 - 2\exp[a \log z](z+i)}{(z+i)^4} \\ &= \frac{a \exp[(a-1)\log z](z+i) - 2\exp[a \log z]}{(z+i)^3}.\end{aligned}$$

Thus

$$\begin{aligned}\operatorname{Res}_{z=i} f(z) = \phi'(i) &= \frac{a \exp[(a-1)(\ln|i| + i \arg i)]2i - 2\exp[a(\ln|i| + i \arg i)]}{(2i)^3} \\ &= \frac{a \exp[(a-1)i\frac{\pi}{2}]2i - 2\exp[ia\frac{\pi}{2}]}{-8i} \\ &= \frac{ia[\cos\frac{\pi(a-1)}{2} + i\sin\frac{\pi(a-1)}{2}] - [\cos\frac{\pi a}{2} + i\sin\frac{\pi a}{2}]}{-4i} \\ &= \frac{ia[\sin\frac{\pi a}{2} - i\cos\frac{\pi a}{2}] - [\cos\frac{\pi a}{2} + i\sin\frac{\pi a}{2}]}{-4i} \\ &= \frac{(a-1)\cos\frac{\pi a}{2} + i(a-1)\sin\frac{\pi a}{2}}{-4i}\end{aligned}$$

and hence

$$2\pi i \operatorname*{Res}_{z=i} f(z) = \frac{\pi(1-a)\cos\frac{a\pi}{2}}{2} + i\frac{\pi(1-a)\sin\frac{a\pi}{2}}{2}.$$

Since for $0 < \rho < 1 < R$, f is analytic on and inside C except only at $z = i$ which is an interior point of C, by the Residue Theorem, we have

$$\int_{C_\rho} f(z)\,dz + \int_{C_R} f(z)\,dz + \int_{L_1} f(z)\,dz + \int_{L_2} f(z)\,dz = 2\pi i \operatorname{Res}_{z=i} f(z).$$

This means

$$\int_{C_\rho} f(z)\,dz + \int_{C_R} f(z)\,dz + \int_{L_1} f(z)\,dz - \int_{-L_2} f(z)\,dz = \frac{\pi(1-a)\cos\frac{a\pi}{2}}{2} + i\frac{\pi(1-a)\sin\frac{a\pi}{2}}{2}.$$

Note that L_1 and $-L_2$ have parametric representations

$$\begin{aligned} L_1 &: z = re^{i0} = r \quad (\rho \le r \le R) \\ -L_2 &: z = re^{i\pi} = -r \quad (\rho \le r \le R). \end{aligned}$$

Then

$$\begin{aligned} \int_{L_1} f(z)dz - \int_{-L_2} f(z)dz &= \int_\rho^R \frac{\exp(a\ln r)}{(r^2+1)^2}dr + \int_\rho^R \frac{\exp(a\ln r + ai\pi)}{(r^2+1)^2}dr \\ &= \int_\rho^R \frac{r^a}{(r^2+1)^2}dr + \int_\rho^R \frac{r^a\exp(ai\pi)}{(r^2+1)^2}dr \\ &= (1+\exp(ai\pi))\int_\rho^R \frac{r^a}{(r^2+1)^2}dr \\ &= (1+(\cos a\pi + i\sin a\pi))\int_\rho^R \frac{r^a}{(r^2+1)^2}dr \\ &= (1+\cos a\pi)\int_\rho^R \frac{r^a}{(r^2+1)^2}dr + i\sin a\pi \int_\rho^R \frac{r^a}{(r^2+1)^2}dr. \end{aligned}$$

Thus

$$\operatorname{Re}\int_{C_\rho} f(z)dz + \operatorname{Re}\int_{C_R} f(z)dz + (1+\cos a\pi)\int_\rho^R \frac{r^a}{(r^2+1)^2}dr = \frac{\pi(1-a)\cos\frac{a\pi}{2}}{2}.$$

Note that for $z \in C_\rho$, we have $|z^2+1|^2 \ge (|z|^2-1)^2 = (1-\rho^2)^2$ then

$$\left|\operatorname{Re}\int_{C_\rho} f(z)dz\right| \le \frac{\rho^a}{(1-\rho^2)^2}\pi\rho = \frac{\pi\rho^{a+1}}{(1-\rho^2)^2}.$$

Since $a > -1$, we have $a+1 > 0$, and thus

$$\left|\operatorname{Re}\int_{C_\rho} f(z)dz\right| \le \frac{\pi\rho^{a+1}}{(1-\rho^2)^2} \to \frac{\pi\cdot 0}{(1-0)^2} = 0, \text{ as } \rho \to 0.$$

Note that for $z \in C_R$, we have $|z^2+1|^2 \geq (|z|^2-1)^2 = (R^2-1)^2$ then

$$\left|\mathrm{Re}\int_{C_R} f(z)dz\right| \leq \frac{R^a}{(R^2-1)^2}\pi R = \frac{\pi R^{a+1}}{(R^2-1)^2}.$$

Since $a<3$, we have $a+1<4$, and thus

$$\left|\mathrm{Re}\int_{C_R} f(z)dz\right| \leq \frac{\pi R^{a+1}}{(R^2-1)^2} \to 0, \text{ as } R\to\infty.$$

Thus we have

$$(1+\cos a\pi)\int_0^\infty \frac{x^a}{(x^2+1)^2}dx = \frac{\pi(1-a)\cos\frac{a\pi}{2}}{2}$$

Since $1+\cos a\pi = 2\cos^2\frac{a\pi}{2}$, we have

$$\int_0^\infty \frac{x^a}{(x^2+1)^2}dx = \frac{\pi(1-a)}{4\cos\frac{a\pi}{2}}.$$

□

Problem 9.11. By using the Residue Theorem, calculate the following integral

$$\int_0^\infty \frac{\ln x}{x^2+4}\,dx.$$

Solution. Consider the branch

$$f(z) = \frac{\log z}{z^2+4} \quad (-\frac{\pi}{2} < \arg z < \frac{3\pi}{2})$$

or the multivalued function $\log z/(z^2+4)$ and the contour $C = L_1 + C_R + L_2 + C_\rho$, where $L_1 = [\rho, R]$, $L_2 = [-R, -\rho]$ and $0<\rho<2<R$. Then f has a simple pole at $z_0 = 2i$ and

$$\mathrm{Res}_{z=z_0} f(z) = \frac{\log z_0}{z_0+2i} = \frac{\ln 2 + i\pi/2}{4i}.$$

Therefore,

$$2\pi i\,\mathrm{Res}_{z=z_0} f(z) = \frac{\pi\ln 2}{2} + \frac{i\pi^2}{4}.$$

By the Residue theorem, we have

$$\int_{C_R} f(z)\,dz + \int_{L_1} f(z)\,dz + \int_{L_2} f(z)\,dz + \int_{C_\rho} f(z)\,dz = \frac{\pi\ln 2}{2} + \frac{i\pi^2}{4}.$$

This is equivalent to

$$\int_{L_1} f(z)\,dz - \int_{-L_2} f(z)\,dz = \frac{\pi \ln 2}{2} + \frac{i\pi^2}{4} - \int_{C_R} f(z)\,dz - \int_{C_\rho} f(z)\,dz.$$

Note that L_1 and $-L_2$ has parametric representations

$$\begin{aligned} L_1 &: z = re^{i0} = r \quad (\rho \le r \le R) \\ -L_2 &: z = re^{i\pi} = -r \quad (\rho \le r \le R). \end{aligned}$$

Thus we have

$$\int_\rho^R \frac{\ln r}{r^2+4}\,dr + \int_\rho^R \frac{\ln r + i\pi}{r^2+4}\,dr = \frac{\pi \ln 2}{2} + \frac{i\pi^2}{4} - \int_{C_R} f(z)\,dz - \int_{C_\rho} f(z)\,dz.$$

Thus

$$2\int_\rho^R \frac{\ln r}{r^2+4}\,dr = \frac{\pi \ln 2}{2} - \operatorname{Re}\int_{C_R} f(z)\,dz - \operatorname{Re}\int_{C_\rho} f(z)\,dz.$$

For $z \in C_R$, we have $|z^2+4| \ge |z|^2 - 4 = R^2 - 4$, and

$$|\log z| \le |\ln |z|| + |\arg z| = \ln R + \pi.$$

Thus

$$\left|\operatorname{Re}\int_{C_R} f(z)\,dz\right| \le \frac{(\ln R + \pi)\pi R}{R^2-4} \to 0, \text{ as } R \to \infty.$$

For $z \in C_\rho$ with $\rho < 1$, we have $|z^2+4| \ge 4 - \rho^2$ and

$$|\log z| \le |\ln |z|| + |\arg z| = -\ln \rho + \pi.$$

Thus

$$\left|\operatorname{Re}\int_{C_\rho} f(z)\,dz\right| \le \frac{(-\ln \rho + \pi)\pi\rho}{4-\rho^2} \to 0, \text{ as } \rho \to 0.$$

Therefore,

$$\int_0^\infty \frac{\ln x}{x^2+4}\,dx = \frac{\pi \ln 2}{4}.$$

□

10 Analytic functions

Problem 10.1. Let $p(z)$ be a polynomial of degree n and take $R > 0$ so large that the circle $\gamma(t) = Re^{it}, 0 \leq t \leq 2\pi$, encloses all the zeros of p in its interior. Prove that

$$\int_\gamma \frac{p'(z)}{p(z)}\, dz = 2\pi n i$$

Solution. Suppose we have k zeros $z_1, \cdots, z_k$ with each z_i with multiplicity e_i, then

$$p(z) = \prod_{i=1}^{k} (z - z_i)^{e_i} \Rightarrow p'(z) = \sum_{i=1}^{k} e_i (z - z_i)^{e_i - 1} \prod_{j=1, j\neq i}^{k} (z - z_j)^{e_j}$$

Thus

$$\frac{p'(z)}{p(z)} = \sum_{i=1}^{k} e_i (z - z_i)^{-1}$$

Then by the reisude theorem we get

$$\int \frac{p'(z)}{p(z)} = 2\pi i \sum e_i = 2\pi i n$$

since $\int \frac{e_i}{z - z_i}\, dz = 2\pi i e_i$ and the sum of multiplicities is the ploynomial degree. □

Problem 10.2. Let $p(z) = (z - a_1)(z - a_2)\cdots(z - a_n)$ be a polynomial of degree n with distinct complex zeros $a_1, a_2, \cdots, a_n$ and $n \geq 2$. Take $R > 0$ so large that the circle $|z| = R$ encloses all the zeros of p in its interior. Prove that

(a)

$$\int_{|z|=R} \frac{dz}{p(z)} = 2\pi i \sum_{k=1}^{n} \frac{1}{\prod_{j\neq k}(a_k - a_j)}$$

(b)

$$\sum_{k=1}^{n} \frac{1}{\prod_{j\neq k}(a_k - a_j)} = 0$$

Solution. First part residue theorem. The second part, parametrize the contour $|z| = R$ by $z = Re^{i\theta}$ then

$$\left| \int_0^{2\pi} \frac{1}{\prod_{i=1}^{n} (Re^{i\theta} - a_i)}\, Rie^{i\theta}\, d\theta \right| \leq 2\pi R \frac{1}{(R - |a_j|)^n} \qquad (*)$$

Where a_j is the farthest zero of $p(z)$ from the origin, so that $|Re^{i\theta} - a_i| \geq R - |a_i| \geq R - |a_j|$ for every i. By the residue theorem

$$\int_{|z|=R} \frac{dz}{p(z)} = 2\pi i \sum_{k=1}^{n} \frac{1}{\prod_{j\neq k}(a_k - a_j)}$$

That holds for any $|z| = R$ that contains all zeros, thus taking the limit as $R \to \infty$ in $(*)$ and $n \geq 2$ give us the desired.
$(\Rightarrow)$Another approach, note that if we use partial fractions we get

$$\frac{1}{\prod(z-a_i)} = \sum_{i=1}^{n} \frac{b_i}{z-a_i} \qquad (*)$$

To find $b_i's$ we choose a small circle $C_i : |z-a_i| = \varepsilon$ such that a_j is not inside C_i for $j \neq i$. Then using the residue theorem we get

$$\int_{C_i} \frac{1}{p(z)} dz = \int_{C_i} \sum_{k=1}^{n} \frac{b_k}{z-a_k} dz$$

Hence

$$2\pi i \frac{1}{\prod_{k\neq i}(a_i - a_k)} = 2\pi i b_i$$

So

$$b_i = \frac{1}{\prod_{k\neq i}(a_i - a_k)}$$

But from $(*)$ we get

$$1 = \sum_{i=1}^{n} \frac{\prod_{j=1}^{n}(z-a_j) b_i}{z-a_i}$$

Thus the coefficient of z^{n-1} on the right hand side would be $\sum_{i=1}^{n} b_i = \sum_{i=1}^{n} \frac{1}{\prod_{k\neq i}(a_i-a_k)} = 0$ because the coefficient of z^{n-1} on the left hand side is 0. □

Problem 10.3. Suppose that the function $f(z)$ has a simple pole at $z = a$, and let $g(z)$ be an analytic function defined in an open set containing a. Show that $\operatorname{Res}(fg; a) = g(a)\operatorname{Res}(f; a)$, where $\operatorname{Res}(f; a)$ means the residue of f at $z = a$.

Solution. Since f has a simple pole at a and g is analytic at a,

$$f(z) = \sum_{i=-1}^{\infty} b_i (z-a)^i,$$
$$g(z) = \sum_{i=0}^{\infty} c_i (z-a)^i$$

for all z in a small enough neighborhood of a. In that same neighborhood, we can directly compute

$$(fg)(z) = \frac{b_{-1}c_0}{z-a} + (b_{-1}c_1 + b_0c_0) + (b_{-1}c_2 + b_0c_1 + b_1c_0)(z-a) + \cdots.$$

Therefore,

$$\operatorname{Res}(fg; a) = b_{-1}c_0 = \operatorname{Res}(f; a)g(a).$$

□

Problem 10.4. Suppose that f is an entire function, and it satisfies

$$|f(z)| \leq A + B|z|^k$$

for all complex numbers z and $k > 0$. Prove that f is a polynomial.

Solution. Let us recall the corollary of Cauchy integral theorem.

> **Corollary**. *If a function f is analytic at a point z_0 then*
>
> $$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(s)ds}{(s - z_0)^{n+1}}$$
>
> *where C is a simple, closed contour and z_0 is a interior point of C.*

Define $C_r = \{z \in \mathbb{C} \mid |z| = r\}$. We have

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(s)ds}{s^{n+1}}.$$

Since

$$\sup_{|s|=r} |f(s)| \leq \sup_{|s|=r} (A + B|s|^k) = A + B|r|^k,$$

for every integer $n > k$ we have

$$|f^{(n)}(0)| \leq \frac{n!}{2\pi} \frac{A + B|r|^k}{|r|^{n+1}} \cdot 2\pi r = n! \frac{A + B|r|^k}{|r|^n} \to 0, \text{ as } r \to \infty.$$

Therefore,

$$f(z) = \sum_{0 \leq n \leq k} a_n z^n, \text{ for all } z \in \mathbb{C}, \text{ where } a_n = \frac{f^{(n)}(0)}{n!}.$$

□

Problem 10.5. Suppose $f(z) = u(x, y) + iv(x, y)$ $(z = x + iy)$ is an entire function, and it satisfies

$$|Du(x, y)| = (u_x^2 + u_y^2)^{1/2} \leq A + B|z|$$

for all complex numbers z, where A, B are positive numbers. Prove that f is either a quadratic polynomial or an affine function.

Solution. Note that $u_y = -v_x$ and $f'(z) = u_x + iv_x$, thus $(u_x^2 + u_y^2)^{1/2} = |f'(z)| \leq A + B|z|$. Then since $f(z)$ is entire by the Cauchy integral formula we have for any $a \in \mathbb{C}$

$$\int_{|z|=R} \frac{f'(z)}{(z - a)^3} = \pi i f'''(a)$$

Hence we will have for any R

$$\pi|f'''(a)| \leq \int_0^{2\pi} \left| \frac{f'(Re^{i\theta})}{R^3(e^{i\theta} - \frac{a}{R})^3} iRe^{i\theta}\, d\theta \right| \leq \int_0^{2\pi} \frac{A + BR}{R^3 m^3} R\, d\theta = \frac{2\pi}{R} \frac{A/R + B}{m^3} \to 0, \text{ as } R \to \infty$$

where $m = \min_\theta \left|e^{i\theta} - \frac{a}{R}\right| \geq 1 - \frac{|a|}{R} \geq \frac{1}{2}$ for $R \geq 2|a|$. Hence $f'''(a) = 0$ for every $a \in \mathbb{C}$, so f' is a polynomial of degree at most 1, and f is a polynomial of degree at most 2: a quadratic polynomial or an affine function. □

Problem 10.6. Let f be an entire function and assume that for each $n \in \mathbb{N}$ we have

$$|f(z)| \leq \left|f\left(\frac{z}{2}\right)\right| + \frac{1}{2^n} \quad \text{if} \quad 2^n \leq |z| \leq 2^{n+1}.$$

Prove that f is a constant function.

Solution. Since f is entire, it is in particular continuous, so it is bounded on the compact set $D = \{z \in \mathbb{C} \mid |z| \leq 2\}$. Let $|f(z)| \leq M$ for all $z \in D$.
Let $2^n \leq |z| \leq 2^{n+1}$. Then

$$\begin{aligned}
|f(z)| &\leq \left|f\left(\frac{z}{2}\right)\right| + \frac{1}{2^n} \\
&\leq \left|f\left(\frac{z}{4}\right)\right| + \frac{1}{2^{n-1}} + \frac{1}{2^n} \\
&\leq \left|f\left(\frac{z}{8}\right)\right| + \frac{1}{2^{n-2}} + \frac{1}{2^{n-1}} + \frac{1}{2^n} \\
&\;\;\vdots \\
&\leq \left|f\left(\frac{z}{2^n}\right)\right| + \sum_{i=1}^{n} \frac{1}{2^i} \\
&\leq M + 1.
\end{aligned}$$

Every z with $|z| \geq 2$ lies in one of these annuli, and $|f| \leq M$ for $|z| \leq 2$. Therefore f is a bounded entire function, so by Liouville's theorem, it is constant. □

Problem 10.7.

(a) If f is an entire function which possesses no or finitely many roots, is it true that f is a polynomial function? Justify your answer.

(b) If g is an entire function such that there exists a $R > 0$ such that $\operatorname{Re} g(z) > 0$ for all z with $|z| > R$, prove g is a constant function.

Solution.

(a) Let $f(z) = e^z$. Indeed, we have

$$|f(a + bi)| = |e^{a+bi}| = e^a |e^{bi}| = e^a |\cos b + i \sin b| = e^a > 0.$$

Alternatively, every polynomial function is surjective, because $\mathbb{C}$ is algebraically closed. So a non-constant function with no roots cannot be a polynomial.

(b) Let $f(z) = e^{-g(z)}$. Notice that f is entire because it is the composite of two entire functions. Then, for $|z| > R$,

$$|f(z)| = |e^{-g(z)}| = e^{-\operatorname{Re} g(z)} < e^0 = 1.$$

We also have that f is bounded on $\{z \mid |z| \leq R\}$ because it is a continuous function evaluated on a compact set. Therefore, f is bounded on all of $\mathbb{C}$; being an entire function, it is constant. This implies that g is also constant.

□

Problem 10.8. Let f be an entire function and suppose that $|f(z)| = |z|$ for $|z| < 1$. Prove that $f(z) = \lambda z$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$.

Solution. Define $D = \{z \in \mathbb{C} \mid |z| < 1\}$. Then for any $z \in D$, we have $|f(z)| < 1$ and $f(0) = 0$. Since f is entire and $f(0) = 0$, we define the function

$$g(z) = \begin{cases} \dfrac{f(z)}{z}, & z \neq 0, \\ f'(0), & z = 0 \end{cases}.$$

Then we have g is analytic at all z. Then for $r < 1$ and $\overline{D}_r := \{z \in \mathbb{C} \mid |z| \leq r\}$, by a corollary of the Maximum Modulus Principle, there is some $0 \neq z_r \in \overline{D}_r$ such that

$$|g(z)| \leq \frac{|f(z_r)|}{|z_r|} = 1, \text{ for all } z \in \overline{D}_r.$$

Let $r \to 1$, we have $|g(z)| \leq 1$ for all $z \in D$. Moreover as $|g(z)| = |f(z)|/|z| = 1$ for some $z \in D$, $z \neq 0$, we deduce that $|g|$ attains its maximum on D. Then by the Maximum Modulus Principle, for all $z \in D$ we have $g(z) = \lambda$ in which $|\lambda| = 1$. Therefore, $f(z) = \lambda z$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$. □

Problem 10.9. Suppose that f is holomorphic on the unit disk D and $|f(z)| \leq 1$ if $z \in D$. Suppose also that f has two fixed points in $D, f(\alpha) = \alpha, f(\beta) = \beta$. Show that f is the identity function.

Solution. Firstly we consider the case of $\alpha = 0$. Then by Schwarz Lemma, we have $f(z) = az$ for all $z \in D$. Since $f(\beta) = \beta$, we have f is a rotation, i.e. $f(z) = e^{i\theta}z$. By $f(\beta) = \beta$ again, we have $f(z) = z$.
Next suppose that $\alpha \neq 0$. Then consider the biholomorphic mapping Blaschke Factor of α

$$\psi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z}.$$

Note that $|z-\alpha| < |1-\bar{\alpha}z|$ is equivalent to

$$\begin{aligned} |z-\alpha|^2 < |1-\bar{\alpha}z|^2 &\iff (z-\alpha)(\bar{z}-\bar{\alpha}) < (1-\bar{\alpha}z)(1-\alpha\bar{z}) \\ &\iff z\bar{z} + \alpha\bar{\alpha} < 1 + \alpha z\bar{\alpha}\bar{z} \\ &\iff (|z|^2-1)(|\alpha|^2-1) > 0. \end{aligned}$$

The latter is always true, which implies that ψ_α maps the unit disk to itself. Then define $g = \psi \circ f \circ \psi^{-1}$. Then g is holomorphic, $|g(z)| \leq 1$ for all z and since $f(\alpha) = \alpha$, we have $g(0) = 0$. Moreover $g(\psi(\beta)) = (\psi \circ f)(\beta) = \psi(f(\beta)) = \psi(\beta)$. Therefore, $g(z) = z$ and thus $f(\psi^{-1}(z)) = \psi^{-1}(z)$ for all $z \in D$. Therefore, $f(z) = z$ for all $z \in D$. □

Problem 10.10. Suppose Ω is a bounded open set in the complex plane, f is an analytic function in Ω, and

$$\limsup_{n\to\infty} |f(z_n)| \leq M$$

for every sequence $\{z_n\}$ in Ω which converges to a boundary point of Ω. Prove that $|f(z)| \leq M$ for all $z \in \Omega$.

Solution. We define the function $g : \overline{\Omega} \to \mathbb{C}$ by

$$g(z) = \begin{cases} f(z) & , z \in \Omega \\ M & , z \in \partial\Omega. \end{cases}$$

From this construction, we guarantee that:

$$\limsup_{n\to\infty} |g(z_n)| \leq |g(\bar{z})|,$$

for every sequence $\{z_n\}$ in Ω converges to $\bar{z} \in \overline{\Omega}$. Then we show that $|g|$ reaches its supremum over $\overline{\Omega}$. Indeed, let $N := \sup_{z\in\overline{\Omega}} |g(z)| = \max\{\sup_{z\in\Omega} |g(z)|; M\}$. If $\sup_{z\in\Omega} |g(z)| = \sup_{z\in\Omega} |f(z)| \leq M$ then our proof completes. Otherwise, assume that $N = \sup_{z\in\Omega} |g(z)|$. By the definition of a supremum, one finds a sequence $\{z_n\} \subset \Omega \subset \overline{\Omega}$ with $\lim_{n\to\infty} |g(z_n)| = N$. As $\overline{\Omega}$ is compact, there is a subsequence $\{z_{n_k}\}$ of $\{z_n\}$ with $z_{n_k} \xrightarrow{k\to\infty} \bar{z} \in \overline{\Omega}$. Then by

$$N = \lim_{k\to\infty} |g(z_{n_k})| = \limsup_{k\to\infty} |g(z_{n_k})| \leq |g(\bar{z})| \leq N,$$

we conclude that $N = |g(\bar{z})|$.

Our next goal is to prove that $|g|$ reaches its supremum N (over $\overline{\Omega}$) at some point in $\partial\Omega$. Without loss of generality we may assume that $\bar{z} \in \Omega$. Let D be the connected component of Ω that contains $\bar{z}$ (i.e. D is the largest connected subset of Ω that contains $\bar{z}$). Consider any $z \in D$, we have z also belongs to Ω open. Thus there is $r > 0$ such that the open disk $D_r(z) \subset \Omega$. But since $D_r(z)$ is connected and containing z, $D_r(z)$ must be contained in

the connected component D ($D \ni z$). That means $D_r(z) \subset D$, and D is an open set. As D is open and connected, it is a domain in $\mathbb{C}$. The function g is analytic in the domain $D \subset \Omega$ and $|g|$ reaches its local (actually, global) maximum at $\overline{z} \in D$, we deduce from the maximum modulus principle that g is constant (with the value $g(\overline{z})$) throughout D. We prove later that $\partial D \subset \partial\Omega$; note that $\partial D \neq \emptyset$, since D is a nonempty bounded open subset of $\mathbb{C}$. For any $u \in \partial D \subset \partial\Omega$, one finds a sequence $\{u_n\} \subset D \subset \Omega$ and $u_n \to u$. Thus

$$N = \lim_{n\to\infty} |g(u_n)| = \limsup_{n\to\infty} |g(u_n)| \leq |g(u)| \leq N,$$

and $|g(u)| = N$. Since $u \in \partial\Omega$ and $|g(u)| = M$, we must have $M = N = \sup_{z\in\overline{\Omega}} |g(z)|$, and

$$|f(z)| = |g(z)| \leq M \quad \text{for all } z \in \Omega.$$

Now we finish the proof by verifying that: $\partial D \subset \partial\Omega$. Let $x \in \overline{D}$, and suppose that $x \in \Omega$ open. Then one finds $\rho > 0$ and $D_\rho(x) \subset \Omega$. As the intersection of the open disk $D_\rho(x)$ and D is non-empty (because $x \in \overline{D}$); and they are both connected, we deduce that $D_\rho(x) \cup D$ is also connected, and is contained in Ω. Thus $D_\rho(x) \cup D \subset D$, and $D_\rho(x) \subset D$. Hence one has $x \in D$. That is, for any $x \in \overline{D}$ we have the implication:

$$x \in \Omega \Longrightarrow x \in D.$$

Therefore,

$$\partial D = \overline{D} \setminus D \subset \overline{D} \setminus \Omega \subset \overline{\Omega} \setminus \Omega = \partial\Omega.$$

□

Problem 10.11. Let $D = \{z \mid |z| < 1\}$ and $\overline{D} = \{z \mid |z| \leq 1\}$. Suppose that $f : \overline{D} \setminus \{0\} \to \mathbb{C}$ is a continuous function that satisfies

1. f is analytic in $D \setminus \{0\}$;
2. $$\limsup_{z\to 0} |f(z)| < \infty;$$
   and
3. $f(z)$ is equal to a finite constant c on the unit circle $|z| = 1$.

Prove that f is a constant function.

Solution. First, we observe from the second assumption that $z = 0$ is a removable singularity of f. Indeed, assume that for all $r > 0$ and all $\delta > 0$, one finds $z \in D(0;r) \setminus \{0\}$

such that $|f(z)| > \delta$. Then for

$$\begin{aligned} &r = 1/2,\ \delta = 2:\ \text{find } 0 \neq z_2 \in D(0;1/2) \text{ and } |f(z_2)| > 2, \\ &r = 1/3,\ \delta = 3:\ \text{find } 0 \neq z_3 \in D(0;1/3) \text{ and } |f(z_3)| > 3, \\ &\vdots \end{aligned}$$

Thus we have constructed a sequence $z_k \to 0$ with $|f(z_k)| \to +\infty$, which contradicts the fact that $\limsup_{z\to 0} |f(z)| < \infty$. Hence there must be some $r > 0$ and $M > 0$ such that $|f(z)| \leq M$ for all $z \in D(0;r) \setminus \{0\}$. As f is analytic and bounded in the deleted neighborhood $0 < |z| < r$ of 0 and not analytic at 0, the Riemann's theorem gives us that $z = 0$ is a removable singularity of f. Therefore we have the representation:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \quad \text{in} \quad 0 < |z| < 1.$$

Then if one sets

$$g(z) = \begin{cases} f(z) & , z \in D \setminus \{0\} \\ a_0 & , z = 0 \end{cases},$$

g is indeed analytic in D and continuous in $\overline{D}$. Let $h(z) := g(z) - c$ for all $z \in \overline{D}$, we have $h = 0$ on the circle $|z| = 1$. If h is constant in D then it is also constant in $\overline{D}$ by its continuity, and we get our desired result. Otherwise we utilize a corollary of the Maximum Modulus Principle and see that for all $z \in \overline{D}$:

$$|h(z)| \leq |h(z_0)| \ \text{ for some } \ z_0 \text{ with } \ |z_0| = 1.$$

And thus $h(z) = 0$ in the whole $\overline{D}$ and we complete our proof here. □

Problem 10.12. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ where $\lim_{n\to\infty} |a_n|^{1/n} = 1$. Show that there exists a $\zeta \in \mathbb{C}$ with $|\zeta| = 1$ such that f has no analytic continuation to $\{z : |z - \zeta| < \varepsilon\}$ for any $\varepsilon > 0$.

Solution. By the Cauchy-Hadamard theorem, the radius of convergence R satisfies the equation

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n},$$

so we have $R = 1$ in our case. In particular, f is defined on $D = \{z \mid |z| < 1\}$.
Suppose that no such ζ exists; in particular, $f(z)$ is defined on all of ∂D. For every $z \in \partial D$, there exists an $\varepsilon_z > 0$ such that f can be analytically continued to all of $B(z, \varepsilon_z)$.
By Lebesgue's Covering Lemma, there exists an $\varepsilon > 0$ such that for every $z_0 \in \partial D$, $B(z_0, \varepsilon) \subseteq B(z, \varepsilon_z)$ for some $z \in \partial D$. Therefore, f can be analytically continued to $D' = \{z \mid |z| < 1 + \varepsilon\}$. But this would imply that $R \geq 1 + \varepsilon > 1$, which is a contradiction. □

Problem 10.13. Suppose u and v are real-valued nonconstant harmonic functions defined on a domain U in the complex plane and the product uv is also harmonic on U. Prove that there exists a real number t such that $u + itv$ is holomorphic on U.

Solution. Notice that

$$\begin{aligned}(uv)_{xx} + (uv)_{yy} &= (u_x v + uv_x)_x + (u_y v + uv_y)_y \\ &= u_{xx}v + 2u_x v_x + uv_{xx} + u_{yy}v + 2u_y v_y + uv_{yy} \\ &= (u_{xx} + u_{yy})v + u(v_{xx} + v_{yy}) + 2(u_x v_x + u_y v_y)\end{aligned}$$

From the assumption that u, v and uv are harmonic, we can then deduce that

$$u_x v_x + u_y v_y = 0.$$

In other words, at any point $z \in U$, the real vectors $(u_x(z), u_y(z))$ and $(v_x(z), v_y(z))$ are perpendicular. Therefore, at the points where $(v_x, v_y) \neq (0,0)$ (all points of U except isolated ones, because $v_x - iv_y$ is analytic and not identically zero, v being nonconstant), the vectors $(u_x(z), u_y(z))$ and $(v_y(z), -v_x(z))$ are collinear, so there exists a real-valued function $c(z)$ such that $(u_x, u_y) = c(z)(v_y, -v_x)$. We claim that c is constant.
By differentiating the defining relation(s) we get the following:

$$\begin{aligned}u_{xx} &= c_x v_y + cv_{xy} \\ u_{yy} &= -c_y v_x - cv_{xy} \\ u_{xy} &= c_y v_y + cv_{yy} \\ u_{xy} &= -c_x v_x - cv_{xx}.\end{aligned}$$

It follows that

$$\begin{aligned}0 = u_{xy} - u_{xy} &= c_y v_y + cv_{yy} - (-c_x v_x - cv_{xx}) = c_y v_y + c_x v_x, \\ 0 = u_{xx} + u_{yy} &= c_x v_y + cv_{xy} - c_y v_x - cv_{xy} = c_x v_y - c_y v_x.\end{aligned}$$

In other words, the vector (c_x, c_y) is perpendicular to (v_x, v_y) as well as $(-v_y, v_x)$. Therefore $(c_x, c_y) = (0,0)$ for every $z \in U$.
Notice that the Cauchy-Riemann equations for u and cv at z can be expressed as the single equation

$$(u_x(z), u_y(z)) = ((cv)_y(z), -(cv)_x(z)) = c(v_y(z), -v_x(z)),$$

so $u + itv$ is holomorphic on U with $t = c$. □

Problem 10.14. Let Ω be a connected open subset of $\mathbb{C}$. Suppose u is harmonic and real-valued in Ω.

(a) Show that $u_x - iu_y$ is analytic in Ω.
(b) If u^2 is also harmonic in Ω, show that u is constant in Ω.

Solution.

(a) Since u is harmonic, we have $u_{xx} + u_{yy} = 0$. We simply check the Cauchy-Riemann equations:

$$(u_x)_x = u_{xx} - (u_{xx} + u_{yy}) = (-u_y)_y$$
$$(u_x)_y = (u_y)_x = -(-u_y)_x.$$

(b) Notice that

$$(u^2)_{xx} = (2u \cdot u_x)_x = 2((u_x)^2 + u \cdot u_{xx}),$$
$$(u^2)_{yy} = (2u \cdot u_y)_y = 2((u_y)^2 + u \cdot u_{yy}).$$

Thus the assumption that u^2 is harmonic implies that

$$0 = (u^2)_{xx} + (u^2)_{yy} = 2((u_x)^2 + (u_y)^2 + u \cdot (u_{xx} + u_{yy})) = 2((u_x)^2 + (u_y)^2).$$

But $(u_x)^2$ and $(u_y)^2$ are squares of real numbers, so non-negative. Therefore we must have $(u_x)^2 = (u_y)^2 = 0$, which implies that $u_x = u_y = 0$, i.e. u is constant.

□

Problem 10.15. Suppose that f is a complex-valued function defined on $D = \{z : |z| < 1\}$ and $g = f^2$, $h = f^3$ are analytic in D. Show that f is analytic in D.

Solution. Suppose that f^2 and f^3 have zeroes of order n and m at z_0, we have $f(z)^2 = (z - z_0)^n \varphi(z), f(z)^3 = (z - z_0)^m \psi(z)$, where φ, ψ are analytic and $\varphi(z_0) \neq 0, \psi(z_0) \neq 0$. Then

$$(f(z))^6 = (f(z)^2)^3 = (z - z_0)^{3n} \varphi(z)^3$$

and

$$(f(z))^6 = (f(z)^3)^2 = (z - z_0)^{2m} \psi(z)^2.$$

Then

$$(z - z_0)^{3n-2m} = \frac{\psi(z)^2}{\varphi(z)^3}.$$

Since ψ^2/φ^3 is not zero near z_0, we must have $3n = 2m$. Then $m = \frac{3}{2}n > n$ and thus $f = f^3/f^2$ is analytic at z_0. Therefore, f is analytic. □